

Machine Learning with R and Python

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About

This book is a companion to the Machine Learning Class. It will be a repository of R and Python Code. The chapters will be updated as we progress through the class.

1 Introduction

1.1 Introduction to Machine Learning

Machine Learning (ML) is a branch of artificial intelligence (AI) that focuses on building systems that **learn patterns from data** and make predictions or decisions without being explicitly programmed to perform specific tasks. Instead of writing rules by hand, a machine learning algorithm “learns” from examples in the data.

1.2 What is Machine Learning?

Formally, machine learning is the process of using data to **train a model** to make accurate predictions or decisions. The model identifies patterns in the training data and generalizes these patterns to new, unseen data.

ML can be thought of as:

- **Predictive modeling** – learning a function that maps inputs (features) to outputs (targets).
 - **Automated decision-making** – systems that improve their performance over time without explicit reprogramming.
-

1.3 Major Types of Machine Learning

Machine learning is commonly categorized into several types based on the **nature of the task** and **availability of labeled data**.

1.3.1 1. Supervised Learning

- Uses **labeled data**, meaning that each training example has an input and a known output.
- The goal is to **learn a function** that maps inputs to outputs accurately.
- Typical tasks:
 - **Regression**: Predict a continuous variable (e.g., house prices).
 - **Classification**: Predict a categorical variable (e.g., spam vs. non-spam email).

1.3.2 2. Unsupervised Learning

- Works with **unlabeled data**, where the output is unknown.
- The goal is to **discover patterns or structure** in the data.
- Typical tasks:
 - **Clustering**: Group similar observations together (e.g., customer segmentation).
 - **Dimensionality reduction**: Reduce the number of features while retaining important information (e.g., PCA).

1.3.3 3. Semi-Supervised Learning

- Combines a small amount of labeled data with a large amount of unlabeled data.
- Useful when labeling data is expensive or time-consuming.

1.3.4 4. Reinforcement Learning

- Focuses on **learning through trial and error**.
- An agent learns to take actions in an environment to **maximize cumulative reward**.
- Examples: Robotics, game AI, recommendation systems.

1.4 Objectives of Machine Learning

The main objective of machine learning is to **build models that generalize well** from historical data to make accurate predictions or decisions on **new, unseen data**. Key considerations include:

1. **Accuracy** – How well does the model predict outcomes?

2. **Generalization** – Does the model perform well on unseen data, or is it overfitting the training data?
 3. **Interpretability** – Can humans understand how the model makes predictions?
-

1.5 Bias-Variance Tradeoff

A fundamental concept in machine learning is the **bias-variance tradeoff**, which explains the balance between:

- **Bias**: Error due to overly simplistic models that **underfit** the data.
- **Variance**: Error due to overly complex models that **overfit** the training data.

The goal is to find a model that **minimizes the total prediction error**:

[Total Error = Bias² + Variance + Irreducible Error]

- **Underfitting** → High bias, low variance
- **Overfitting** → Low bias, high variance

The optimal model balances bias and variance for the best predictive performance.

1.6 The Machine Learning Process

A typical workflow in machine learning consists of the following steps:

1. Data Collection

- Gather data from experiments, databases, or external sources.

2. Data Preprocessing

- Handle missing values, outliers, and feature engineering.

3. Model Selection

- Choose an appropriate algorithm (e.g., linear regression, random forest, neural networks).

4. Model Training

- Fit the model to the training data.

5. Model Evaluation

- Assess performance using metrics such as RMSE, accuracy, precision, recall, or AUC.

6. Model Tuning

- Adjust hyperparameters to improve performance.

7. Model Deployment

- Use the trained model to make predictions on new data.

8. Monitoring and Maintenance

- Continuously track model performance and retrain if necessary.

1.7 Summary

Machine learning allows us to build predictive models that **learn from data**. Understanding the different types of ML, their objectives, and the tradeoffs between bias and variance is critical to applying ML effectively. The remainder of this book explores practical algorithms, their implementation in **R and Python**, and strategies for model selection and interpretation.

2 Model Selection and Regularization - R Version

2.1 Overview

In this section, we'll use the **Ames Housing** dataset to demonstrate model selection and regularization.

We'll cover:

1. Splitting data into training and testing sets
 2. Performing **stepwise regression** with cross-validation
 3. Running **Ridge** and **Lasso** regression, and visualizing how λ affects RMSE and model complexity
 4. Exploring **Elastic Net**, varying λ to balance Ridge and Lasso
 5. Comparing all models on the test set
-

2.2 Step 0: Setup and Data Preparation

We'll use a reduced set of variables for speed and clarity.

```
# Suppress messages and warnings globally
knitr::opts_chunk$set(echo = TRUE, message = FALSE, warning = FALSE)

library(tidyverse)
library(caret)
library(glmnet)
library(AmesHousing)
```

```

library(GGally)

set.seed(123)

# Load data and split into 70% training, 30% testing
ames <- make_ordinal_ames() %>% mutate(id = row_number())
train <- ames %>% sample_frac(0.7)
test  <- anti_join(ames, train, by = "id")

# Select a manageable subset of predictors
keep <- c("Sale_Price", "Bedroom_AbvGr", "Year_Built", "Mo_Sold", "Lot_Area",
          "Street", "Central_Air", "First_Flr_SF", "Second_Flr_SF", "Full_Bath",
          "Half_Bath", "Fireplaces", "Garage_Area", "Gr_Liv_Area", "TotRms_AbvGrd")

train <- train %>% select(all_of(keep))
test  <- test  %>% select(all_of(keep))

# Convert categorical variables to factors
train <- train %>% mutate(across(c(Street, Central_Air), as.factor))
test  <- test  %>% mutate(across(c(Street, Central_Air), as.factor))

```

2.3 Step 1: Stepwise Regression with 10-Fold Cross Validation

We'll use backward stepwise regression as a traditional benchmark.

```

ctrl <- trainControl(method = "cv", number = 10)

step_model <- train(
  Sale_Price ~ ., data = train,
  method = "lmStepAIC",
  trControl = ctrl,
  trace = FALSE,
  direction = "backward"
)

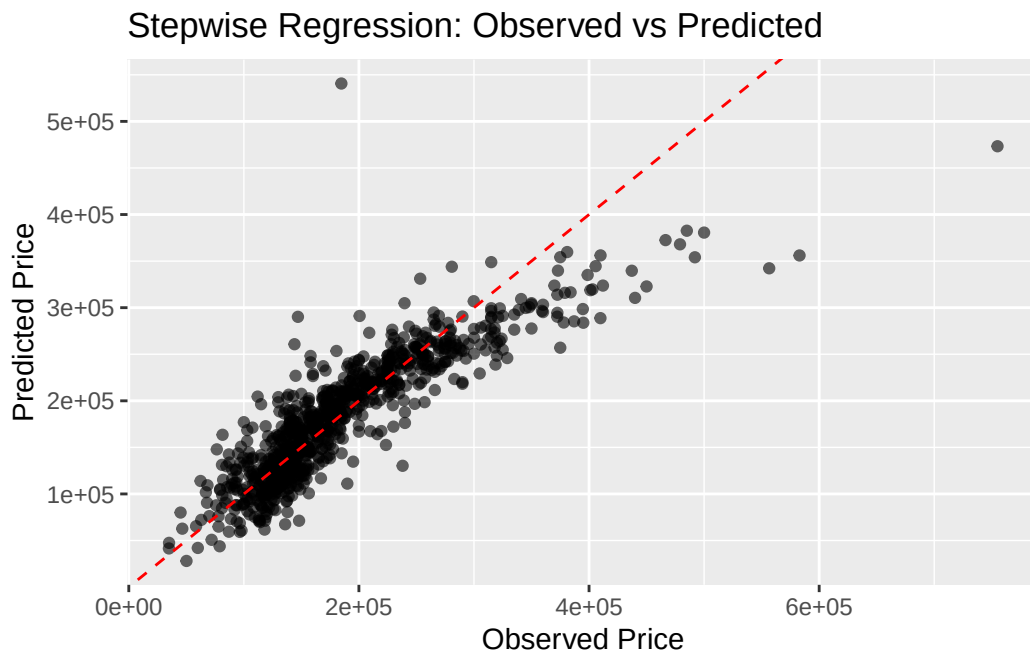
# Evaluate on the test set
step_pred <- predict(step_model, newdata = test)

```

```
step_perf <- postResample(step_pred, test$Sale_Price)
step_perf
```

```
      RMSE      Rsquared      MAE
3.820776e+04 7.670269e-01 2.613098e+04
```

```
# Visualize predicted vs observed
ggplot(data.frame(obs = test$Sale_Price, pred = step_pred), aes(obs, pred)) +
  geom_point(alpha = 0.6) +
  geom_abline(linetype = "dashed", color = "red") +
  labs(title = "Stepwise Regression: Observed vs Predicted",
       x = "Observed Price", y = "Predicted Price")
```



Note:

Stepwise regression is intuitive and fast, but it can be unstable.

Stepwise regression (forward, backward, or both) selects predictors one at a time based on how much they improve a criterion (like AIC or adjusted R^2).

2.4 Step 2: Ridge and Lasso Regression

Ridge and Lasso both shrink coefficients, but in different ways: - Ridge shrinks all coefficients toward zero. - Lasso can set some coefficients *exactly* to zero, performing variable selection.

```
x_train <- model.matrix(Sale_Price ~ ., train)[, -1]
y_train <- train$Sale_Price
x_test  <- model.matrix(Sale_Price ~ ., test)[, -1]
y_test  <- test$Sale_Price

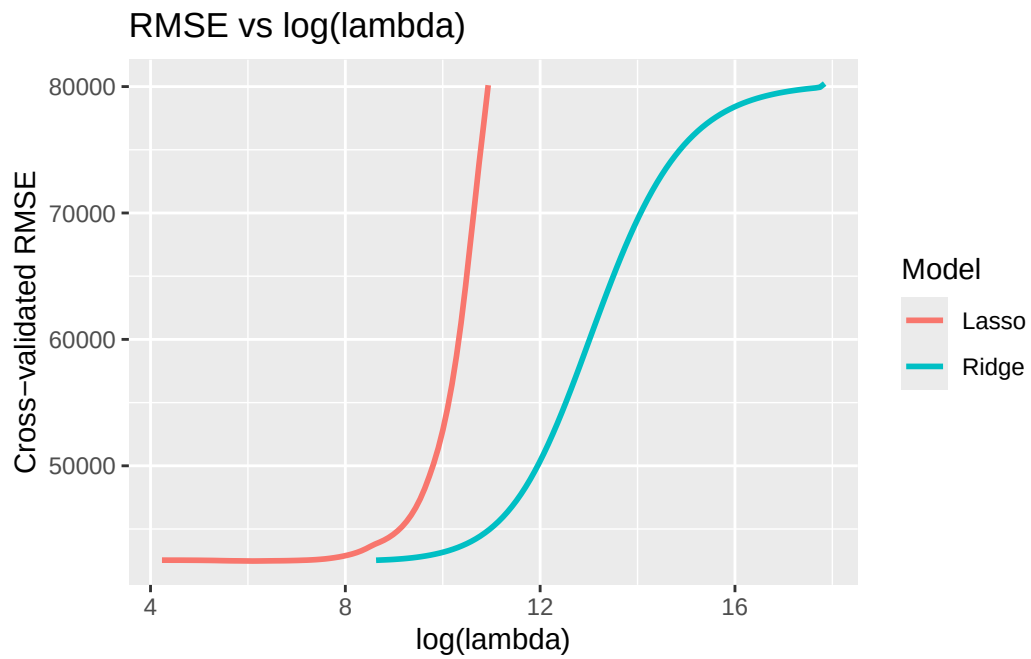
set.seed(123)
cv_ridge <- cv.glmnet(x_train, y_train, alpha = 0, nfolds = 10)
cv_lasso <- cv.glmnet(x_train, y_train, alpha = 1, nfolds = 10)
```

2.4.1 Visualizing RMSE and Model Complexity

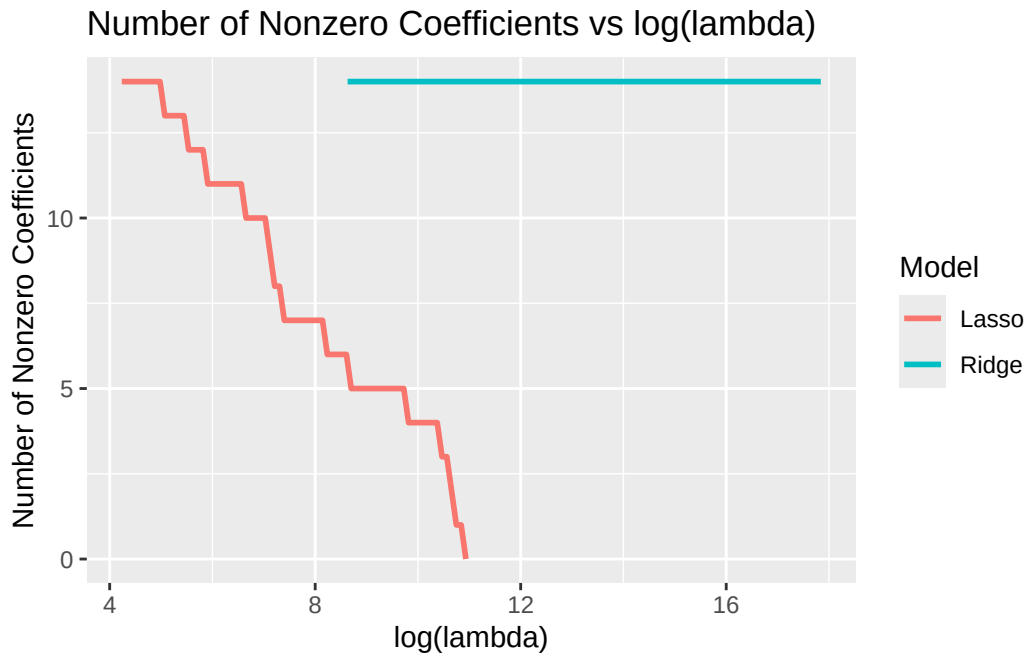
```
build_path_df <- function(cvfit, label) {
  fit <- cvfit$glmnet.fit
  tibble(
    lambda = fit$lambda,
    log_lambda = log(fit$lambda),
    RMSE = sqrt(cvfit$cvm),
    nonzero = colSums(abs(fit$beta) > 0),
    Model = label
  )
}

ridge_df <- build_path_df(cv_ridge, "Ridge")
lasso_df <- build_path_df(cv_lasso, "Lasso")
df <- bind_rows(ridge_df, lasso_df)

ggplot(df, aes(log_lambda, RMSE, color = Model)) +
  geom_line(size = 1) +
  labs(title = "RMSE vs log(lambda)", y = "Cross-validated RMSE", x = "log(lambda)")
```

```
ggplot(df, aes(log_lambda, nonzero, color = Model)) +  
  geom_line(size = 1) +  
  labs(title = "Number of Nonzero Coefficients vs log(lambda)",  
        y = "Number of Nonzero Coefficients", x = "log(lambda)")
```



Note:

- Increasing λ increases regularization.
- Ridge never eliminates variables, Lasso can.
- There's a sweet spot where RMSE is minimized.

2.5 Step 3: Elastic Net (Balancing Ridge and Lasso)

Elastic Net introduces α to control the mix between Ridge ($\alpha = 0$) and Lasso ($\alpha = 1$).

```
alpha_grid <- seq(0, 1, by = 0.25)

elastic_results <- map_df(alpha_grid, function(a) {
  cv_fit <- cv.glmnet(x_train, y_train, alpha = a, nfolds = 10)
  tibble(
    alpha = a,
    best_lambda = cv_fit$lambda.min,
    best_RMSE = sqrt(min(cv_fit$cvm)),
    nonzero = sum(abs(coef(cv_fit, s = "lambda.min")[-1]) > 0)
  )
})
```

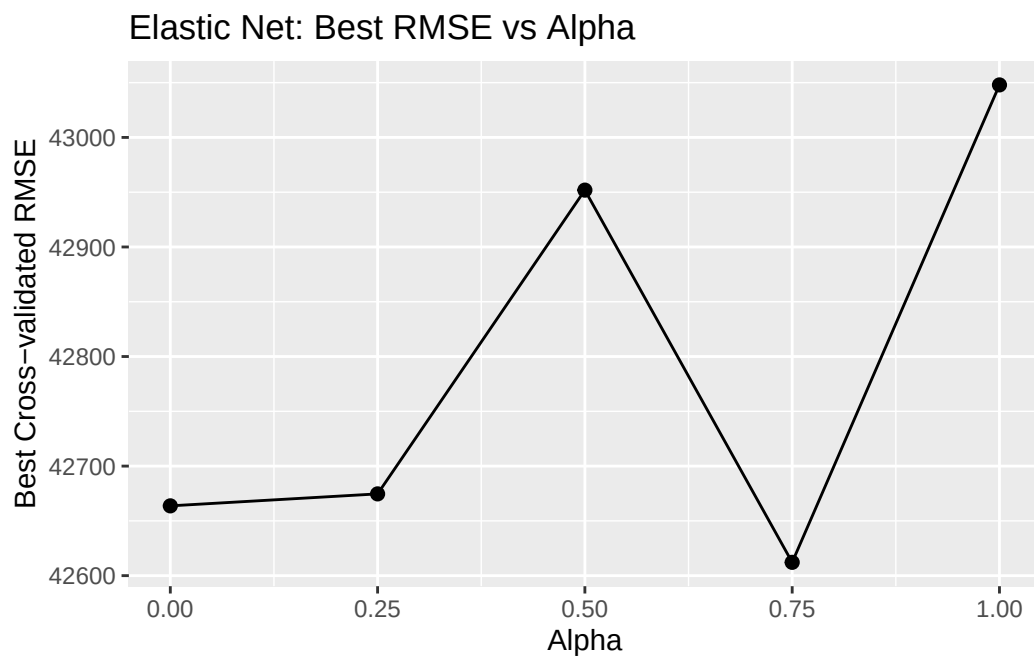
```
})
```

```
elastic_results
```

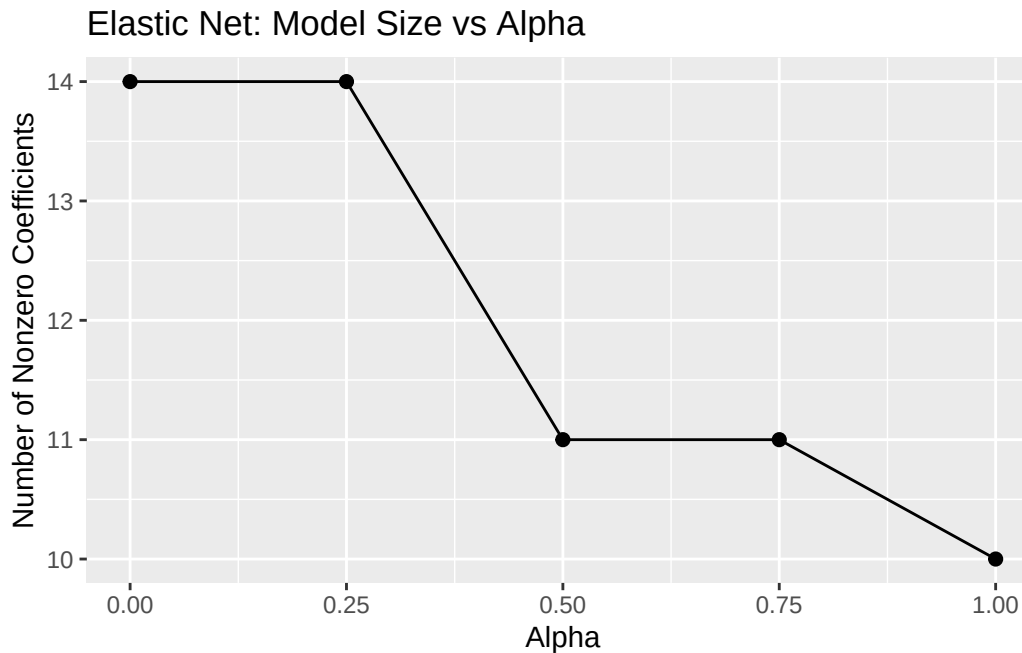
```
# A tibble: 5 x 4
  alpha best_lambda best_RMSE nonzero
  <dbl>     <dbl>     <dbl>   <int>
1 0       5603.     42664.    14
2 0.25    1224.     42675.    14
3 0.5     1552.     42952.    11
4 0.75     650.     42612.    11
5 1       1126.     43048.    10
```

2.5.1 Visualizing Effects

```
ggplot(elastic_results, aes(alpha, best_RMSE)) +
  geom_line() + geom_point(size = 2) +
  labs(title = "Elastic Net: Best RMSE vs Alpha",
       y = "Best Cross-validated RMSE", x = "Alpha")
```



```
ggplot(elastic_results, aes(alpha, nonzero)) +
  geom_line() + geom_point(size = 2) +
  labs(title = "Elastic Net: Model Size vs Alpha",
       y = "Number of Nonzero Coefficients", x = "Alpha")
```



Note:

Elastic Net combines the strengths of Ridge and Lasso: - Ridge for stability. - Lasso for sparsity. - Often performs best at intermediate values.

2.6 Step 4: Comparing All Models on the Test Set

We'll compare Stepwise, Ridge, Lasso, and the best Elastic Net.

```
ridge_pred <- predict(cv_ridge, newx = x_test, s = "lambda.min")
lasso_pred <- predict(cv_lasso, newx = x_test, s = "lambda.min")

best_alpha <- elastic_results %>% arrange(best_RMSE) %>% slice(1) %>% pull(alpha)
cv_en_best <- cv.glmnet(x_train, y_train, alpha = best_alpha, nfolds = 10)
elastic_pred <- predict(cv_en_best, newx = x_test, s = "lambda.min")
```

```

compare_tbl <- bind_rows(
  Stepwise = as_tibble_row(postResample(step_pred, test$Sale_Price)),
  Ridge = as_tibble_row(postResample(ridge_pred, y_test)),
  Lasso = as_tibble_row(postResample(lasso_pred, y_test)),
  Elastic = as_tibble_row(postResample(elastic_pred, y_test)),
  .id = "Model"
)

colnames(compare_tbl) <- c("Model", "RMSE", "Rsquared", "MAE")
compare_tbl <- compare_tbl %>% arrange(RMSE)
compare_tbl

```

```

# A tibble: 4 x 4
  Model      RMSE Rsquared    MAE
  <chr>    <dbl>   <dbl>  <dbl>
1 Stepwise 38208.   0.767 26131.
2 Lasso    38297.   0.767 26045.
3 Elastic  38301.   0.767 26039.
4 Ridge    38412.   0.766 26022.

```

```

# -----

# Display the best cross-validated Lasso, Ridge, and Elastic Net models

# -----

library(glmnet)
library(knitr)

# Helper function to summarize a fitted cv.glmnet model

summarize_glmnet <- function(cvfit, X_train, y_train, X_test, y_test, model_name) {
  cat("\n===== \n")
  cat("Model:", model_name, "\n")
  cat("===== \n")

  # Extract best lambda
  best_lambda <- cvfit$lambda.min

```

```

cat("Best Lambda (lambda.min):", round(best_lambda, 6), "\n")

# Coefficients (non-zero)
coefs <- as.matrix(coef(cvfit, s = "lambda.min"))
nonzero <- coefs[coefs != 0, , drop = FALSE]

# Print nonzero coefficients
cat("\nNonzero Coefficients:\n")
print(knitr::kable(as.data.frame(nonzero), digits = 4))

# Predictions and performance
pred_train <- predict(cvfit, newx = X_train, s = "lambda.min")
pred_test  <- predict(cvfit, newx = X_test, s = "lambda.min")

rmse <- function(y, yhat) sqrt(mean((y - yhat)^2))
rsq  <- function(y, yhat) 1 - sum((y - yhat)^2) / sum((y - mean(y))^2)

train_rmse <- rmse(y_train, pred_train)
test_rmse  <- rmse(y_test, pred_test)
test_r2    <- rsq(y_test, pred_test)

# Return one-row summary
data.frame(
  Model = model_name,
  Lambda = round(best_lambda, 6),
  Train_RMSE = round(train_rmse, 2),
  Test_RMSE = round(test_rmse, 2),
  Test_R2 = round(test_r2, 3),
  Nonzero_Coeff = nrow(nonzero)
)
}

# Summarize and display all three models
results <- rbind(
  summarize_glmnet(cv_lasso, x_train, y_train, x_test, y_test, "Lasso"),
  summarize_glmnet(cv_ridge, x_train, y_train, x_test, y_test, "Ridge"),
  summarize_glmnet(cv_en_best, x_train, y_train, x_test, y_test, "Elastic Net")
)

```

```

=====
Model: Lasso

```

=====
Best Lambda (lambda.min): 444.0297

Nonzero Coefficients:

	lambda.min
:-----	:-----
(Intercept)	-1536712.4502
Bedroom_AbvGr	-12871.7248
Year_Built	786.0097
Lot_Area	0.3320
StreetPave	14782.6929
Central_AirY	3463.4742
First_Flr_SF	22.9872
Half_Bath	-1509.7005
Fireplaces	11647.3233
Garage_Area	62.2549
Gr_Liv_Area	74.6167
TotRms_AbvGrd	1301.9029

=====
Model: Ridge

=====
Best Lambda (lambda.min): 5603.019

Nonzero Coefficients:

	lambda.min
:-----	:-----
(Intercept)	-1398127.9251
Bedroom_AbvGr	-13051.9562
Year_Built	706.9534
Mo_Sold	105.9352
Lot_Area	0.4001
StreetPave	22908.4375
Central_AirY	6951.0203
First_Flr_SF	45.3168
Second_Flr_SF	21.9030
Full_Bath	4203.4612
Half_Bath	-26.1950
Fireplaces	12666.1199

```
|Garage_Area      |      64.5695|
|Gr_Liv_Area     |      42.3845|
|TotRms_AbvGrd  |     3067.4622|
```

```
=====
```

```
Model: Elastic Net
```

```
=====
```

```
Best Lambda (lambda.min): 592.0396
```

```
Nonzero Coefficients:
```

```
|          |      lambda.min|
|:-----|:-----:|
|(Intercept) | -1532932.2050|
|Bedroom_AbvGr | -12808.7530|
|Year_Built   |      783.9250|
|Lot_Area     |       0.3319|
|StreetPave   |     14821.6671|
|Central_AirY |     3497.9772|
|First_Flr_SF |      23.2544|
|Half_Bath    |    -1319.3051|
|Fireplaces   |     11686.0135|
|Garage_Area  |       62.4230|
|Gr_Liv_Area  |       73.9778|
|TotRms_AbvGrd |     1388.0576|
```

```
# Combined performance table
cat("\n\n***Model Performance Comparison***\n")
```

```
***Model Performance Comparison***
```

```
kable(results, caption = "Performance of Cross-Validated Regularized Models")
```

Table 2.1: Performance of Cross-Validated Regularized Models

Model	Lambda	Train_RMSE	Test_RMSE	Test_R2	Nonzero_Coeff
Lasso	444.0297	41745.42	38296.94	0.764	12

Model	Lambda	Train_RMSE	Test_RMSE	Test_R2	Nonzero_Coeff
Ridge	5603.0194	41830.59	38411.78	0.763	15
Elastic Net	592.0396	41748.11	38300.92	0.764	12

2.7 Key Takeaways

- Stepwise selection is simple but can lead to unstable models.
- Ridge adds stability via coefficient shrinkage.
- Lasso enforces sparsity by removing weak predictors.
- Elastic Net balances both effects.
- Regularization often produces models that generalize better than purely stepwise ones.

3 Model Selection and Regularization - Python Version

3.1 Overview

This section parallels the R version of Model Selection and Regularization. We will:

1. Load the Ames Housing data
 2. Split into training and test sets
 3. Perform Ridge, Lasso, and Elastic Net regressions with 10-fold CV
 4. Visualize RMSE and coefficient shrinkage patterns
 5. Compare test set performance
-

3.2 Step 0: Setup and Data

We'll use scikit-learn for modeling.

If the CSVs `ames_training.csv` and `ames_testing.csv` are not present, you can load Ames Housing via the `fetch_openml` function.

```
import warnings
warnings.filterwarnings("ignore")

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import fetch_openml
from sklearn.model_selection import train_test_split, KFold, cross_val_score
```

```

from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.linear_model import RidgeCV, LassoCV, ElasticNetCV
from sklearn.metrics import mean_squared_error

# Load Ames dataset directly
ames = fetch_openml(name="house_prices", as_frame=True)
df = ames.frame

# Subset variables to match the R version
keep = [
    "SalePrice", "BedroomAbvGr", "YearBuilt", "MoSold", "LotArea", "Street", "CentralAir",
    "1stFlrSF", "2ndFlrSF", "FullBath", "HalfBath", "Fireplaces", "GarageArea",
    "GrLivArea", "TotRmsAbvGrd"
]
df = df[keep].copy()

# Clean up names to match Python variable rules
df.columns = ["Sale_Price", "Bedroom_AbvGr", "Year_Built", "Mo_Sold", "Lot_Area", "Street",
              "Central_Air", "First_Flr_SF", "Second_Flr_SF", "Full_Bath", "Half_Bath",
              "Fireplaces", "Garage_Area", "Gr_Liv_Area", "TotRms_AbvGrd"]

# Drop missing rows
df = df.dropna()

# Train/test split (70/30)
train, test = train_test_split(df, test_size=0.3, random_state=123)

# Identify predictors and target
y_train = train["Sale_Price"]
y_test = test["Sale_Price"]
X_train = train.drop(columns="Sale_Price")
X_test = test.drop(columns="Sale_Price")

cat_vars = ["Street", "Central_Air"]
num_vars = [c for c in X_train.columns if c not in cat_vars]

# Preprocess: scale numeric, one-hot encode categorical
preprocessor = ColumnTransformer([
    ("num", StandardScaler(), num_vars),
    ("cat", OneHotEncoder(drop="first"), cat_vars)
])

```

```
])
```

3.3 Step 1: Ridge Regression

We'll use cross-validation to tune α .

Then we visualize RMSE vs $\log(\lambda)$ and the number of nonzero coefficients.

```
alphas = np.logspace(4, -4, 80)
ridge = Pipeline([
    ("prep", preprocessor),
    ("model", RidgeCV(alphas=alphas, scoring="neg_mean_squared_error", cv=10))
])
ridge.fit(X_train, y_train)

ridge_best = ridge.named_steps["model"].alpha_
print("Best Ridge alpha:", ridge_best)

# Evaluate RMSE on test set
ridge_pred = ridge.predict(X_test)
ridge_rmse = np.sqrt(mean_squared_error(y_test, ridge_pred))
print("Test RMSE (Ridge):", ridge_rmse)
```

```
Best Ridge alpha: 94.33732216299774
Test RMSE (Ridge): 37055.84511170759
```

3.4 Step 2: Lasso Regression

Lasso adds variable selection — some coefficients go exactly to zero.

```
lasso = Pipeline([
    ("prep", preprocessor),
    ("model", LassoCV(alphas=alphas, cv=10, max_iter=20000))
])
lasso.fit(X_train, y_train)
```

```

lasso_best = lasso.named_steps["model"].alpha_
print("Best Lasso alpha:", lasso_best)

lasso_pred = lasso.predict(X_test)
lasso_rmse = np.sqrt(mean_squared_error(y_test, lasso_pred))
print("Test RMSE (Lasso):", lasso_rmse)

```

```

Best Lasso alpha: 11.568875283162821
Test RMSE (Lasso): 37074.74978063246

```

3.5 Step 3: Visualize Coefficient Shrinkage and RMSE

Let's visualize how RMSE and model sparsity change with λ .

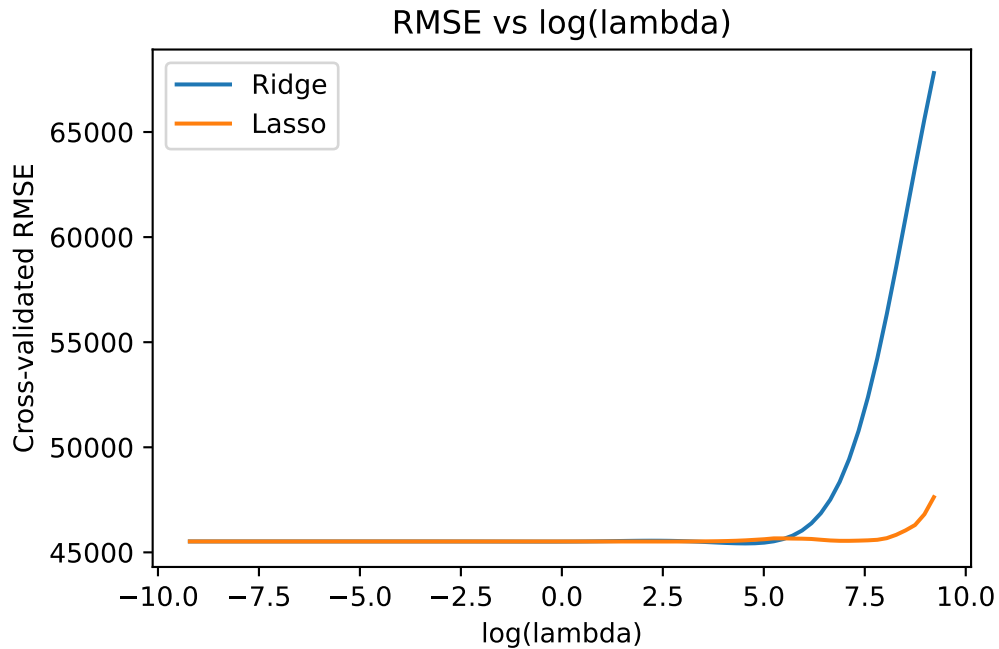
```

# Compute RMSE path manually
def cv_rmse(model_class, X, y, alphas):
    rmse = []
    for a in alphas:
        model = Pipeline([
            ("prep", preprocessor),
            ("model", model_class(alpha=a, max_iter=20000))
        ])
        scores = cross_val_score(model, X, y, scoring="neg_mean_squared_error", cv=KFold(10,
        rmse.append(np.sqrt(-scores.mean()))
    return np.array(rmse)

from sklearn.linear_model import Ridge, Lasso
ridge_rmse_path = cv_rmse(Ridge, X_train, y_train, alphas)
lasso_rmse_path = cv_rmse(Lasso, X_train, y_train, alphas)

plt.figure()
plt.plot(np.log(alphas), ridge_rmse_path, label="Ridge")
plt.plot(np.log(alphas), lasso_rmse_path, label="Lasso")
plt.xlabel("log(lambda)")
plt.ylabel("Cross-validated RMSE")
plt.title("RMSE vs log(lambda)")
plt.legend()
plt.show()

```



3.6 Step 4: Elastic Net

Elastic Net blends Ridge and Lasso.
We'll vary α (the mix) and visualize the tradeoff.

```
l1_ratios = np.linspace(0, 1, 6)
en_cv = Pipeline([
    ("prep", preprocessor),
    ("model", ElasticNetCV(l1_ratio=l1_ratios, alphas=alphas, cv=10, max_iter=50000))
])
en_cv.fit(X_train, y_train)

print("Best Elastic Net alpha:", en_cv.named_steps["model"].alpha_)
print("Best Elastic Net l1_ratio:", en_cv.named_steps["model"].l1_ratio_)

en_pred = en_cv.predict(X_test)
en_rmse = np.sqrt(mean_squared_error(y_test, en_pred))
print("Test RMSE (Elastic Net):", en_rmse)
```

Best Elastic Net alpha: 0.55825862688627
Best Elastic Net l1_ratio: 0.8
Test RMSE (Elastic Net): 37134.497858434275

3.7 Step 5: Compare Models

```
results = pd.DataFrame({
    "Model": ["Ridge", "Lasso", "Elastic Net"],
    "Test_RMSE": [ridge_rmse, lasso_rmse, en_rmse]
})
print(results.sort_values("Test_RMSE"))
```

	Model	Test_RMSE
0	Ridge	37055.845112
1	Lasso	37074.749781
2	Elastic Net	37134.497858

3.8 Summary

- Ridge stabilizes coefficients by shrinking them.
- Lasso enforces sparsity by zeroing weak predictors.
- Elastic Net blends both for balance.
- Regularization often beats pure stepwise regression on unseen data.

4 GAM – Piecewise, MARS, LOESS, and Splines - R Version

Goal: Predict `Sale_Price` from a focused set of Ames features using increasingly flexible methods in the GAM framework.

4.1 1. Setup and Data

```
set.seed(4321)

library(tidyverse)
library(dplyr)
library(tidyr)
library(AmesHousing) # make_ordinal_ames()
library(earth)       # MARS
library(segmented)   # piecewise (segmented) regression
library(splines)     # regression splines (bs, ns) - kept for reference
library(mgcv)        # smoothing splines via GAM
library(caret)       # data splitting / utilities
library(Metrics)     # rmse, mse helpers (plus we'll compute R2 manually)
theme_set(theme_minimal())

# Limit to requested columns
cols <- c("Sale_Price", "Bedroom_AbvGr", "Year_Built", "Mo_Sold", "Lot_Area", "Street",
          "Central_Air", "First_Flr_SF", "Second_Flr_SF", "Full_Bath", "Half_Bath",
          "Fireplaces", "Garage_Area", "Gr_Liv_Area", "TotRms_AbvGrd")

ames <- make_ordinal_ames() |>
  dplyr::select(dplyr::all_of(cols)) |>
  tidyr::drop_na()

# Train/test split
set.seed(4321)
```



```

idx <- sample.int(nrow(ames), size = floor(0.7*nrow(ames)))
train <- ames[idx,]
test  <- ames[-idx,]

glimpse(train)

```

```

Rows: 2,051
Columns: 15
$ Sale_Price      <int> 274000, 75200, 329900, 145400, 108000, 184000, 176000, 1~
$ Bedroom_AbvGr  <int> 3, 2, 4, 3, 2, 2, 3, 3, 3, 3, 3, 3, 3, 4, 3, 2, 3, 3, 2,~
$ Year_Built     <int> 2001, 1922, 2005, 1926, 1949, 1999, 1962, 1915, 1999, 19~
$ Mo_Sold        <int> 1, 9, 8, 5, 5, 6, 5, 6, 9, 6, 8, 5, 7, 7, 4, 8, 5, 11, 5~
$ Lot_Area       <int> 9720, 3672, 11643, 7000, 8777, 5858, 19296, 8094, 3768, ~
$ Street         <fct> Pave, Pave, Pave, Pave, Pave, Pave, Pave, Pave, Pave, Pa~
$ Central_Air    <fct> Y, Y, Y, Y, Y, Y, Y, Y, Y, Y, Y, Y, Y, Y, Y, Y, Y, Y,~
$ First_Flr_SF   <int> 1366, 816, 1544, 861, 1126, 1337, 1382, 1048, 713, 792, ~
$ Second_Flr_SF  <int> 581, 0, 814, 424, 0, 0, 0, 720, 739, 725, 0, 1151, 695, ~
$ Full_Bath      <int> 2, 1, 2, 1, 2, 2, 1, 2, 2, 1, 1, 2, 2, 2, 1, 2, 2, 1, 1,~
$ Half_Bath      <int> 1, 0, 1, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 0, 1, 0, 1, 0, 0,~
$ Fireplaces     <int> 1, 0, 1, 0, 0, 1, 1, 0, 0, 2, 0, 1, 2, 1, 0, 1, 0, 0, 1,~
$ Garage_Area    <dbl> 725, 100, 784, 506, 520, 511, 884, 576, 506, 400, 0, 434~
$ Gr_Liv_Area    <int> 1947, 816, 2358, 1285, 1126, 1337, 1382, 1768, 1452, 151~
$ TotRms_AbvGrd <int> 7, 5, 10, 6, 5, 5, 6, 8, 6, 7, 5, 8, 7, 5, 6, 7, 6, 6, 5~

```

```
summary(train$Sale_Price)
```

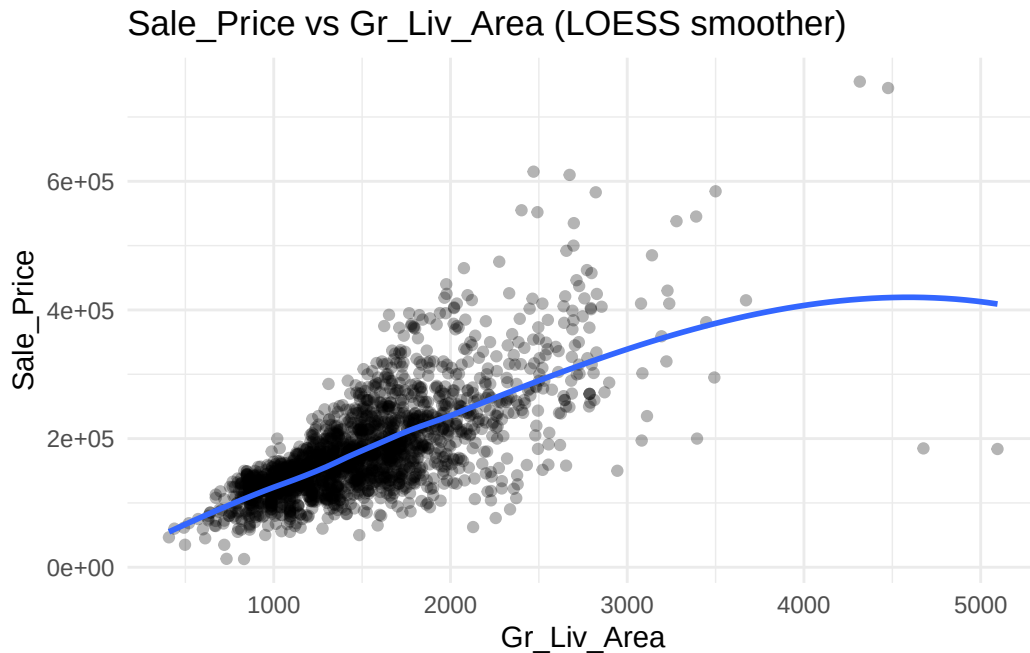
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
12789	130000	161000	180897	215000	755000

4.2 2. Quick Visual: Nonlinearity is Common

```

ggplot(train, aes(Gr_Liv_Area, Sale_Price)) +
  geom_point(alpha=.3) +
  geom_smooth(method="loess", se=FALSE) +
  labs(title="Sale_Price vs Gr_Liv_Area (LOESS smoother)")

```



Takeaway: The relationship bends—nonlinear methods can help.

4.3 3. Piecewise Regression

```
m_lin <- lm(Sale_Price ~ Gr_Liv_Area, data=train)
m_seg_init <- lm(Sale_Price ~ Gr_Liv_Area, data=train)
m_seg <- segmented(m_seg_init, seg.Z = ~ Gr_Liv_Area, psi = list(Gr_Liv_Area = 2000))
summary(m_seg)
```

Regression Model with Segmented Relationship(s)

Call:

```
segmented.lm(obj = m_seg_init, seg.Z = ~Gr_Liv_Area, psi = list(Gr_Liv_Area = 2000))
```

Estimated Break-Point(s):

	Est.	St.Err
psi1.Gr_Liv_Area	2466	332.881

Coefficients of the linear terms:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	7901.186	4698.641	1.682	0.0928 .
Gr_Liv_Area	115.705	3.171	36.488	<2e-16 ***
U1.Gr_Liv_Area	-23.479	12.436	-1.888	NA

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 55110 on 2047 degrees of freedom

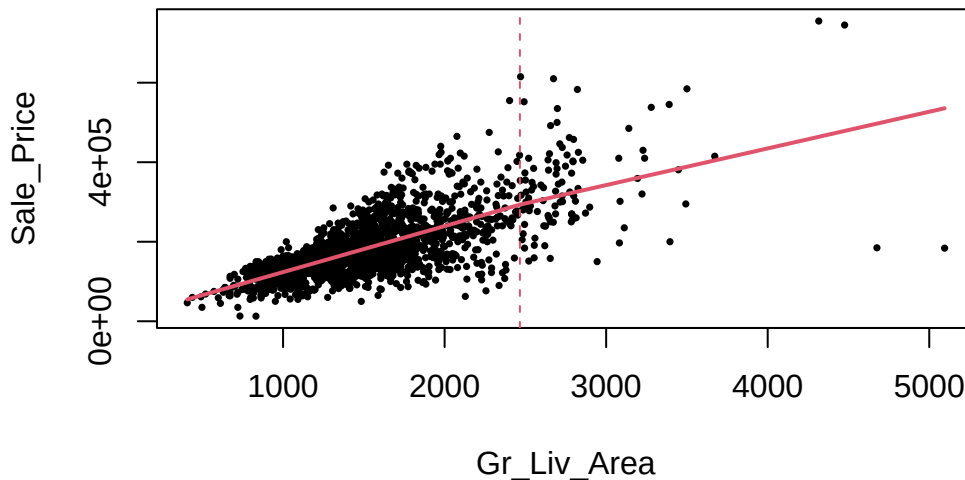
Multiple R-Squared: 0.5124, Adjusted R-squared: 0.5116

Boot restarting based on 6 samples. Last fit:

Convergence attained in 1 iterations (rel. change 0)

```
plot(train$Gr_Liv_Area, train$Sale_Price, pch=16, cex=.5,  
      xlab="Gr_Liv_Area", ylab="Sale_Price", main="Piecewise fit at estimated knot")  
plot(m_seg, add=TRUE, col=2, lwd=2)  
abline(v = m_seg$psi[,"Est."], lty=2, col=2)
```

Piecewise fit at estimated knot



Interpretation (piecewise): Two linear slopes before/after a knot. Useful when you expect a threshold effect.

4.4 4. MARS (earth) – From Simple to Interactions

MARS = *Multivariate Adaptive Regression Splines*. It builds **hinge** basis functions like $h(x - c) = \max(0, x - c)$ and can **interact** them.

- A term such as $h(\text{Gr_Liv_Area} - 1500)$ means *once* Gr_Liv_Area exceeds 1500, the fitted line's slope can change.

- An interaction like $h(\text{Gr_Liv_Area} - 1500) * \text{Central_AirY}$ means the slope change applies when Central_Air == "Y".

4.4.1 4A. Univariate MARS (Garage_Area)

```
mars1 <- earth(Sale_Price ~ Garage_Area, data=train)
summary(mars1)
```

Call: earth(formula=Sale_Price~Garage_Area, data=train)

	coefficients
(Intercept)	124159.039
$h(286 - \text{Garage_Area})$	-60.257
$h(\text{Garage_Area} - 286)$	297.277
$h(\text{Garage_Area} - 521)$	-483.642
$h(\text{Garage_Area} - 576)$	733.859
$h(\text{Garage_Area} - 758)$	-356.460
$h(\text{Garage_Area} - 1043)$	-490.873

Selected 7 of 7 terms, and 1 of 1 predictors

Termination condition: RSq changed by less than 0.001 at 7 terms

Importance: Garage_Area

Number of terms at each degree of interaction: 1 6 (additive model)

GCV 3427475346 RSS 6.94092e+12 GRSq 0.4492014 RSq 0.4556309

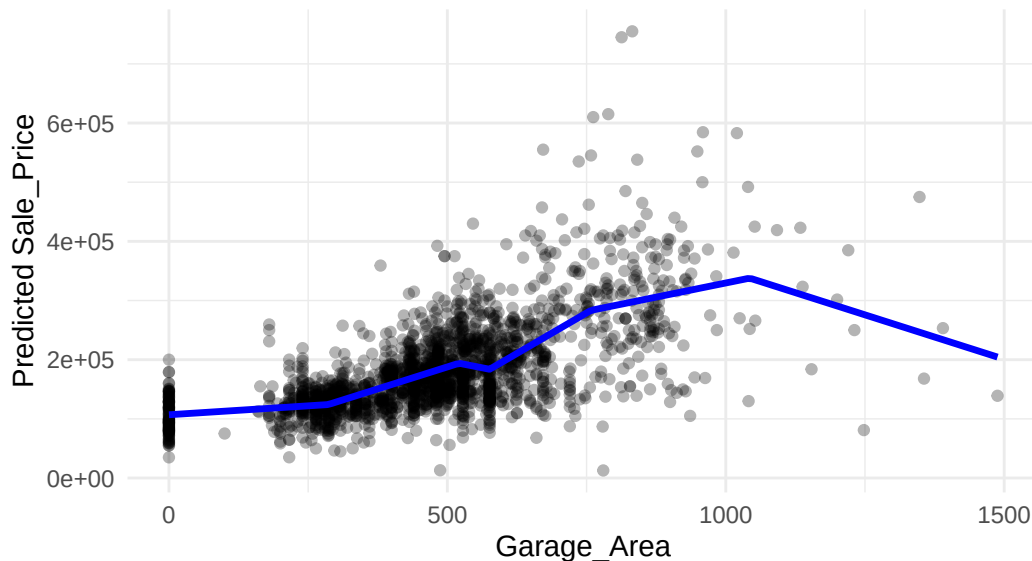
```
grid <- tibble(Garage_Area = seq(min(train$Garage_Area), max(train$Garage_Area), length.out=100))
grid$pred <- predict(mars1, newdata=grid)

ggplot(train, aes(Garage_Area, Sale_Price)) +
  geom_point(alpha=.3) +
  geom_line(data=grid, aes(Garage_Area, pred), color="blue", linewidth=1.2) +
```

```
labs(title="Step 1: MARS with One Predictor (Garage_Area)",
      subtitle="Piecewise linear fit with automatically chosen knots",
      y="Predicted Sale_Price")
```

Step 1: MARS with One Predictor (Garage_Area)

Piecewise linear fit with automatically chosen knots



4.4.2 4B. Additive MARS (degree = 1; no interactions)

```
mars2 <- earth(Sale_Price ~ Bedroom_AbvGr + Year_Built + Mo_Sold + Lot_Area +
                Street + Central_Air + First_Flr_SF + Second_Flr_SF + Full_Bath +
                Half_Bath + Fireplaces + Garage_Area + Gr_Liv_Area + TotRms_AbvGrd,
                data=train, degree=1, nfold=5)
summary(mars2)
```

Call: earth(formula=Sale_Price~Bedroom_AbvGr+Year_Built+Mo_Sold+Lot_Ar...),
data=train, degree=1, nfold=5)

	coefficients
(Intercept)	319493.46
Central_AirY	20289.49
h(4-Bedroom_AbvGr)	9214.66
h(Bedroom_AbvGr-4)	-23009.05

h(Year_Built-1977)	1275.57
h(2004-Year_Built)	-336.64
h(Year_Built-2004)	5315.57
h(13869-Lot_Area)	-2.09
h(Lot_Area-13869)	0.22
h(First_Flr_SF-1600)	104.91
h(2402-First_Flr_SF)	-71.56
h(First_Flr_SF-2402)	-176.61
h(1523-Second_Flr_SF)	-53.13
h(Second_Flr_SF-1523)	426.63
h(Half_Bath-1)	-45378.31
h(2-Fireplaces)	-14408.56
h(Fireplaces-2)	-26072.58
h(Garage_Area-539)	101.97
h(Garage_Area-1043)	-294.30
h(Gr_Liv_Area-2049)	65.21
h(Gr_Liv_Area-3194)	-159.79

Selected 21 of 24 terms, and 10 of 14 predictors

Termination condition: Reached nk 29

Importance: First_Flr_SF, Second_Flr_SF, Year_Built, Garage_Area, ...

Number of terms at each degree of interaction: 1 20 (additive model)

GCV 1033819964 RSS 2.036439e+12 GRSq 0.8338641 RSq 0.8402842 CVRSq 0.7978671

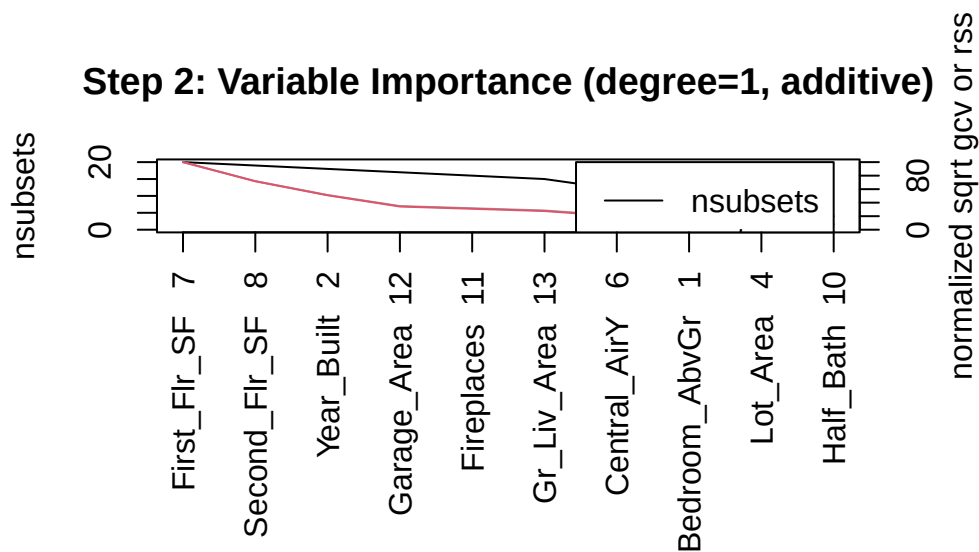
Note: the cross-validation sd's below are standard deviations across folds

Cross validation: nterms 21.00 sd 0.71 nvars 9.20 sd 0.84

CVRSq	sd	MaxErr	sd
0.798	0.043	-428941	282429

```
ev2 <- evimp(mars2); plot(ev2, main="Step 2: Variable Importance (degree=1, additive)")
```

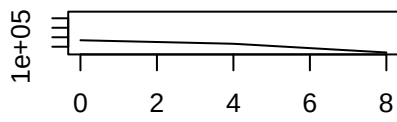
Step 2: Variable Importance (degree=1, additive)



```
par(mfrow=c(2,2)); plotmo(mars2, type="response", nresponse=1, do.par=FALSE); par(mfrow=c(1,1))
```

```
plotmo grid:   Bedroom_AbvGr Year_Built Mo_Sold Lot_Area Street Central_Air
               3      1974      6    9480   Pave      Y
First_Flr_SF Second_Flr_SF Full_Bath Half_Bath Fireplaces Garage_Area
    1092          0          2          0          1          480
Gr_Liv_Area TotRms_AbvGrd
    1442          6
```

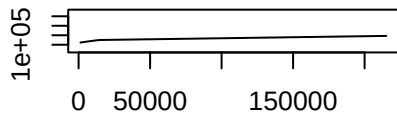
1 Bdrm_AbvG



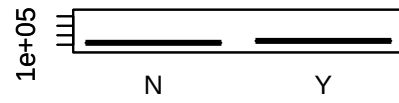
2 Year_Bult



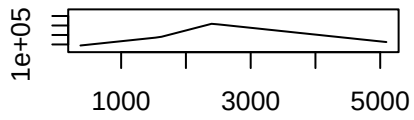
3 Lot_Area



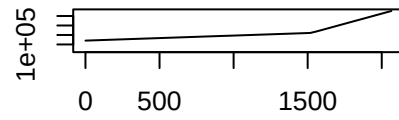
4 Centrl_Ar



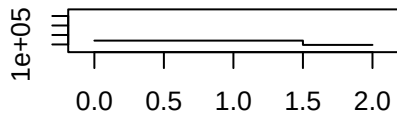
5 Frst_F_SF



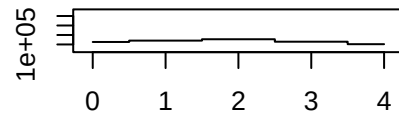
6 Scnd_F_SF

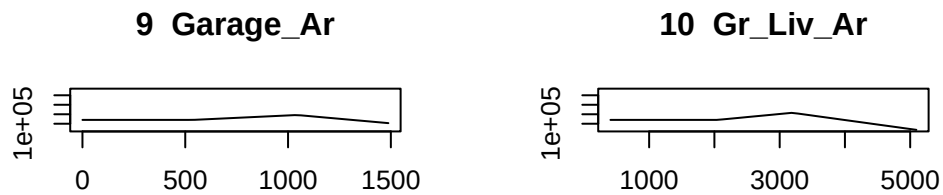


7 Half_Bath



8 Fireplacs





4.4.3 4C. MARS with 2-way Interactions (degree = 2)

```
mars3 <- earth(Sale_Price ~ Bedroom_AbvGr + Year_Built + Mo_Sold + Lot_Area +
               Street + Central_Air + First_Flr_SF + Second_Flr_SF + Full_Bath +
               Half_Bath + Fireplaces + Garage_Area + Gr_Liv_Area + TotRms_AbvGrd,
               data=train, degree=2, nfold=5)
summary(mars3)
```

Call: earth(formula=Sale_Price~Bedroom_AbvGr+Year_Built+Mo_Sold+Lot_Ar...),
data=train, degree=2, nfold=5)

	coefficients
(Intercept)	411949.27
h(Year_Built-2004)	169869.37
h(14803-Lot_Area)	-1.75
h(Lot_Area-14803)	0.48
h(1570-First_Flr_SF)	-150.79
h(First_Flr_SF-1570)	224.90
h(1523-Second_Flr_SF)	-86.91
h(Second_Flr_SF-1523)	156.02
h(1-Half_Bath)	-7743.19

h(Half_Bath-1)	-54819.24
h(2-Fireplaces)	-10516.92
h(1043-Garage_Area)	-70.97
h(Garage_Area-1043)	-141.86
h(Gr_Liv_Area-3194)	106.27
h(3-Bedroom_AbvGr) * h(1523-Second_Flr_SF)	7.01
h(Bedroom_AbvGr-3) * h(1523-Second_Flr_SF)	-9.61
h(2004-Year_Built) * h(First_Flr_SF-876)	-1.86
h(2004-Year_Built) * h(3194-Gr_Liv_Area)	-0.17
h(Year_Built-2004) * h(Gr_Liv_Area-2320)	-159.80
h(Year_Built-2004) * h(2320-Gr_Liv_Area)	156.64
h(Year_Built-2004) * h(3194-Gr_Liv_Area)	-173.05
h(First_Flr_SF-1778) * h(2-Fireplaces)	-78.19
h(1043-Garage_Area) * h(2122-Gr_Liv_Area)	0.05

Selected 23 of 29 terms, and 9 of 14 predictors

Termination condition: Reached nk 29

Importance: First_Flr_SF, Second_Flr_SF, Year_Built, Gr_Liv_Area, ...

Number of terms at each degree of interaction: 1 13 9

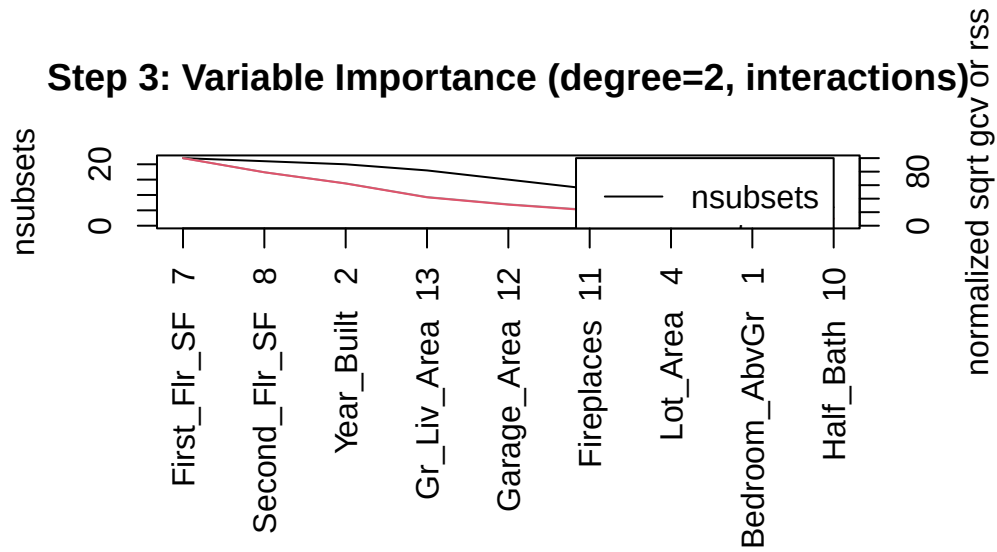
GCV 795873462 RSS 1.544416e+12 GRSq 0.8721024 RSq 0.8788731 CVRSq 0.7824362

Note: the cross-validation sd's below are standard deviations across folds

Cross validation: nterms 22.60 sd 1.82 nvars 9.40 sd 1.14

CVRSq	sd	MaxErr	sd
0.782	0.114	-945502	422348

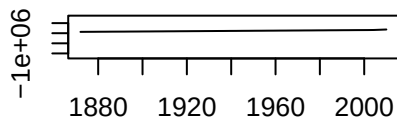
```
ev3 <- evimp(mars3); plot(ev3, main="Step 3: Variable Importance (degree=2, interactions)")
```



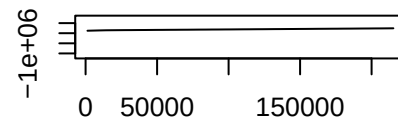
```
par(mfrow=c(2,2)); plotmo(mars3, type="response", nresponse=1, do.par=FALSE); par(mfrow=c(1,1));
```

```
plotmo grid:  Bedroom_AbvGr  Year_Built  Mo_Sold  Lot_Area  Street  Central_Air
              3      1974      6      9480  Pave      Y
First_Flr_SF Second_Flr_SF Full_Bath Half_Bath Fireplaces Garage_Area
1092          0          2          0          1          480
Gr_Liv_Area TotRms_AbvGrd
1442          6
```

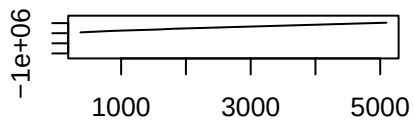
1 Year_Bult



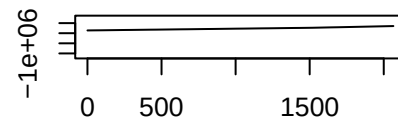
2 Lot_Area



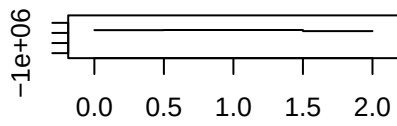
3 Frst_F_SF



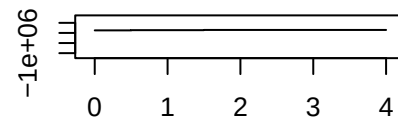
4 Scnd_F_SF



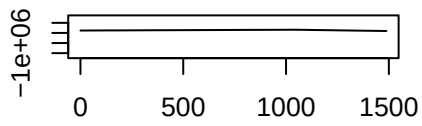
5 Half_Bath



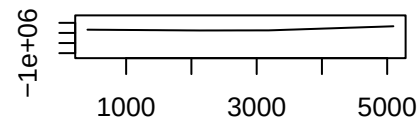
6 Fireplaces



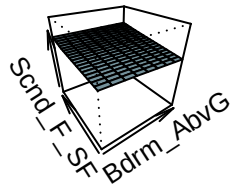
7 Garage_Ar



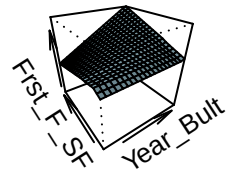
8 Gr_Liv_Ar



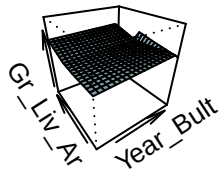
1 Bdrm_AbvG: Scnd_F_SF



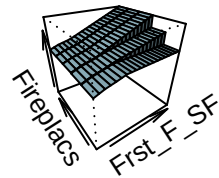
2 Year_Bult: Frst_F_SF



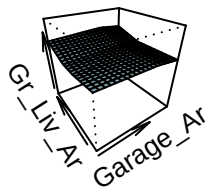
3 Year_Bult: Gr_Liv_Ar



4 Frst_F_SF: Fireplacs



5 Garage_Ar: Gr_Liv_Ar



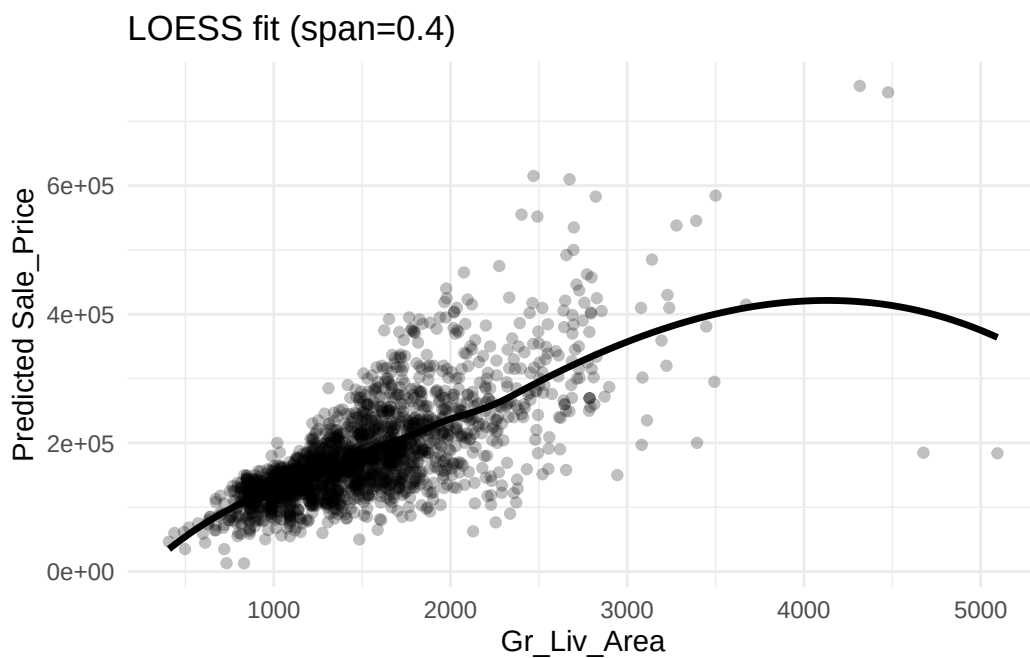
Interpreting MARS terms:

Look for $h(\cdot)$ pieces and products. Coefficients on $h(x-c)$ indicate how slope changes after the knot c . Interaction products mean “the slope change depends on another variable”.

4.5 5. LOESS (Visual local regression)

```
loess_fit <- loess(Sale_Price ~ Gr_Liv_Area, data=train, span=0.4)
grid_lo <- tibble(Gr_Liv_Area = seq(min(train$Gr_Liv_Area), max(train$Gr_Liv_Area), length.out=100))
grid_lo$pred <- predict(loess_fit, newdata=grid_lo)

ggplot() +
  geom_point(data=train, aes(Gr_Liv_Area, Sale_Price), alpha=.25) +
  geom_line(data=grid_lo, aes(Gr_Liv_Area, pred), linewidth=1.2) +
  labs(title="LOESS fit (span=0.4)", y="Predicted Sale_Price")
```



We keep LOESS as a visualization-oriented smoother (not used in the multivariate comparison below).

4.6 6. GAM Splines with mgcv – Simple → Complex

GAM smooths are written $s(x)$ and estimated with penalized splines. The **edf** (effective degrees of freedom) in the summary tells you the smooth's complexity (larger edf = wigglier function).

4.6.1 6A. Univariate GAM: Sale_Price ~ s(Garage_Area)

```
gam1 <- gam(Sale_Price ~ s(Garage_Area, k=9), data=train, method="REML")
summary(gam1) # inspect edf and significance
```

Family: gaussian

Link function: identity

Formula:

Sale_Price ~ s(Garage_Area, k = 9)

Parametric coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	180897	1292	140	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:

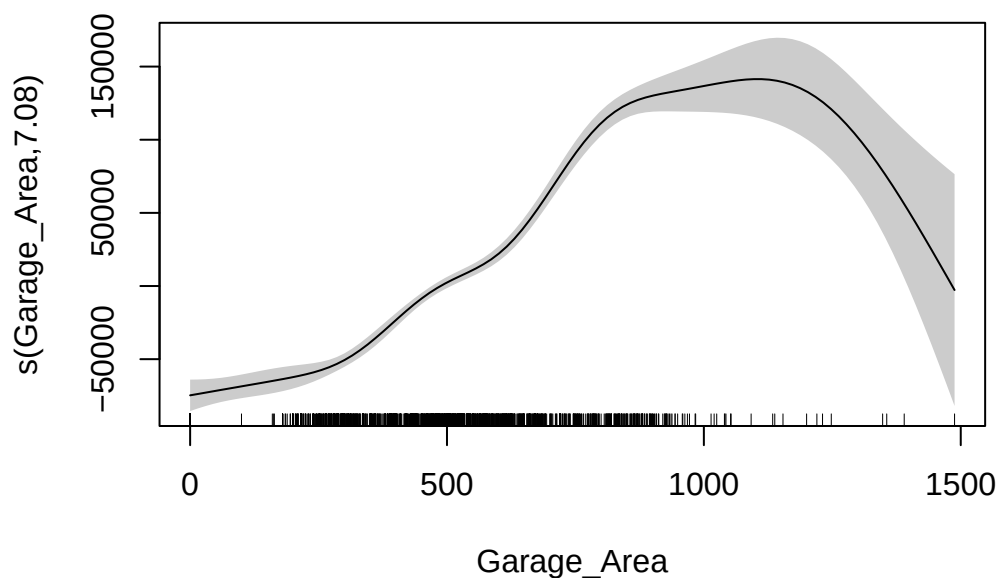
	edf	Ref.df	F	p-value
s(Garage_Area)	7.078	7.719	217.1	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) = 0.45 Deviance explained = 45.2%

-REML = 25418 Scale est. = 3.4232e+09 n = 2051

```
plot(gam1, pages=1, shade=TRUE, scheme=1, scale=0)
```



4.6.2 6B. Two-smooth Additive GAM: $s(\text{Garage_Area}) + s(\text{Gr_Liv_Area})$

```
gam2 <- gam(Sale_Price ~ s(Garage_Area, k=9) + s(Gr_Liv_Area, k=9), data=train, method="REML")
summary(gam2)
```

Family: gaussian

Link function: identity

Formula:

$\text{Sale_Price} \sim s(\text{Garage_Area}, k = 9) + s(\text{Gr_Liv_Area}, k = 9)$

Parametric coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	180897	1026	176.2	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:

	edf	Ref.df	F	p-value
$s(\text{Garage_Area})$	6.446	7.293	103.3	<2e-16 ***


```
s(Gr_Liv_Area) 7.585 7.941 152.6 <2e-16 ***
```

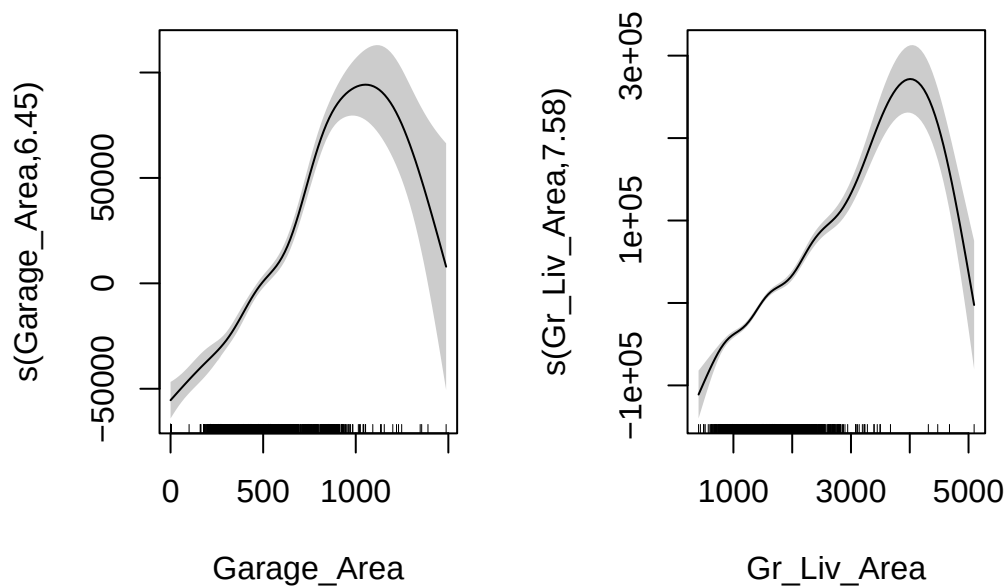
```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
R-sq.(adj) = 0.653 Deviance explained = 65.5%
```

```
-REML = 24949 Scale est. = 2.1608e+09 n = 2051
```

```
plot(gam2, pages=1, shade=TRUE, scheme=1, scale=0)
```



4.6.3 6C. Full Multivariate GAM (selected smooths + factors)

```
gam3 <- gam(Sale_Price ~  
  s(Garage_Area, k=9) +  
  s(Gr_Liv_Area, k=9) +  
  s(Year_Built, k=7) +  
  s(Lot_Area, k=7) +  
  Bedroom_AbvGr + Mo_Sold +  
  Street + Central_Air +  
  First_Flr_SF + Second_Flr_SF +  
  Full_Bath + Half_Bath +
```

```

      Fireplaces + TotRms_AbvGrd,
      data=train, method="REML")
summary(gam3)

```

Family: gaussian
Link function: identity

Formula:

```

Sale_Price ~ s(Garage_Area, k = 9) + s(Gr_Liv_Area, k = 9) +
  s(Year_Built, k = 7) + s(Lot_Area, k = 7) + Bedroom_AbvGr +
  Mo_Sold + Street + Central_Air + First_Flr_SF + Second_Flr_SF +
  Full_Bath + Half_Bath + Fireplaces + TotRms_AbvGrd

```

Parametric coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-21237.92	30106.84	-0.705	0.4806	
Bedroom_AbvGr	-10350.20	1382.38	-7.487	1.05e-13	***
Mo_Sold	-409.12	270.24	-1.514	0.1302	
StreetPave	30589.21	13486.61	2.268	0.0234	*
Central_AirY	17189.68	3366.03	5.107	3.59e-07	***
First_Flr_SF	140.27	17.44	8.044	1.47e-15	***
Second_Flr_SF	101.93	17.36	5.871	5.06e-09	***
Full_Bath	-4548.94	2211.69	-2.057	0.0398	*
Half_Bath	2355.98	2175.40	1.083	0.2789	
Fireplaces	13066.61	1371.59	9.527	< 2e-16	***
TotRms_AbvGrd	-1640.87	948.67	-1.730	0.0838	.

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

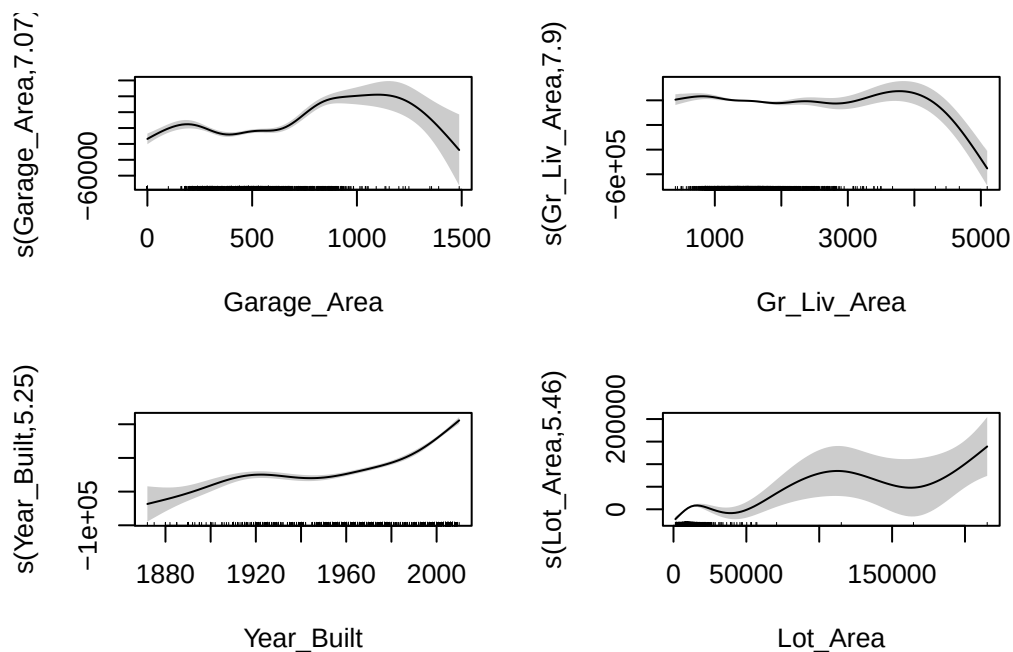
Approximate significance of smooth terms:

	edf	Ref.df	F	p-value	
s(Garage_Area)	7.067	7.712	24.96	<2e-16	***
s(Gr_Liv_Area)	7.899	7.996	46.44	<2e-16	***
s(Year_Built)	5.255	5.764	128.05	<2e-16	***
s(Lot_Area)	5.465	5.883	16.44	<2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) = 0.827 Deviance explained = 83%
-REML = 24167 Scale est. = 1.0786e+09 n = 2051

```
plot(gam3, pages=1, shade=TRUE, scheme=1, scale=0)
```



Interpreting GAM splines:

A smooth $s(x)$ shows how the expected outcome varies with x *holding other terms constant*. The edf indicates complexity; wide confidence bands suggest greater uncertainty in those regions.

4.7 7. Model Comparison (R^2 , RMSE, MSE)

```
# Helper for R^2 on test
rsq <- function(y, yhat) {
  1 - sum((y - yhat)^2) / sum((y - mean(y))^2)
}

# Predictions
p_piece <- predict(m_seg, newdata=test)
p_mars  <- predict(mars3, newdata=test) # final MARS with interactions
p_gam   <- predict(gam3, newdata=test)  # full multivariate GAM
```

```
# Metrics
tbl <- tibble(
  Model = c("Piecewise (segmented)", "MARS (degree=2)", "GAM splines (mgcv full)"),
  RMSE = c(rmse(test$Sale_Price, p_piece),
           rmse(test$Sale_Price, p_mars),
           rmse(test$Sale_Price, p_gam)),
  MSE = c(mse(test$Sale_Price, p_piece),
           mse(test$Sale_Price, p_mars),
           mse(test$Sale_Price, p_gam)),
  R2 = c(rsq(test$Sale_Price, p_piece),
          rsq(test$Sale_Price, p_mars),
          rsq(test$Sale_Price, p_gam))
) |> arrange(desc(R2))

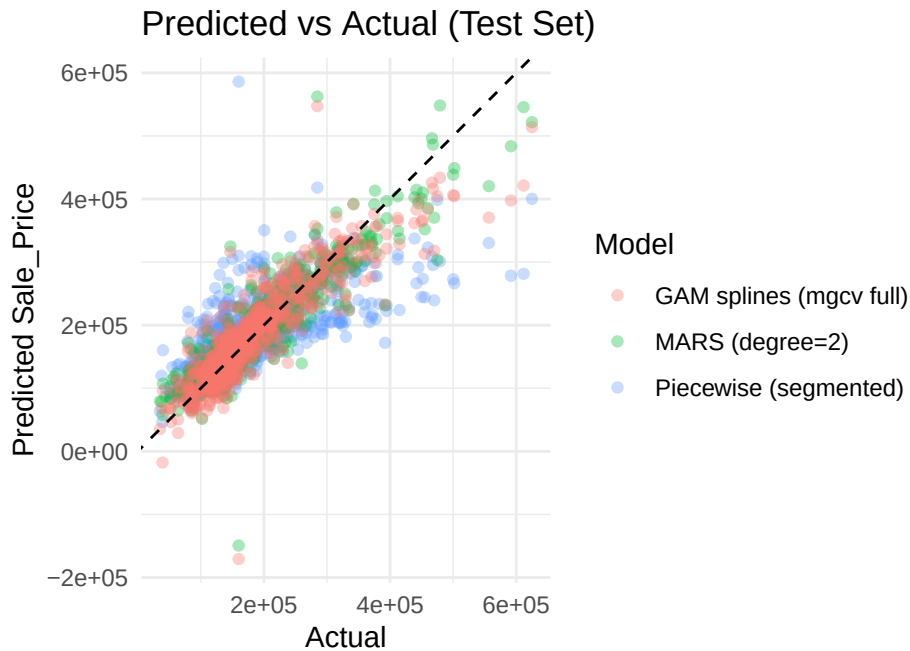
tbl
```

```
# A tibble: 3 x 4
  Model          RMSE      MSE    R2
  <chr>          <dbl>    <dbl> <dbl>
1 MARS (degree=2) 33028. 1090840972. 0.839
2 GAM splines (mgcv full) 36796. 1353918715. 0.800
3 Piecewise (segmented) 59034. 3484963067. 0.484
```

4.8 8. Final Plot: Predicted vs Actual (Test Set)

```
plot_df <- bind_rows(
  tibble(Model="Piecewise (segmented)", Actual=test$Sale_Price, Pred=p_piece),
  tibble(Model="MARS (degree=2)", Actual=test$Sale_Price, Pred=p_mars),
  tibble(Model="GAM splines (mgcv full)", Actual=test$Sale_Price, Pred=p_gam)
)

ggplot(plot_df, aes(Actual, Pred, color=Model)) +
  geom_point(alpha=.35) +
  geom_abline(intercept=0, slope=1, linetype="dashed") +
  labs(title="Predicted vs Actual (Test Set)", y="Predicted Sale_Price") +
  coord_equal()
```



4.9 9. How to Explain

- **Piecewise:** “There’s a threshold where the effect changes.” Simple story, limited flexibility.
- **MARS:** “The model finds breakpoints and sometimes *when* they matter (interactions).” Explain $h(x-c)$ as *extra slope after c*.
- **GAM (splines):** “We model smooth curves for key variables; edf shows wiggleness. More edf more flexibility.”
- **Tradeoff:** More flexibility usually improves error (RMSE/MSE) and R^2 , but reduce interpretability. Validate with cross-validation and keep the story aligned with business intuition.

5 GAM— Piecewise, LOESS, and GAM Splines - Python Version

Note: The original MARS section used the `sklearn-contrib-py-earth` package, which is **no longer actively maintained** and may fail to install on modern Python versions.

The MARS code is left **commented out** for reference — you can un-comment it if you successfully install `py-earth` (try `conda install -c conda-forge sklearn-contrib-py-earth`).

Equivalent flexibility can be achieved through **GAM** or **spline-based models**, which are shown below.

5.1 1. Setup and Data

```
import sys
print("Active Python interpreter:", sys.executable)

import numpy as np, pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.linear_model import LinearRegression, Ridge
import inspect

# --- Fix for SciPy >= 1.13 removing .A from sparse matrices ---
import scipy.sparse as sp

if not hasattr(sp.spmatrix, "A"):
    def _toarray(self):
```

```

        return self.toarray()
    sp.spmatrix.A = property(_toarray)
# -----

np.random.seed(4321)

```

Active Python interpreter: /opt/anaconda3/bin/python

```

from sklearn.datasets import fetch_openml

ames = fetch_openml(name="house_prices", as_frame=True).frame

keep = {
    "SalePrice": "SalePrice",
    "BedroomAbvGr": "Bedroom_AbvGr",
    "YearBuilt": "Year_Built",
    "MoSold": "Mo_Sold",
    "LotArea": "Lot_Area",
    "Street": "Street",
    "CentralAir": "Central_Air",
    "1stFlrSF": "First_Flr_SF",
    "2ndFlrSF": "Second_Flr_SF",
    "FullBath": "Full_Bath",
    "HalfBath": "Half_Bath",
    "Fireplaces": "Fireplaces",
    "GarageArea": "Garage_Area",
    "GrLivArea": "Gr_Liv_Area",
    "TotRmsAbvGrd": "TotRms_AbvGrd"
}

df = ames[list(keep.keys())].rename(columns=keep).dropna().copy()

X = df.drop(columns=["SalePrice"])
y = df["SalePrice"].values

num_cols = X.select_dtypes(include=[np.number]).columns.tolist()
cat_cols = [c for c in X.columns if c not in num_cols]

# Handle OneHotEncoder version changes and ensure dense output
if "sparse_output" in inspect.signature(OneHotEncoder).parameters:
    encoder = OneHotEncoder(drop="first", handle_unknown="ignore", sparse_output=False)

```

```

else:
    encoder = OneHotEncoder(drop="first", handle_unknown="ignore", sparse=False)

pre = ColumnTransformer([
    ("num", StandardScaler(), num_cols),
    ("cat", encoder, cat_cols)
], sparse_threshold=0)

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=4321)
df.head()

```

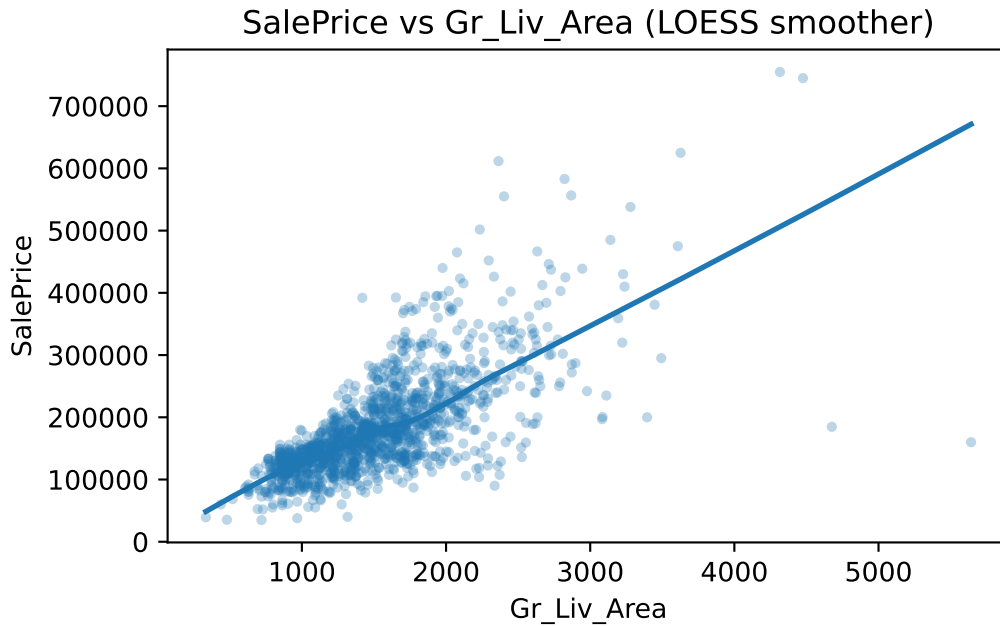
	SalePrice	Bedroom_AbvGr	Year_Built	Mo_Sold	Lot_Area	Street	Central_Air	First_Flr_SF
0	208500	3	2003	2	8450	Pave	Y	856
1	181500	3	1976	5	9600	Pave	Y	1262
2	223500	3	2001	9	11250	Pave	Y	920
3	140000	3	1915	2	9550	Pave	Y	961
4	250000	4	2000	12	14260	Pave	Y	1145

5.2 2. Visual Check for Nonlinearity

```

import statsmodels.api as sm
fig = plt.figure()
plt.scatter(df["Gr_Liv_Area"], df["SalePrice"], s=8, alpha=0.3)
low = sm.nonparametric.lowess(df["SalePrice"], df["Gr_Liv_Area"], frac=0.3, return_sorted=True)
plt.plot(low[:,0], low[:,1], linewidth=2)
plt.title("SalePrice vs Gr_Liv_Area (LOESS smoother)")
plt.xlabel("Gr_Liv_Area"); plt.ylabel("SalePrice")
plt.tight_layout()

```

5.3 3. Piecewise Regression Example

```
class PiecewiseLinear:
    def __init__(self, knot=2000.0):
        self.knot = knot
        self.lin = LinearRegression()

    def _transform(self, x):
        x1 = np.asarray(x).reshape(-1,1)
        hx = np.maximum(0, x1 - self.knot)
        return np.hstack([x1, hx])

    def fit(self, x, y):
        Z = self._transform(x)
        self.lin.fit(Z, y)
        return self

    def predict(self, x):
        Z = self._transform(x)
```

```

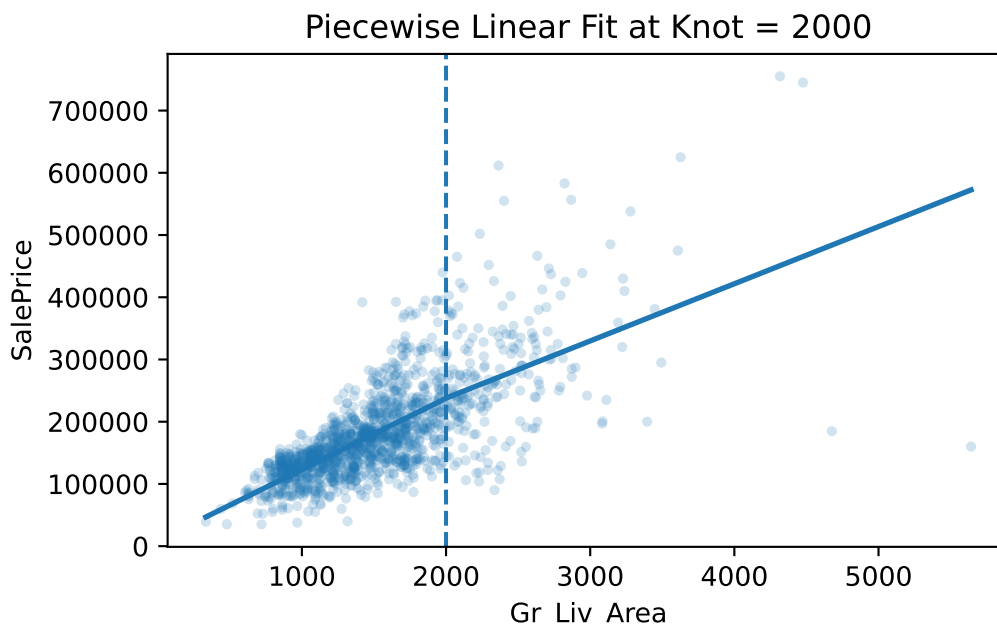
        return self.lin.predict(Z)

pw = PiecewiseLinear(knot=2000).fit(df["Gr_Liv_Area"].values, df["SalePrice"].values)

grid = np.linspace(df["Gr_Liv_Area"].min(), df["Gr_Liv_Area"].max(), 200)
pred = pw.predict(grid)

plt.figure()
plt.scatter(df["Gr_Liv_Area"], df["SalePrice"], s=8, alpha=0.2)
plt.plot(grid, pred, linewidth=2)
plt.axvline(2000, linestyle="--")
plt.title("Piecewise Linear Fit at Knot = 2000")
plt.xlabel("Gr_Liv_Area"); plt.ylabel("SalePrice")
plt.tight_layout()

```



5.4 4. MARS (Commented Out — For Reference Only)

```

# The MARS implementation (py-earth) is not actively maintained and may fail to install on m
# If you wish to try it, install from conda-forge:
#     conda install -c conda-forge sklearn-contrib-py-earth
#
# Example (commented out):
#
# from pyearth import Earth
#
# mars = Pipeline([("prep", pre), ("model", Earth(max_degree=2))]).fit(X_train, y_train)
# print(mars.named_steps["model"].summary())
#
# xs = np.linspace(X["Garage_Area"].min(), X["Garage_Area"].max(), 200)
# yhat = mars.predict(pd.DataFrame({"Garage_Area": xs}))
#
# plt.figure()
# plt.scatter(X_train["Garage_Area"], y_train, s=8, alpha=0.2)
# plt.plot(xs, yhat, linewidth=2)
# plt.title("MARS with Garage_Area (Commented Out Example)")
# plt.xlabel("Garage_Area"); plt.ylabel("SalePrice")
# plt.tight_layout()

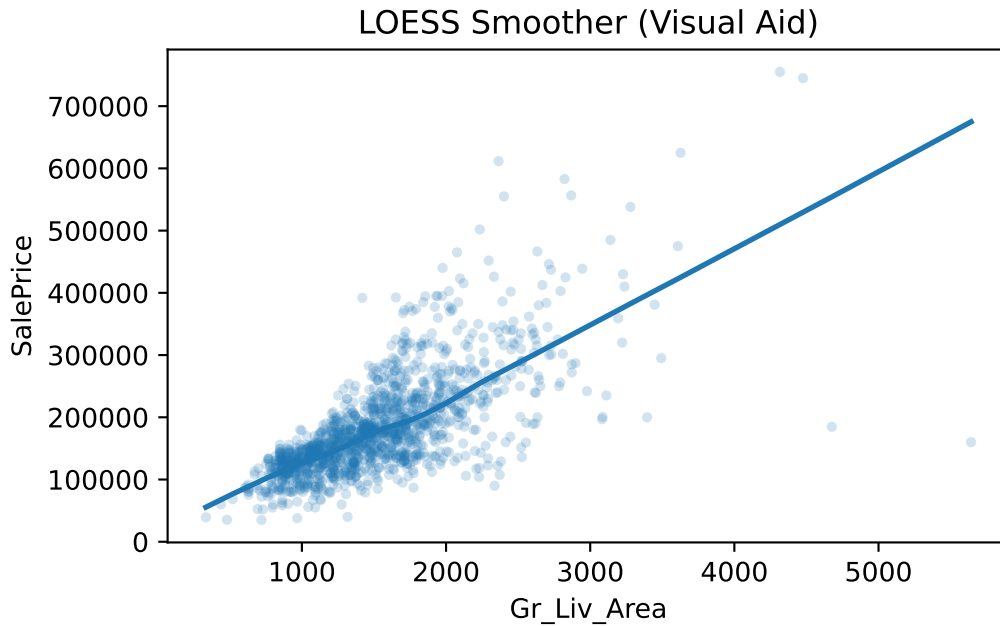
```

5.5 5. LOESS Visualization

```

fig = plt.figure()
plt.scatter(df["Gr_Liv_Area"], df["SalePrice"], s=8, alpha=0.2)
low = sm.nonparametric.lowess(df["SalePrice"], df["Gr_Liv_Area"], frac=0.4, return_sorted=True)
plt.plot(low[:,0], low[:,1], linewidth=2)
plt.title("LOESS Smoother (Visual Aid)")
plt.xlabel("Gr_Liv_Area"); plt.ylabel("SalePrice")
plt.tight_layout()

```



5.6 6. GAM and Spline Approaches (Modern Alternatives)

```
from pygam import LinearGAM, s
gam1 = LinearGAM(s(0)).fit(X_train[["Garage_Area"]].values, y_train)
print(gam1.summary())
gam2 = LinearGAM(s(0) + s(1)).fit(X_train[["Garage_Area", "Gr_Liv_Area"]].values, y_train)
print(gam2.summary())
num_train = X_train[num_cols].values
gam3 = LinearGAM().fit(num_train, y_train)
print(gam3.summary())
```

LinearGAM

```
=====
Distribution:                               NormalDist Effective DoF:
Link Function:                             IdentityLink Log Likelihood:
23263.1874
Number of Samples:                         1022 AIC:
                                           AICc:
```

```

GCV: 3
Scale: 3
Pseudo R-Squared:

=====
Feature Function      Lambda      Rank      EDoF      P > x
=====
s(0)                  [0.6]      20        11.7      1.11e-
16      ***
intercept              1          0.0      1.11e-
16      ***
=====
Significance codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

WARNING: Fitting splines and a linear function to a feature introduces a model identifiability problem
which can cause p-values to appear significant when they are not.

WARNING: p-values calculated in this manner behave correctly for un-penalized models or models with
known smoothing parameters, but when smoothing parameters have been estimated, the p-values
are typically lower than they should be, meaning that the tests reject the null too often.

None
LinearGAM
=====
Distribution:          NormalDist Effective DoF:
Link Function:         IdentityLink Log Likelihood:
22770.8323
Number of Samples:    1022 AIC:
                        AICc:
                        GCV: 19
                        Scale:
                        Pseudo R-Squared:

=====
Feature Function      Lambda      Rank      EDoF      P > x
=====
s(0)                  [0.6]      20        12.6      1.11e-
16      ***
s(1)                  [0.6]      20         8.3      1.11e-
16      ***
intercept              1          0.0      1.11e-
16      ***
=====
Significance codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

WARNING: Fitting splines and a linear function to a feature introduces a model identifiability issue which can cause p-values to appear significant when they are not.

WARNING: p-values calculated in this manner behave correctly for un-penalized models or models with known smoothing parameters, but when smoothing parameters have been estimated, the p-values are typically lower than they should be, meaning that the tests reject the null too often.

None

LinearGAM

```

=====
Distribution:                NormalDist Effective DoF:
Link Function:              IdentityLink Log Likelihood:
21995.395
Number of Samples:          1022 AIC:
                              AICc:
                              GCV:
                              Scale:
                              Pseudo R-Squared:
=====

```

Feature	Function	Lambda	Rank	EDoF	P > x
s(0)		[0.6]	20	8.7	6.58e-
07	***				
s(1)		[0.6]	20	12.8	1.11e-
16	***				
s(2)		[0.6]	20	11.0	5.15e-
01					
s(3)		[0.6]	20	8.4	1.11e-
16	***				
s(4)		[0.6]	20	8.6	1.11e-
16	***				
s(5)		[0.6]	20	10.1	2.17e-
03	**				
s(6)		[0.6]	20	3.2	9.76e-
02	.				
s(7)		[0.6]	20	2.2	4.19e-
01					
s(8)		[0.6]	20	2.9	3.16e-
10	***				
s(9)		[0.6]	20	9.0	1.11e-
16	***				
s(10)		[0.6]	20	5.7	1.11e-
16	***				

s(11)	[0.6]	20	6.5	3.78e-
01				
intercept		1	0.0	1.11e-
16	***			

Significance codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

WARNING: Fitting splines and a linear function to a feature introduces a model identifiability issue which can cause p-values to appear significant when they are not.

WARNING: p-values calculated in this manner behave correctly for un-penalized models or models with known smoothing parameters, but when smoothing parameters have been estimated, the p-values are typically lower than they should be, meaning that the tests reject the null too often.

5.7 7. Regression Spline Example (Fallback)

```
from patsy import dmatrix

design_tr = dmatrix("bs(Garage_Area, df=6) + bs(Gr_Liv_Area, df=6)", data=X_train, return_type='matrix')
design_te = dmatrix("bs(Garage_Area, df=6) + bs(Gr_Liv_Area, df=6)", data=X_test, return_type='matrix')

ridge = Ridge(alpha=1.0).fit(design_tr, y_train)
pred_spline = ridge.predict(design_te)
```

5.8 8. Model Comparison (Piecewise vs GAM vs Spline)

```
def rmse(y, yhat):
    try:
        return mean_squared_error(y, yhat, squared=False)
    except TypeError:
        return np.sqrt(mean_squared_error(y, yhat))
```

```

def mse(y, yhat): return mean_squared_error(y, yhat)
def rsq(y, yhat): return r2_score(y, yhat)

p_piece = pw.predict(X_test["Gr_Liv_Area"].values)

try:
    p_gam = gam3.predict(X_test[num_cols].values)
except Exception:
    p_gam = np.full_like(y_test, np.nan, dtype=float)

p_spline = pred_spline

cmp = pd.DataFrame({
    "Model": ["Piecewise", "GAM (pyGAM)" if not np.isnan(p_gam).all() else "GAM (not available)", "Regression Spline"],
    "RMSE": [rmse(y_test, p_piece), rmse(y_test, p_gam) if not np.isnan(p_gam).all() else np.nan, rmse(y_test, p_spline)],
    "MSE": [mse(y_test, p_piece), mse(y_test, p_gam) if not np.isnan(p_gam).all() else np.nan, mse(y_test, p_spline)],
    "R2": [rsq(y_test, p_piece), rsq(y_test, p_gam) if not np.isnan(p_gam).all() else np.nan, rsq(y_test, p_spline)]
}).sort_values("R2", ascending=False)

cmp

```

	Model	RMSE	MSE	R2
1	GAM (pyGAM)	40805.867162	1.665119e+09	0.766556
0	Piecewise	58108.537553	3.376602e+09	0.526612
2	Regression Spline	60289.000524	3.634764e+09	0.490419

5.9 9. Predicted vs Actual Comparison

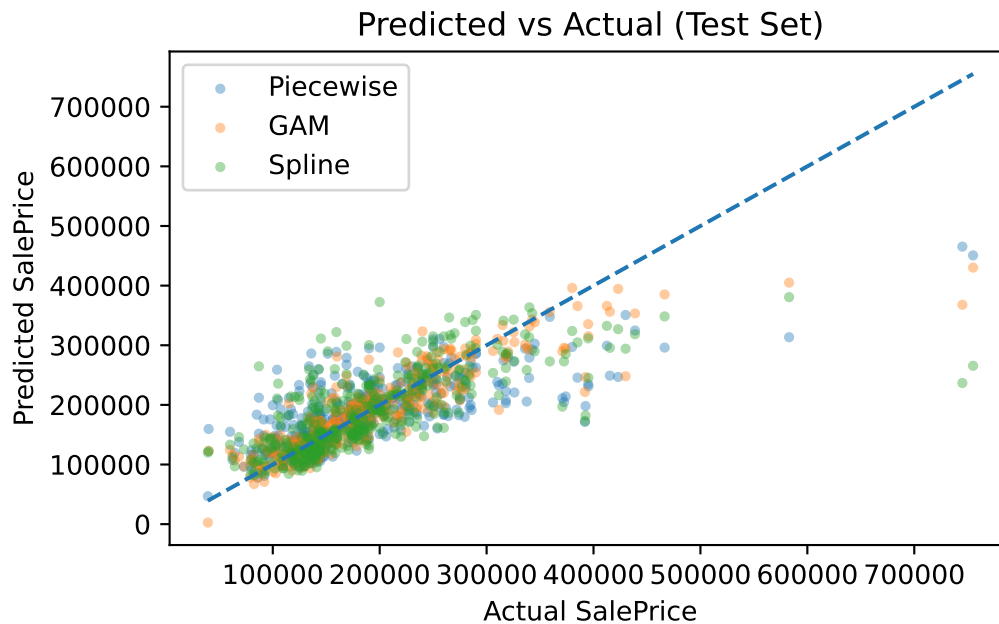
```

plt.figure()
plt.scatter(y_test, p_piece, s=8, alpha=0.4, label="Piecewise")
if not np.isnan(p_gam).all():
    plt.scatter(y_test, p_gam, s=8, alpha=0.4, label="GAM")
plt.scatter(y_test, p_spline, s=8, alpha=0.4, label="Spline")
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()], linestyle="--")
plt.legend()

```



```
plt.title("Predicted vs Actual (Test Set)")
plt.xlabel("Actual SalePrice"); plt.ylabel("Predicted SalePrice")
plt.tight_layout()
```



5.10 10. Interpretation Summary

- **Piecewise:** adds simple thresholds, easy to explain.
- **GAMs:** provide smooth, interpretable nonlinearities.
- **Splines:** flexible approximations that behave like local polynomials.
- **MARS (optional):** similar conceptually but less maintained in Python; consider using R's `earth` instead.

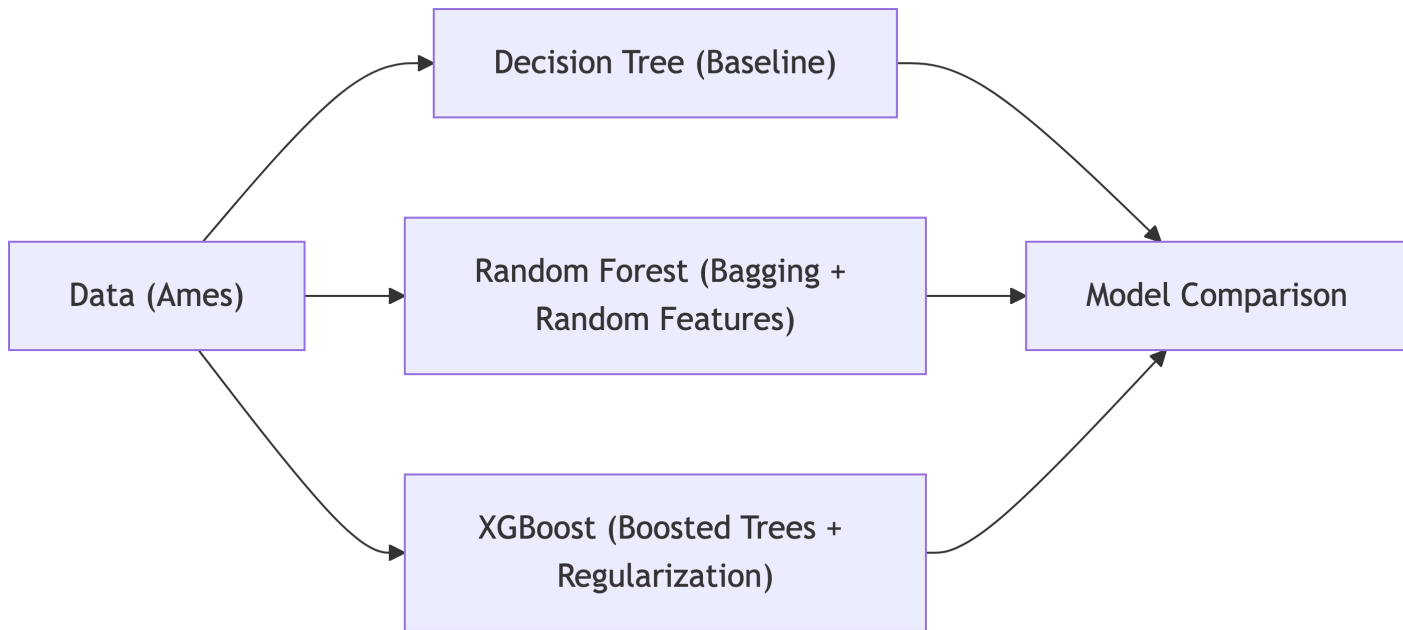
6 Tree Based Models - R Version

Comparison: **Decision Tree**, **Random Forest**, and **XGBoost** on the Ames dataset. Includes random-variable baselines, variable importance, partial dependence plots.

6.1 Visual Roadmap

```
flowchart LR
    A["Data (Ames)"]
    DT["Decision Tree (Baseline)"]
    RF["Random Forest (Bagging + Random Features)"]
    XGB["XGBoost (Boosted Trees + Regularization)"]
    Compare["Model Comparison"]

    A --> DT
    A --> RF
    A --> XGB
    DT --> Compare
    RF --> Compare
    XGB --> Compare
```



6.2 Libraries, Data, Split

```
library(tidyverse)
library(caret)
library(rpart)
library(rpart.plot)
library(randomForest)
library(xgboost)
library(Ckmeans.1d.dp)
library(pdp)
library(AmesHousing)

ames <- make_ordinal_ames()
ames <- ames %>% mutate(id = row_number())

set.seed(4321)
training <- ames %>% sample_frac(0.7)
testing <- anti_join(ames, training, by = 'id')

training <- training %>%
  select(Sale_Price,
```

```
Bedroom_AbvGr, Year_Built, Mo_Sold, Lot_Area, Street, Central_Air,
First_Flr_SF, Second_Flr_SF, Full_Bath, Half_Bath, Fireplaces,
Garage_Area, Gr_Liv_Area, TotRms_AbvGrd)
```

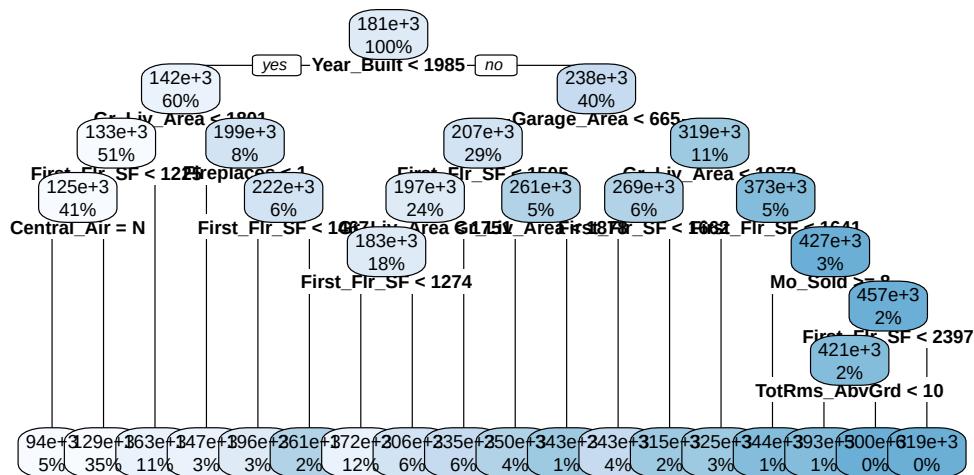
```
training.df <- as.data.frame(training)
```

6.3 Decision Tree (Baseline)

```
set.seed(12345)
tree.ames <- rpart(Sale_Price ~ ., data = training.df, method = "anova",
                    control = rpart.control(cp = 0.005))

rpart.plot(tree.ames, type = 2, fallen.leaves = TRUE, cex = 0.6,
            main = "Decision Tree for Ames Housing")
```

Decision Tree for Ames Housing



```
testing_sel <- testing %>% select(names(training.df))
pred_tree <- predict(tree.ames, newdata = testing_sel)

tree_perf <- testing_sel %>%
```

```

mutate(pred = pred_tree,
       AE = abs(Sale_Price - pred),
       APE = 100*abs((Sale_Price - pred)/Sale_Price)) %>%
summarise(Model = "Decision Tree",
          RMSE = sqrt(mean((Sale_Price - pred)^2)),
          MAE = mean(AE),
          MAPE = mean(APE),
          R2 = 1 - sum((Sale_Price - pred)^2)/sum((Sale_Price - mean(Sale_Price))^2))
tree_perf

```

```

# A tibble: 1 x 5
  Model      RMSE    MAE  MAPE    R2
  <chr>    <dbl> <dbl> <dbl> <dbl>
1 Decision Tree 42938. 28322.  17.5 0.727

```

6.4 Random Forest: Random Variable, PDPs, and Importance

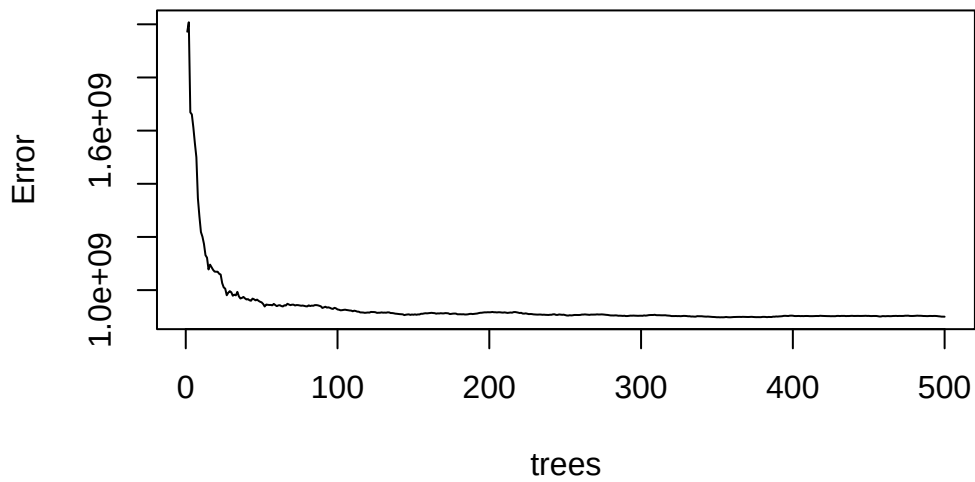
```

# Add random variable for feature selection sanity check
training.df$random <- rnorm(nrow(training.df))

set.seed(12345)
rf.ames <- randomForest(Sale_Price ~ ., data = training.df, ntree = 500, mtry = 4, importance = TRUE)
plot(rf.ames, main = "Random Forest: Error vs Trees")

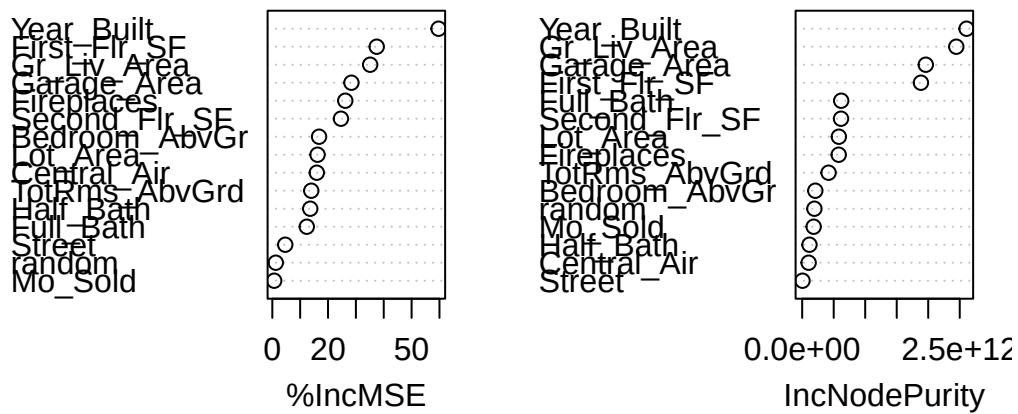
```

Random Forest: Error vs Trees



```
varImpPlot(rf.ames, sort = TRUE, n.var = 15, main = "Random Forest: Variables vs Random Baseline")
```

Random Forest: Variables vs Random Baseline

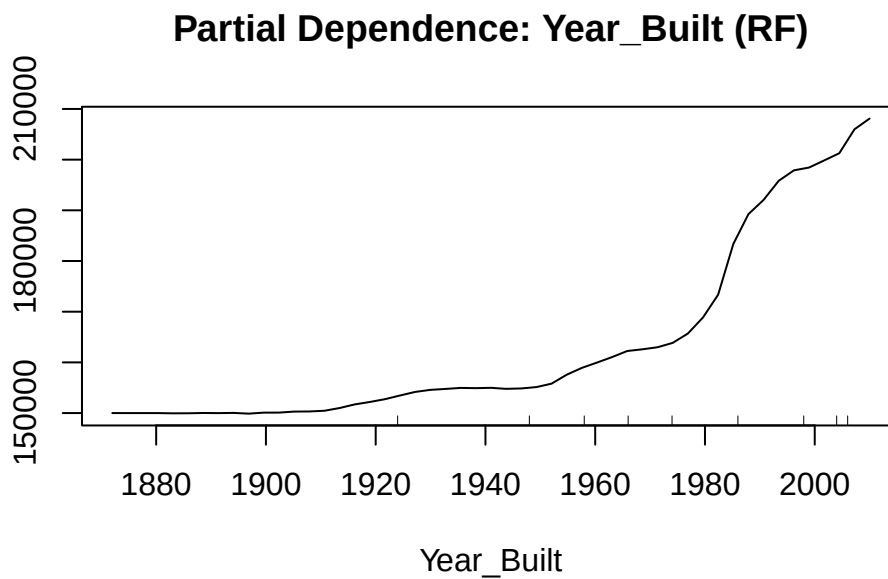


```
importance(rf.ames)
```

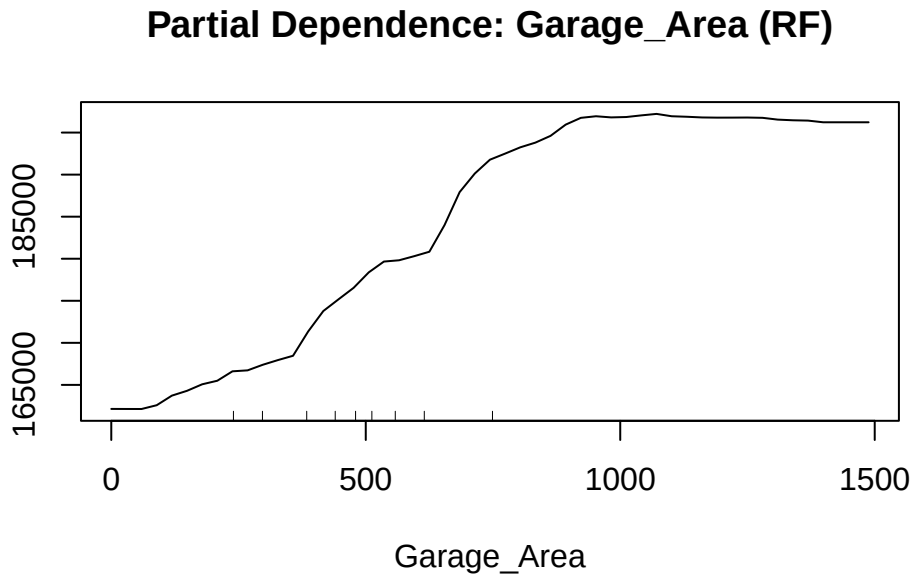
	%IncMSE	IncNodePurity
Bedroom_AbvGr	16.7830622	2.081292e+11
Year_Built	59.6827740	2.608576e+12
Mo_Sold	0.6717216	1.820404e+11
Lot_Area	16.2410098	5.818823e+11
Street	4.6451725	5.141533e+09
Central_Air	15.9709319	1.023533e+11
First_Flr_SF	37.4760827	1.884177e+12
Second_Flr_SF	24.6324055	6.159270e+11
Full_Bath	12.3716046	6.191162e+11
Half_Bath	13.6183708	1.153802e+11
Fireplaces	26.1893181	5.796209e+11
Garage_Area	28.4006898	1.961373e+12
Gr_Liv_Area	35.1279551	2.445543e+12
TotRms_AbvGrd	13.9865453	4.189078e+11
random	1.2436860	1.937137e+11

```
# Partial dependence for key variables
```

```
partialPlot(rf.ames, training.df, Year_Built, main = "Partial Dependence: Year_Built (RF)")
```



```
partialPlot(rf.ames, training.df, Garage_Area, main = "Partial Dependence: Garage_Area (RF)")
```



6.4.1 Random Forest Results

```
testing_sel$random <- rnorm(nrow(testing_sel))
testing_sel$pred_rf <- predict(rf.ames, testing_sel)

rf_perf <- testing_sel %>%
  mutate(AE = abs(Sale_Price - pred_rf),
         APE = 100*abs((Sale_Price - pred_rf)/Sale_Price)) %>%
  summarise(Model = "Random Forest",
            RMSE = sqrt(mean((Sale_Price - pred_rf)^2)),
            MAE = mean(AE),
            MAPE = mean(APE),
            R2 = 1 - sum((Sale_Price - pred_rf)^2)/sum((Sale_Price - mean(Sale_Price))^2))
rf_perf
```

```
# A tibble: 1 x 5
  Model      RMSE    MAE  MAPE    R2
  <chr>    <dbl>  <dbl> <dbl> <dbl>
1 Random Forest 30147. 19223.  11.6 0.866
```


6.5 XGBoost: Random Variable, CV, Tuning, PDPs

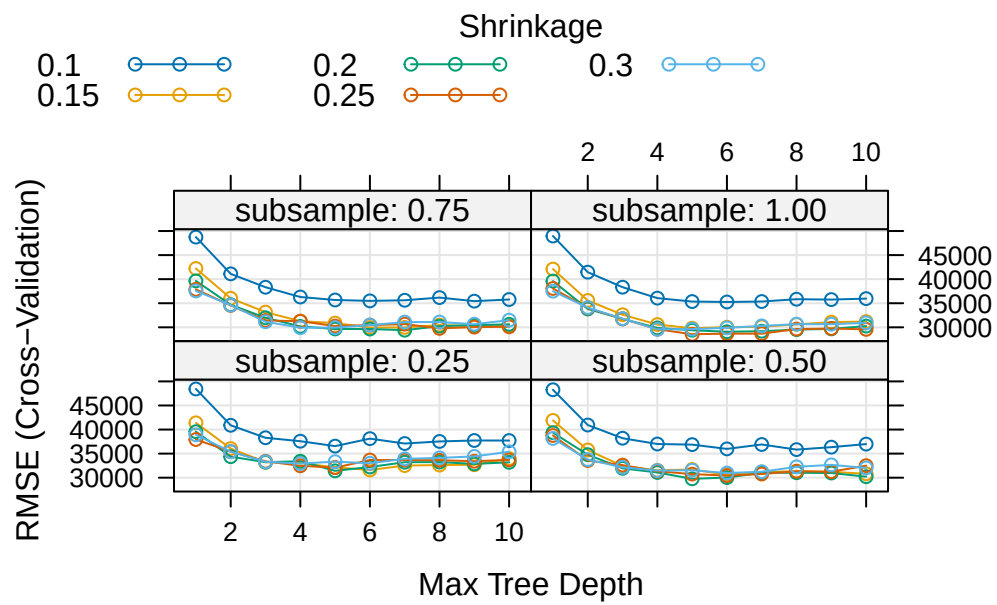
```
training$random <- rnorm(nrow(training))
train_x <- model.matrix(Sale_Price ~ ., data = training)[, -1]
train_y <- training$Sale_Price

set.seed(12345)
xgb.ames <- xgboost(
  data = train_x, label = train_y, subsample = 0.5, nrounds = 100,
  objective = "reg:squarederror", verbose = 0
)

xgbcv.ames <- xgb.cv(
  data = train_x, label = train_y, subsample = 0.5, nrounds = 100,
  nfold = 10, objective = "reg:squarederror", verbose = 0
)

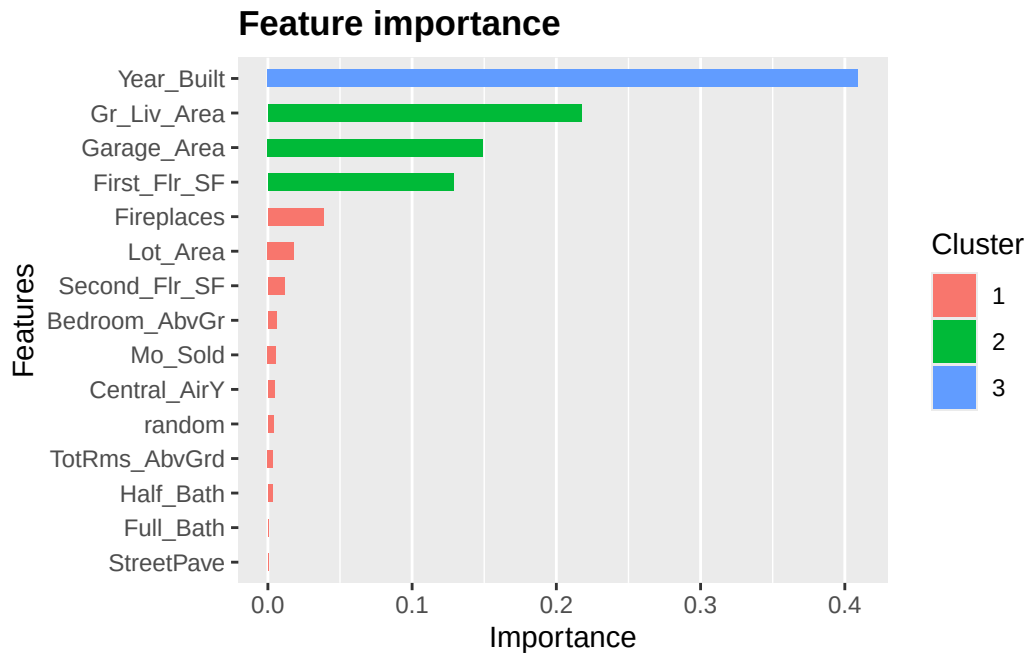
tune_grid <- expand.grid(
  nrounds = 24,
  eta = c(0.1, 0.15, 0.2, 0.25, 0.3),
  max_depth = c(1:10),
  gamma = c(0),
  colsample_bytree = 1,
  min_child_weight = 1,
  subsample = c(0.25, 0.5, 0.75, 1)
)

set.seed(12345)
xgb.ames.caret <- train(
  x = train_x, y = train_y,
  method = "xgbTree",
  tuneGrid = tune_grid,
  trControl = trainControl(method = 'cv', number = 10)
)
plot(xgb.ames.caret)
```



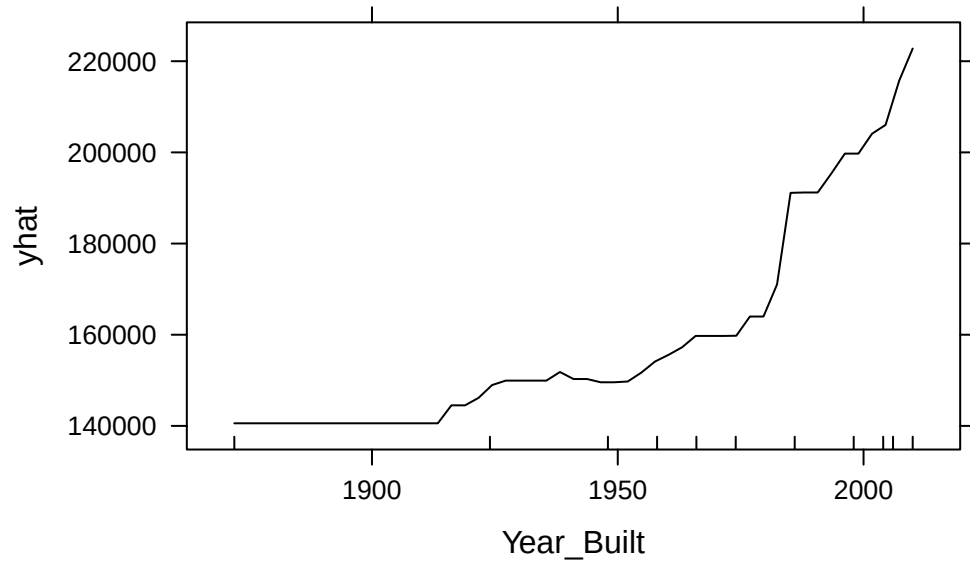
```
# Fit tuned model
xgb.ames <- xgboost(
  data = train_x, label = train_y, subsample = 1,
  nrounds = 24, eta = 0.25, max_depth = 5,
  objective = "reg:squarederror", verbose = 0
)

# Variable importance including random variable
xgb.ggplot.importance(xgb.importance(feature_names = colnames(train_x), model = xgb.ames))
```

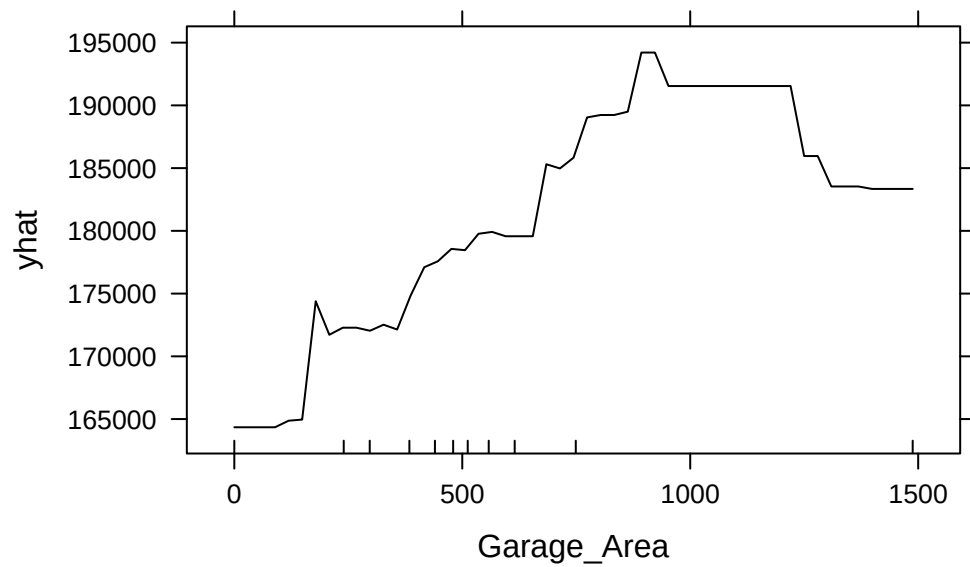


```
# Partial dependence plots
train_df <- as.data.frame(train_x)

partial(xgb.ames, pred.var = "Year_Built",
        plot = TRUE, rug = TRUE, alpha = 0.1, plot.engine = "lattice",
        train = train_df, pdp.color = "red")
```



```
partial(xgb.ames, pred.var = "Garage_Area",  
        plot = TRUE, rug = TRUE, alpha = 0.1, plot.engine = "lattice",  
        train = train_df)
```



6.5.1 XGBoost Results (Aligned Columns)

```
testing$random <- rnorm(nrow(testing))
testing_x <- model.matrix(Sale_Price ~ ., data = testing)[, -1]

# Align test columns to training columns
missing_cols <- setdiff(colnames(train_x), colnames(testing_x))
for (col in missing_cols) {
  testing_x <- cbind(testing_x, 0)
  colnames(testing_x)[ncol(testing_x)] <- col
}
testing_x <- testing_x[, colnames(train_x)]

testing_sel$pred_xgb <- predict(xgb.ames, testing_x)

xgb_perf <- testing_sel %>%
  mutate(AE = abs(Sale_Price - pred_xgb),
         APE = 100*abs((Sale_Price - pred_xgb)/Sale_Price)) %>%
  summarise(Model = "XGBoost",
            RMSE = sqrt(mean((Sale_Price - pred_xgb)^2)),
            MAE = mean(AE),
            MAPE = mean(APE),
            R2 = 1 - sum((Sale_Price - pred_xgb)^2)/sum((Sale_Price - mean(Sale_Price))^2))
xgb_perf
```

A tibble: 1 x 5

Model	RMSE	MAE	MAPE	R2
<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1 XGBoost	30355.	19820.	11.8	0.864

6.6 Final Model Comparison

```
bind_rows(tree_perf, rf_perf, xgb_perf) %>%
  arrange(RMSE)
```

A tibble: 3 x 5

Model	RMSE	MAE	MAPE	R2
<chr>	<dbl>	<dbl>	<dbl>	<dbl>

1	Random Forest	30147.	19223.	11.6	0.866
2	XGBoost	30355.	19820.	11.8	0.864
3	Decision Tree	42938.	28322.	17.5	0.727

Summary: The Decision Tree provides interpretability, Random Forest and XGBoost attempt to further reduce bias and manage variance error through randomization and ensembles of weak learners. The issue is that they become less and less interpretable, so we have to introduce other tools to understand what variables are important and how they are relating to the outcome.

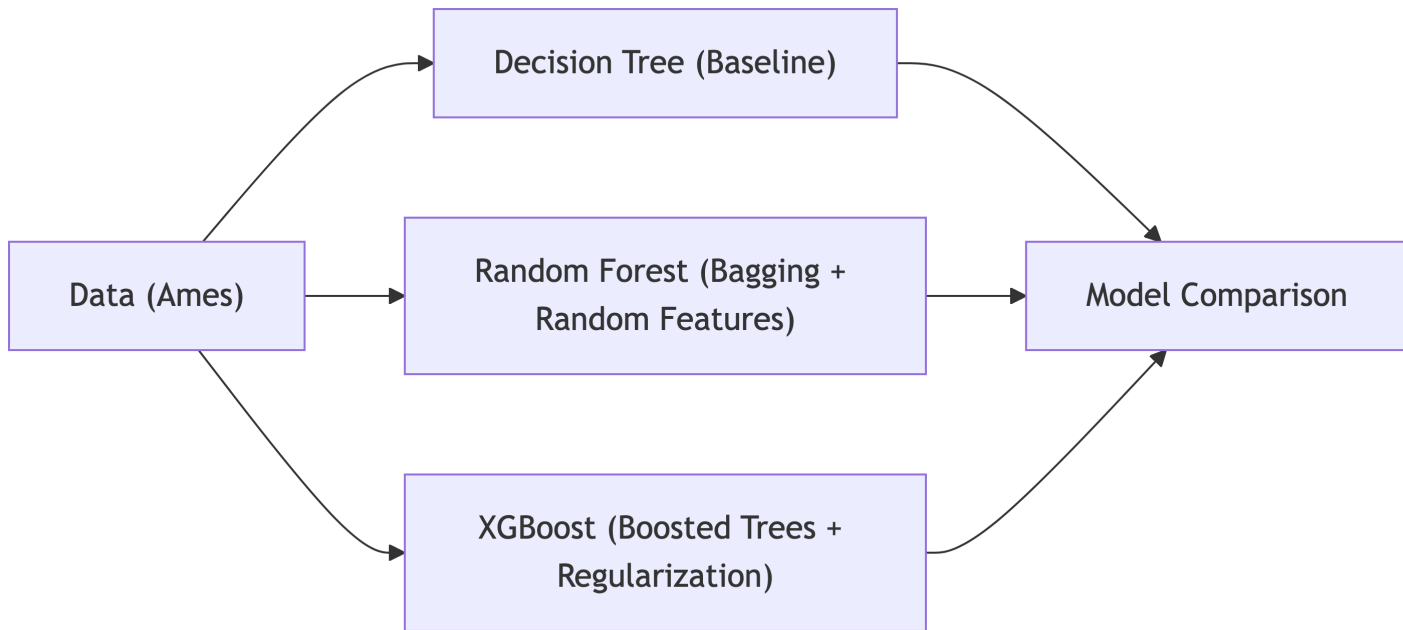
7 Tree Based Models - Python Version

Tree Models: **Decision Tree**, **Random Forest**, and **XGBoost** on the Ames Housing dataset. Includes random-variable baselines, variable importance, partial dependence plots, and hyperparameter tuning (random search).

7.1 Visual Roadmap

```
flowchart LR
    A["Data (Ames)"]
    DT["Decision Tree (Baseline)"]
    RF["Random Forest (Bagging + Random Features)"]
    XGB["XGBoost (Boosted Trees + Regularization)"]
    Compare["Model Comparison"]

    A --> DT
    A --> RF
    A --> XGB
    DT --> Compare
    RF --> Compare
    XGB --> Compare
```



7.2 Setup and Data Preparation

```
import pandas as pd
import numpy as np
from sklearn.datasets import fetch_openml
from sklearn.model_selection import train_test_split, RandomizedSearchCV, KFold
from sklearn.tree import DecisionTreeRegressor, plot_tree
from sklearn.ensemble import RandomForestRegressor
from sklearn.inspection import PartialDependenceDisplay
from xgboost import XGBRegressor, plot_importance
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings("ignore")

# Set random seed for reproducibility
seed = 12345
np.random.seed(seed)

# Load Ames dataset
ames = fetch_openml(name="house_prices", as_frame=True)
```



```

df = ames.frame

# Feature selection matching R version
features = [
    'BedroomAbvGr', 'YearBuilt', 'MoSold', 'LotArea', 'Street', 'CentralAir',
    '1stFlrSF', '2ndFlrSF', 'FullBath', 'HalfBath', 'Fireplaces',
    'GarageArea', 'GrLivArea', 'TotRmsAbvGrd'
]
df = df[features + ['SalePrice']].dropna()

# Encode categoricals
df = pd.get_dummies(df, drop_first=True)

# Train/test split
train, test = train_test_split(df, test_size=0.3, random_state=seed)

# Add random variable for variable-importance baseline
train['random'] = np.random.randn(len(train))
test['random'] = np.random.randn(len(test))

X_train = train.drop(columns=['SalePrice'])
y_train = train['SalePrice']
X_test = test.drop(columns=['SalePrice'])
y_test = test['SalePrice']

```

7.3 Decision Tree (Baseline)

```

tree = DecisionTreeRegressor(random_state=seed, max_depth=5)
tree.fit(X_train, y_train)

plt.figure(figsize=(16,8))
plot_tree(tree, feature_names=X_train.columns, filled=True, max_depth=3)
plt.title("Decision Tree for Ames Housing")
plt.show()

# Metrics
pred_tree = tree.predict(X_test)
rmse_tree = mean_squared_error(y_test, pred_tree)
mae_tree = mean_absolute_error(y_test, pred_tree)

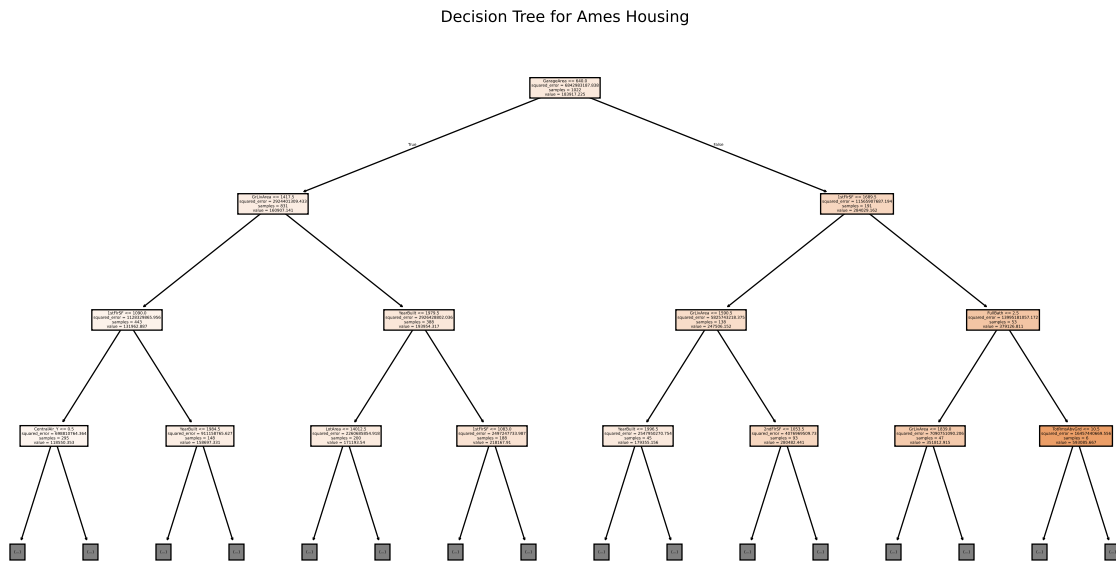
```

```

mape_tree = np.mean(np.abs((y_test - pred_tree) / y_test)) * 100
r2_tree = r2_score(y_test, pred_tree)

pd.DataFrame([["Decision Tree", rmse_tree, mae_tree, mape_tree, r2_tree]],
             columns=["Model", "RMSE", "MAE", "MAPE", "R2"])

```



	Model	RMSE	MAE	MAPE	R2
0	Decision Tree	2.378377e+09	28465.348099	17.701235	0.522975

7.4 Random Forest: Random Variable, Importance, PDPs, and Tuning

```

rf = RandomForestRegressor(random_state=seed)

# Define parameter grid (similar to R tuning)
param_dist_rf = {
    'n_estimators': [100, 200, 300, 400, 500],
    'max_features': ['sqrt', 'log2', None, 4, 6, 8],
    'max_depth': [None, 5, 10, 15, 20],
    'min_samples_split': [2, 5, 10]
}

```

```

}

cv = KFold(n_splits=10, shuffle=True, random_state=seed)

rf_random = RandomizedSearchCV(
    estimator=rf,
    param_distributions=param_dist_rf,
    n_iter=20,
    cv=cv,
    random_state=seed,
    n_jobs=-1,
    scoring='neg_root_mean_squared_error'
)

rf_random.fit(X_train, y_train)
rf_best = rf_random.best_estimator_
print("Best Random Forest Parameters:", rf_random.best_params_)

# Importance with random variable
importances = pd.Series(rf_best.feature_importances_, index=X_train.columns).sort_values(ascending=False)
importances[:15].plot(kind='barh', title='Random Forest: Top Variables (with Random Baseline)')
plt.show()

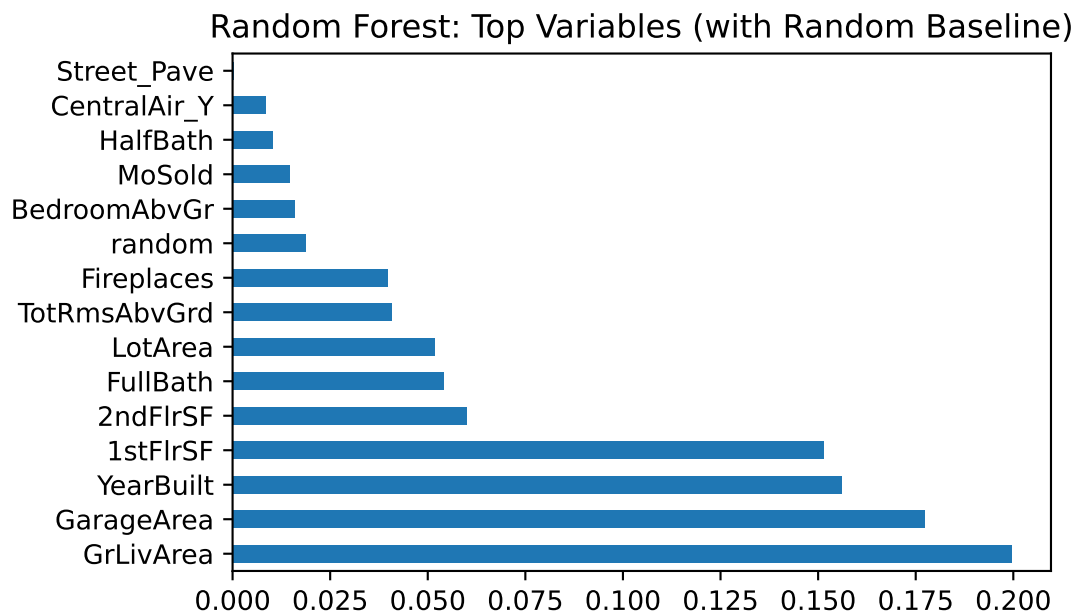
# Partial dependence plots
plt.figure()
PartialDependenceDisplay.from_estimator(rf_best, X_train, ['YearBuilt', 'GarageArea'])
plt.suptitle("Partial Dependence (Random Forest)")
plt.show()

# Performance metrics
pred_rf = rf_best.predict(X_test)
rmse_rf = mean_squared_error(y_test, pred_rf)
mae_rf = mean_absolute_error(y_test, pred_rf)
mape_rf = np.mean(np.abs((y_test - pred_rf) / y_test)) * 100
r2_rf = r2_score(y_test, pred_rf)

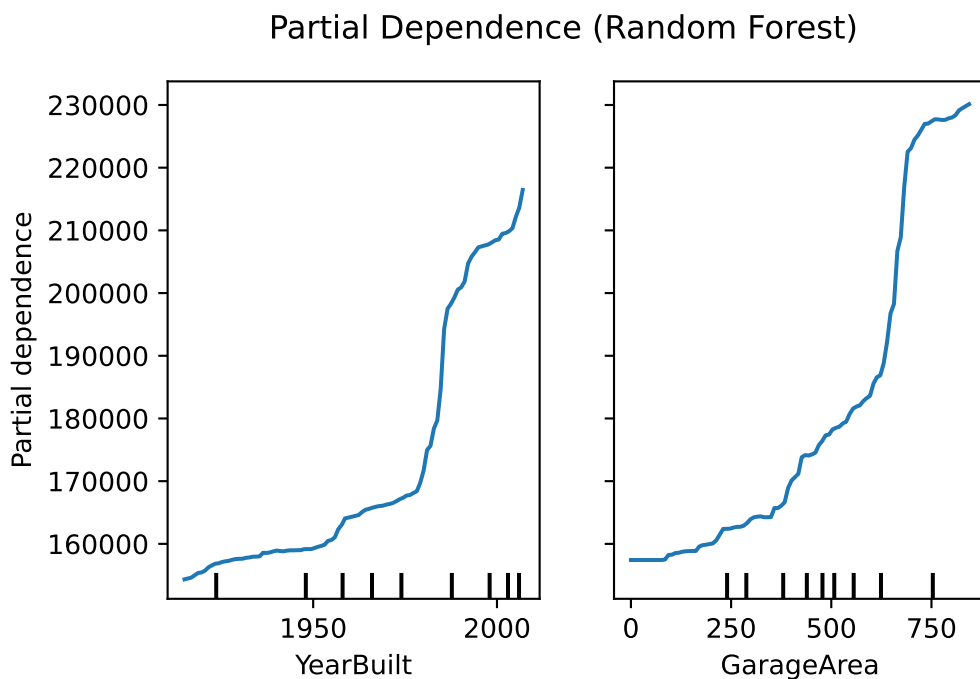
pd.DataFrame([["Random Forest (Tuned)", rmse_rf, mae_rf, mape_rf, r2_rf]],
              columns=["Model", "RMSE", "MAE", "MAPE", "R2"])

```

Best Random Forest Parameters: {'n_estimators': 500, 'min_samples_split': 2, 'max_features':



<Figure size 1650x1050 with 0 Axes>



	Model	RMSE	MAE	MAPE	R2
0	Random Forest (Tuned)	1.093313e+09	19889.772151	12.832756	0.780717

7.5 XGBoost: Random Variable, Importance, PDPs, and Tuning

```

xgb = XGBRegressor(
    objective='reg:squarederror',
    random_state=seed,
    n_jobs=-1
)

# Define tuning grid similar to R (eta = learning_rate, subsample, max_depth)
param_dist_xgb = {
    'n_estimators': [24, 50, 100, 200],
    'learning_rate': [0.1, 0.15, 0.2, 0.25, 0.3],
    'max_depth': [1, 3, 5, 7, 10],
    'subsample': [0.25, 0.5, 0.75, 1],
    'colsample_bytree': [0.5, 0.75, 1]
}

xgb_random = RandomizedSearchCV(
    estimator=xgb,
    param_distributions=param_dist_xgb,
    n_iter=20,
    cv=cv,
    random_state=seed,
    n_jobs=-1,
    scoring='neg_root_mean_squared_error'
)

xgb_random.fit(X_train, y_train)
xgb_best = xgb_random.best_estimator_
print("Best XGBoost Parameters:", xgb_random.best_params_)

# Importance plot with random variable
plot_importance(xgb_best, max_num_features=15, importance_type='gain')
plt.title("XGBoost: Variable Importance (with Random Baseline)")
plt.show()

```

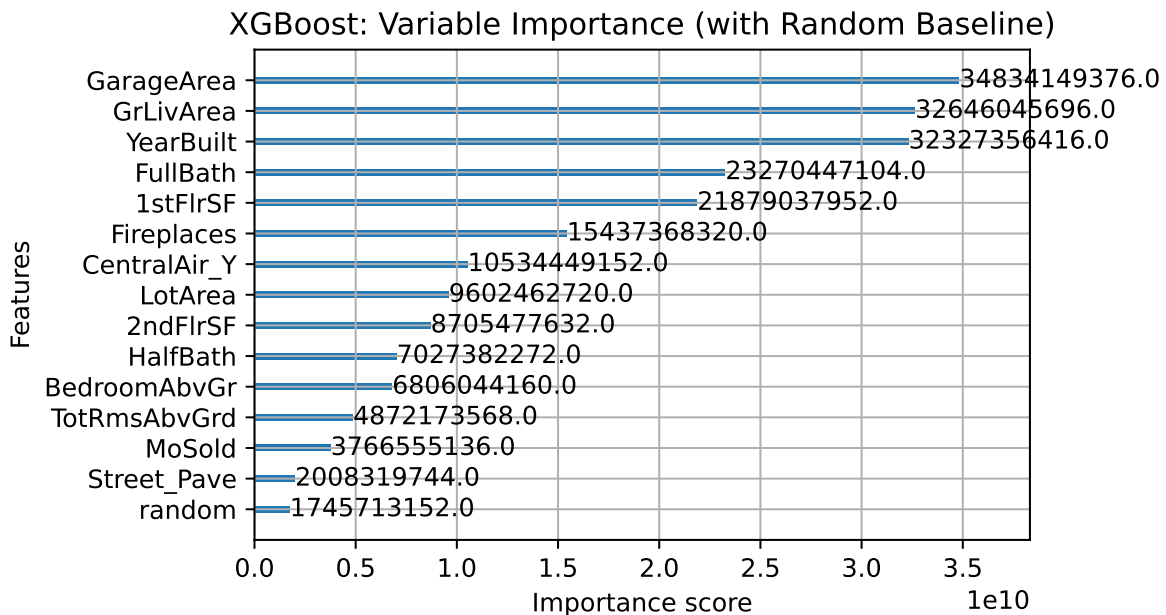
```
# Partial dependence plots
plt.figure()
PartialDependenceDisplay.from_estimator(xgb_best, X_train, ['YearBuilt', 'GarageArea'])
plt.suptitle("Partial Dependence (XGBoost)")
plt.show()

# Metrics
pred_xgb = xgb_best.predict(X_test)
rmse_xgb = mean_squared_error(y_test, pred_xgb)
mae_xgb = mean_absolute_error(y_test, pred_xgb)
mape_xgb = np.mean(np.abs((y_test - pred_xgb) / y_test)) * 100
r2_xgb = r2_score(y_test, pred_xgb)

pd.DataFrame([["XGBoost (Tuned)", rmse_xgb, mae_xgb, mape_xgb, r2_xgb]],
              columns=["Model", "RMSE", "MAE", "MAPE", "R2"])

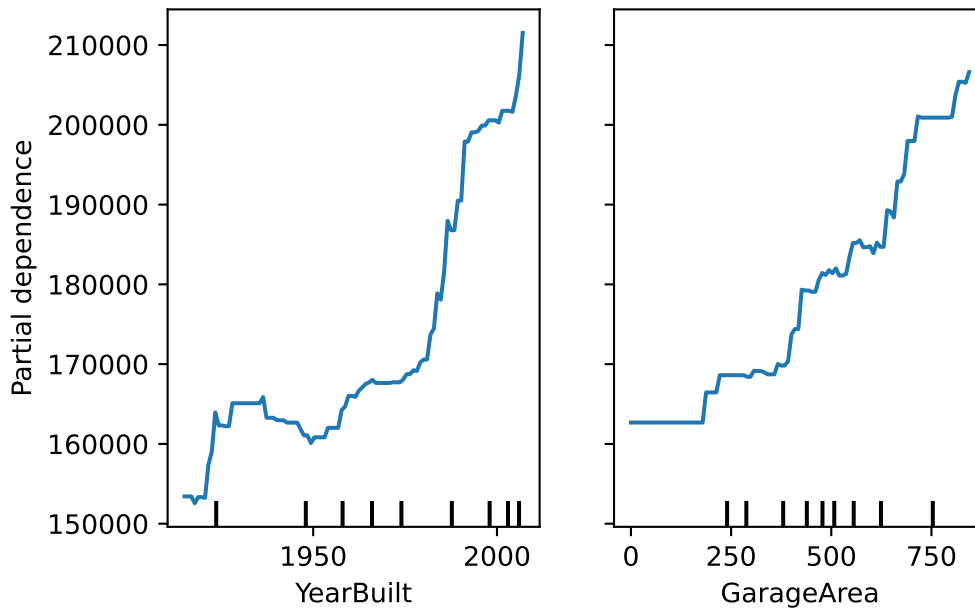
```

Best XGBoost Parameters: {'subsample': 0.75, 'n_estimators': 50, 'max_depth': 5, 'learning_rate': 0.1}



<Figure size 1650x1050 with 0 Axes>

Partial Dependence (XGBoost)



Model	RMSE	MAE	MAPE	R2
0 XGBoost (Tuned)	1.133839e+09	20819.921875	13.330445	0.772589

7.6 Final Model Comparison

```
results = pd.DataFrame([
    ["Decision Tree", rmse_tree, mae_tree, mape_tree, r2_tree],
    ["Random Forest (Tuned)", rmse_rf, mae_rf, mape_rf, r2_rf],
    ["XGBoost (Tuned)", rmse_xgb, mae_xgb, mape_xgb, r2_xgb]
], columns=["Model", "RMSE", "MAE", "MAPE", "R2"]).sort_values("RMSE")
results
```

	Model	RMSE	MAE	MAPE	R2
1	Random Forest (Tuned)	1.093313e+09	19889.772151	12.832756	0.780717
2	XGBoost (Tuned)	1.133839e+09	20819.921875	13.330445	0.772589
0	Decision Tree	2.378377e+09	28465.348099	17.701235	0.522975

Summary: The Decision Tree provides interpretability, Random Forest and XG-Boost attempt to further reduce bias and manage variance error through randomization and ensembles of weak learners. The issue is that they become less and less interpretable, so we have to introduce other tools to understand what variables are important and how they are relating to the outcome.

8 Understanding Random Forest and XGBoost Hyperparameters

8.1 Random Forest Hyperparameters

8.1.1 Switching Between Regression and Classification

Language	How to Specify Regression	How to Specify Classification
R	<code>randomForest(y ~ ., data = df)</code> where y is numeric	<code>randomForest(y ~ ., data = df)</code> where y is a factor
Python	<code>RandomForestRegressor()</code>	<code>RandomForestClassifier()</code>

In **R**, the outcome type (**numeric** vs **factor**) determines whether the model performs **regression** or **classification** automatically.

In **Python**, you explicitly choose the class (`RandomForestRegressor` or `RandomForestClassifier`).

Concept	R (<code>randomForest</code>)	Python (<code>RandomForestRegressor</code> / <code>RandomForestClassifier</code>)	Meaning / Interpretation	Typical Values	Impact / Interpretation
Model type	Determined by outcome type (numeric vs factor)	Different estimator class	Defines whether the forest predicts continuous or categorical outcomes	–	Choose based on target type

Concept	R (randomForest)	Python (RandomForestRegressor / RandomForestClassifier)	Meaning / Classifier	Typical Values	Impact / In- terpretation
Number of trees	n_estimators	n_estimators	Number of trees to grow in the forest	100–1000 (500 common)	More trees reduce variance but increase runtime
Features per split	mtry	max_features	# of variables randomly selected at each split	Regression: $\sqrt{p}$; Classification: $p/3$	Smaller $\rightarrow$ higher bias, more diversity; larger $\rightarrow$ lower bias, more correlation
Max tree depth	(no direct arg, via <i>nodesize</i>)	max_depth	Maximum depth of each tree	None, or 5–30	Deeper trees capture complexity but risk overfitting
Min samples per split	(via <i>nodesize</i> indirectly)	min_samples_split	Minimum # of samples to split a node	2–10	Larger $\rightarrow$ smoother, higher bias; smaller $\rightarrow$ lower bias, higher variance
Min samples per leaf	(via <i>nodesize</i>)	min_samples_leaf	Minimum samples per terminal node	1–5	Prevents tiny leaves, reduces overfitting
Bootstrap sampling	replace	bootstrap	Whether to sample with replacement	TRUE/FALSE	Adds randomness; TRUE recommended
Sample fraction per tree	(rarely tuned)	max_samples	Fraction of training data used per tree	0.5–1.0	Lower $\rightarrow$ more randomness, higher bias

Concept	R (<code>randomForest</code>)	Python (<code>RandomForestRegressor</code> / <code>RandomForestClassifier</code>)	Meaning / Purpose	Typical Values	Impact / In- terpretation
Learning objective	Implicit: minimize MSE (regression) or Gini/entropy (classification)	Implicit: minimize MSE or Gini/entropy	Objective changes automatically with model type	–	Regression averages numeric values; classification votes on class labels
Random seed	<code>set.seed()</code>	<code>random_state</code>	Ensures reproducibility	any integer	Use same seed for consistency

8.1.2 Thinking About Random Forest Tuning

- Start with **many trees** (500).
- Tune `mtry` / `max_features` to balance bias vs variance.
- Use `max_depth`, `min_samples_split`, or `min_samples_leaf` to prevent overfitting.
- RF tuning is relatively forgiving — focus on **efficiency and stability** rather than micro-optimization.
- For **classification**, check **OOB error** or **ROC-AUC** rather than RMSE.

8.2 XGBoost Hyperparameters

Concept	R (xgboost)	Python (XGBRegressor / XGBClassifier)	Meaning / Role	Typical Values	Impact / In- terpretation
Boosting rounds	nrounds	n_estimators	Total # of trees (boosting steps)	100–1000	Too high → overfit unless learning rate small
Learning rate	eta	learning_rate	Shrinks each tree's contribution	0.01–0.3	Lower = slower, safer; higher = faster, riskier
Tree depth	max_depth	max_depth	Max depth per tree	3–10	Shallow = less overfit; deep = more complex
Min child weight	min_child_weight	min_child_weight	Min sum of instance weights per leaf	1–10	Larger = more conservative splits
Subsample (rows)	subsample	subsample	Fraction of training rows used per round	0.5–1.0	Lower adds randomness, improves generalization
Col sample (features)	colsample_bytree	colsample_bytree	Fraction of features per tree	0.5–1.0	Reduces feature correlation
Gamma	gamma	gamma	Min loss reduction to split a node	0–5	Higher = more conservative, prevents overfitting
L1 regularization	alpha	reg_alpha	Penalty on abs(weights)	0–10	Encourages sparsity (feature selection)
L2 regularization	lambda	reg_lambda	Penalty on squared(weights)	0–10	Controls weight magnitudes (stability)

Concept	R (<code>xgboost</code>)	Python (<code>XGBRegressor</code> / <code>XGBClassifier</code>)	Meaning / Role	Typical Values	Impact / In- terpretation
Learning objective	<code>objective</code>	<code>objective</code>	Specifies prediction type	" <code>reg:squarederror</code> ", " <code>binary:logistic</code> ", etc.	Must , match type
Early stopping	via <code>xgb.cv()</code>	<code>early_stopping_rounds</code>	Stops when no improvement	10–50 rounds	Prevents unnecessary rounds
Seed	<code>set.seed()</code>	<code>random_state</code>	Ensures reproducibility	any integer	Consistent results across runs

8.2.1 Thinking About XGBoost Tuning

1. **Start simple** — tune `n_estimators`, `max_depth`, and `learning_rate` first.
2. Add regularization (`gamma`, `alpha`, `lambda`) to control overfitting.
3. Use `subsample` and `colsample_bytree` to add randomness.

8.3 Practical Tips and Heuristics

Situation	Random Forest Strategy	XGBoost Strategy
Overfitting	Increase <code>min_samples_split</code> , reduce <code>max_depth</code>	Lower <code>learning_rate</code> , increase <code>gamma</code> or <code>min_child_weight</code>
Underfitting	Increase <code>n_estimators</code> , allow deeper trees	Lower regularization, increase <code>max_depth</code> or <code>n_estimators</code>
Slow training	Reduce trees or sample fraction	Increase <code>learning_rate</code> , reduce trees

Situation	Random Forest Strategy	XGBoost Strategy
Unstable results	Fix <code>random_state</code> , use more trees	Fix <code>random_state</code> , enable early stopping
High variance	Lower <code>max_features</code> , increase randomness	Increase <code>subsample</code> diversity, lower <code>max_depth</code>

8.4 Rule-of-Thumb Summary

Parameter	Low Value →	High Value →
<code>n_estimators</code>	Underfitting	Stable but slower
<code>max_depth</code>	Simpler model (underfit)	Complex model (overfit)
<code>learning_rate</code>	Slow, steady, robust	Fast, risk of overfit
<code>subsample</code> / <code>colsample_bytree</code>	More bias, less variance	Lower bias, higher variance
<code>mtry</code> / <code>max_features</code>	More randomization	More correlation between trees
<code>min_samples_split</code> / <code>min_child_weight</code>	Flexible, low bias	Conservative, higher bias

8.4.1 Key Takeaways

- **Random Forest**
 - Regression/classification determined by outcome type (R) or model class (Python).
 - Reduces variance through bagging.
 - Tuning focuses on **stability** and **efficiency**.
- **XGBoost**
 - Requires explicit objective selection.
 - Reduces error via boosting and regularization.
- Use interpretability tools (importance, PDPs, SHAP) to understand **why** tuning changes model behavior.

9 Neural Networks in R

9.1 Introduction

This chapter introduces **neural networks in R**, progressing from feed-forward to recurrent (RNN) and convolutional (CNN) architectures.

It uses the Ames housing dataset, a sine wave sequence, and the MNIST digit dataset to demonstrate increasingly complex network structures.

Each example illustrates a different ANN architecture.

9.2 1. Feed-Forward Neural Networks (Ames Data)

Feed-forward networks are the simplest type of neural networks. They map input features (like home size, rooms, or year built) to an output (home price) through **layers of neurons**. Each neuron applies a weighted sum and activation function, allowing the network to learn complex relationships between variables.

```
#Note: This is specific to my environment
library(reticulate)
use_python("/Users/donaldhale/r-arm/bin/python", required = TRUE)

library(keras3)

tf <- reticulate::import("tensorflow", delay_load = FALSE)
cat("TensorFlow version:", tf$`__version__`, "\n")
```

```
TensorFlow version: 2.16.1
```

```

library(tidyverse)
library(caret)
library(nnet)
library(NeuralNetTools)
library(AmesHousing)

set.seed(4321)

# Load and sample data
ames <- make_ordinal_ames() %>% mutate(id = row_number())
train_idx <- sample(1:nrow(ames), 0.7 * nrow(ames))
training <- ames[train_idx, ]
testing <- ames[-train_idx, ]

# Select a manageable subset of variables
vars <- c("Sale_Price", "Bedroom_AbvGr", "Year_Built", "Mo_Sold", "Lot_Area",
          "Street", "Central_Air", "First_Flr_SF", "Second_Flr_SF",
          "Full_Bath", "Half_Bath", "Fireplaces", "Garage_Area",
          "Gr_Liv_Area", "TotRms_AbvGrd")
training <- training[, vars]
testing <- testing[, vars]

# Convert categorical predictors into dummy numeric columns
x_train <- model.matrix(Sale_Price ~ ., data = training)[, -1]
y_train <- training$Sale_Price
x_test <- model.matrix(Sale_Price ~ ., data = testing)[, -1]
y_test <- testing$Sale_Price

# Drop low-variance predictors and scale for stable training
nzv <- nearZeroVar(x_train)
if (length(nzv) > 0) {
  x_train <- x_train[, -nzv, drop = FALSE]
  x_test <- x_test[, -nzv, drop = FALSE]
}

x_train <- scale(x_train)
x_test <- scale(x_test, center = attr(x_train, "scaled:center"),
               scale = attr(x_train, "scaled:scale"))

y_train_s <- scale(y_train)

# Combine into a single training dataframe

```

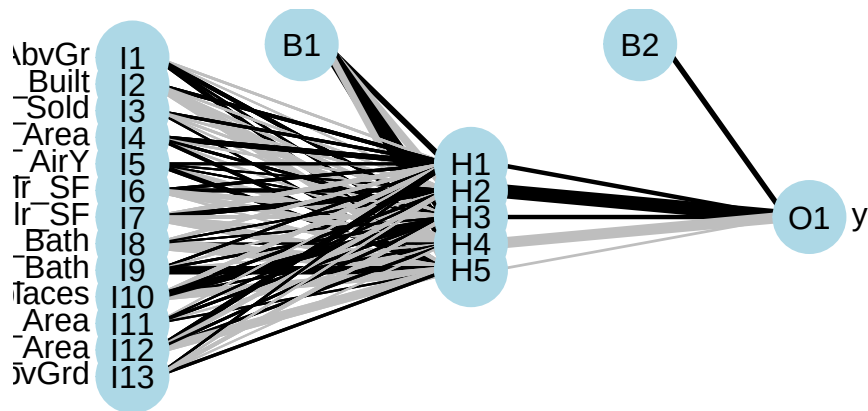


```

train_df <- data.frame(y = as.numeric(y_train_s), x_train)

set.seed(123)
nn.ames <- nnet(y ~ ., data = train_df, size = 5, linout = TRUE, trace = FALSE)
plotnet(nn.ames)

```



9.2.1 1.2 Cross-Validation Tuning

We tune two important parameters: - **size** — number of neurons in the hidden layer
 - **decay** — regularization strength (prevents overfitting)

`caret::train()` automates this search using k-fold cross-validation.

```

tune_grid <- expand.grid(size = c(3,5,7,9), decay = c(0,0.1,0.5,1))
ctrl <- trainControl(method = "cv", number = 5)

set.seed(123)

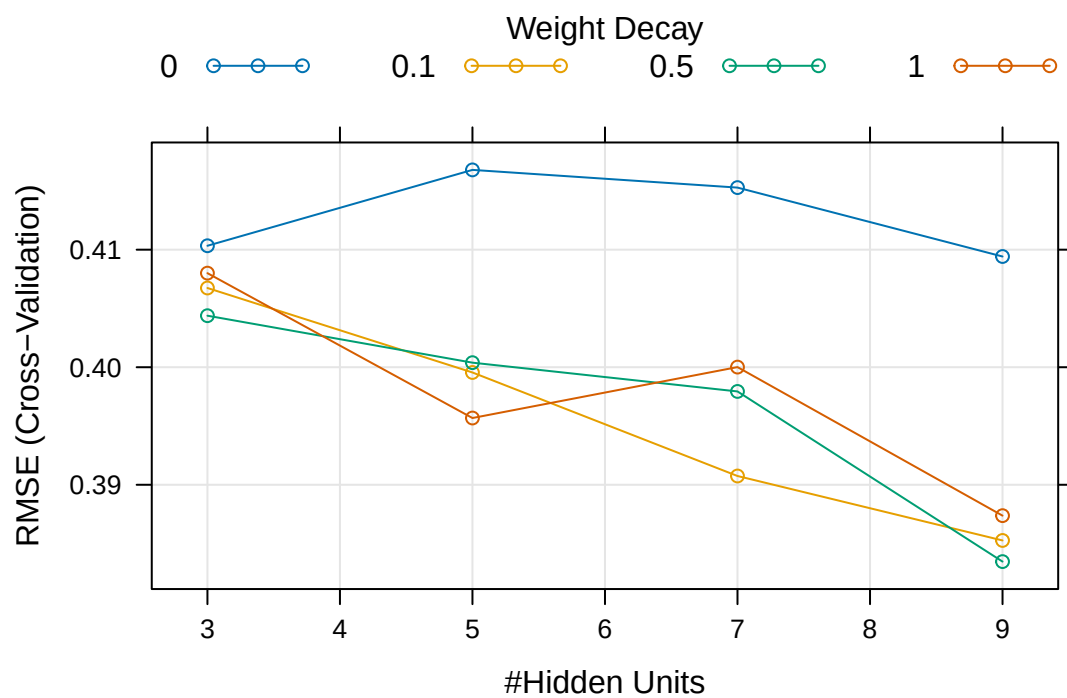
```

```
nn.tuned <- train(
  x = x_train, y = as.numeric(y_train_s),
  method = "nnet",
  tuneGrid = tune_grid,
  trControl = ctrl,
  linout = TRUE,
  trace = FALSE
)

nn.tuned$bestTune
```

```
size decay
15    9    0.5
```

```
plot(nn.tuned)
```



9.2.2 1.3 Evaluate Performance

After identifying the best parameters, we retrain the model and evaluate its predictive accuracy.

```
best_size <- nn.tuned$bestTune$size
best_decay <- nn.tuned$bestTune$decay

set.seed(123)
nn.final <- nnet(
  y ~ .,
  data = train_df,
  size = best_size,
  decay = best_decay,
  linout = TRUE,
  trace = FALSE
)

pred_std <- predict(nn.final, newdata = as.data.frame(x_test))
pred_nn <- pred_std * attr(y_train_s, "scaled:scale") + attr(y_train_s, "scaled:center")

perf <- data.frame(
  MAE = mean(abs(y_test - pred_nn)),
  MAPE = mean(abs((y_test - pred_nn) / y_test)) * 100
)
perf
```

	MAE	MAPE
1	20392.57	12.63864

9.3 2. Feed-Forward Neural Networks (Keras 3)

The **Keras 3** API is a modern deep learning interface. It supports more complex models and GPUs while maintaining a simple syntax.

This example shows the same concept as above using a multi-layer perceptron (MLP).

```

#library(keras3)
set.seed(123)

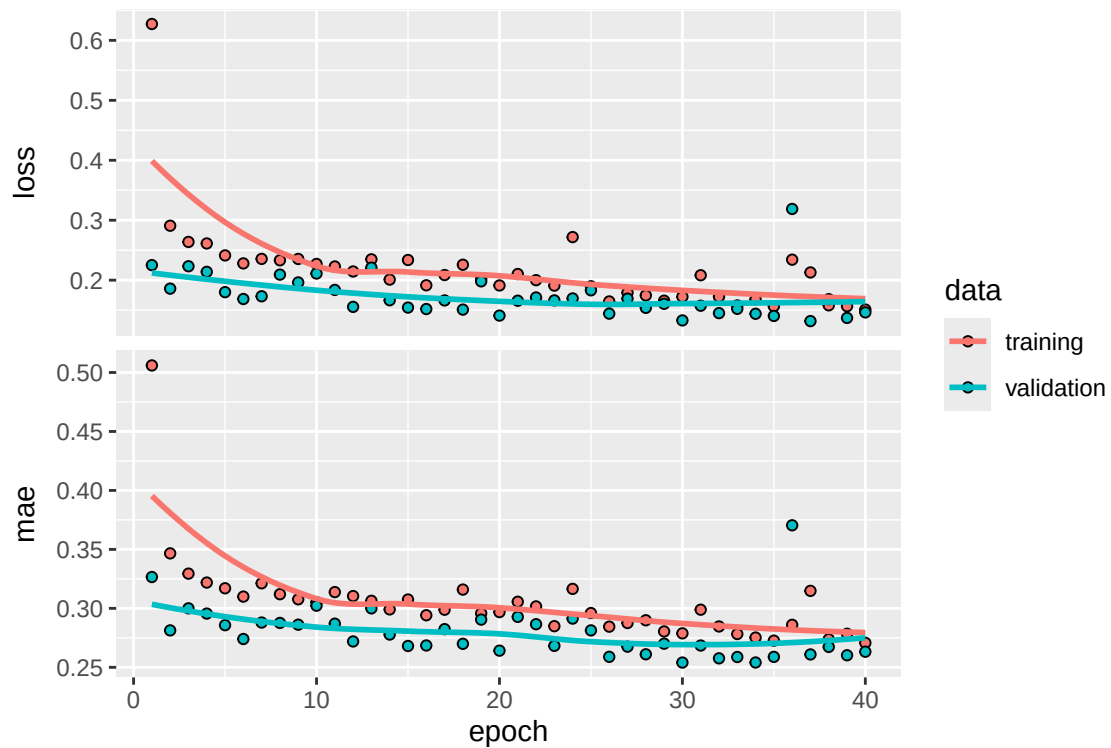
model_ff <- keras_model_sequential() |>
  layer_dense(units = 16, activation = "relu", input_shape = ncol(x_train)) |>
  layer_dropout(rate = 0.1) |>
  layer_dense(units = 1)

compile(model_ff, optimizer = optimizer_adam(learning_rate = 0.01),
        loss = "mse", metrics = "mae")

history <- fit(model_ff, x_train, as.numeric(y_train_s),
              epochs = 40, batch_size = 32, validation_split = 0.2, verbose = 0)

plot(history)

```

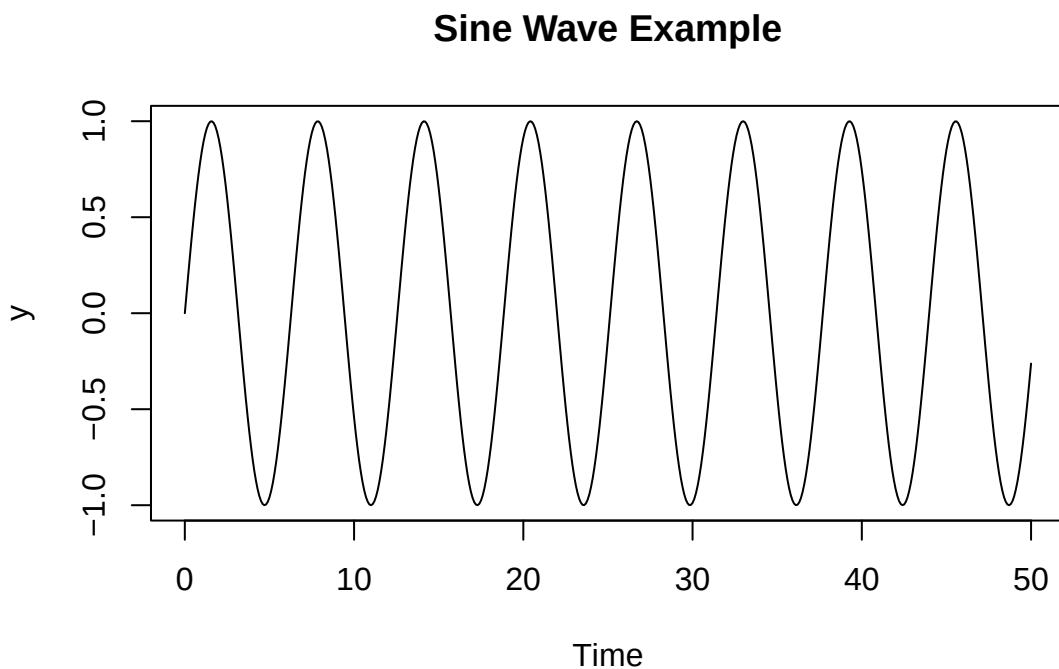


9.4 3. Recurrent Neural Network (RNN)

RNNs are ideal for **sequential data** such as time series, text, or speech.

We use a sine wave as a toy example because its repeating pattern requires the model to “remember” recent values — something feed-forward networks cannot do.

```
set.seed(42)
t <- seq(0, 50, by = 0.1)
y <- sin(t)
plot(t, y, type='l', main="Sine Wave Example", ylab="y", xlab="Time")
```



```
window_size <- 20
X <- sapply(1:(length(y) - window_size), function(i) y[i:(i + window_size - 1)])
X <- t(X)
y_out <- y[(window_size + 1):length(y)]
X <- array(X, dim = c(nrow(X), window_size, 1))

split <- round(0.8 * nrow(X))
x_train <- X[1:split,,]; y_train <- y_out[1:split]
```

```

x_test <- X[(split + 1):nrow(X),,]; y_test <- y_out[(split + 1):length(y_out)]

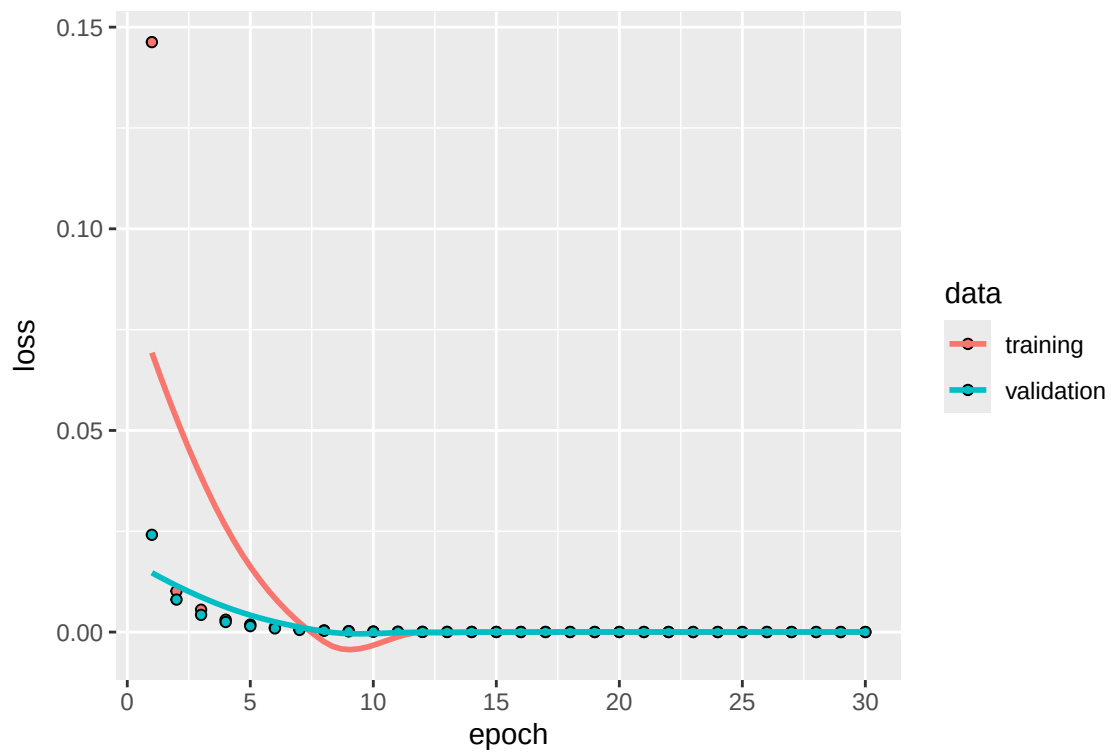
model_rnn <- keras_model_sequential() |>
  layer_simple_rnn(units = 16, activation = "tanh", input_shape = c(window_size, 1)) |>
  layer_dense(units = 1)

compile(model_rnn, loss = "mse", optimizer = "adam")

history_rnn <- fit(model_rnn, x_train, y_train, epochs = 30, batch_size = 16,
  validation_split = 0.2, verbose = 0)

plot(history_rnn)

```



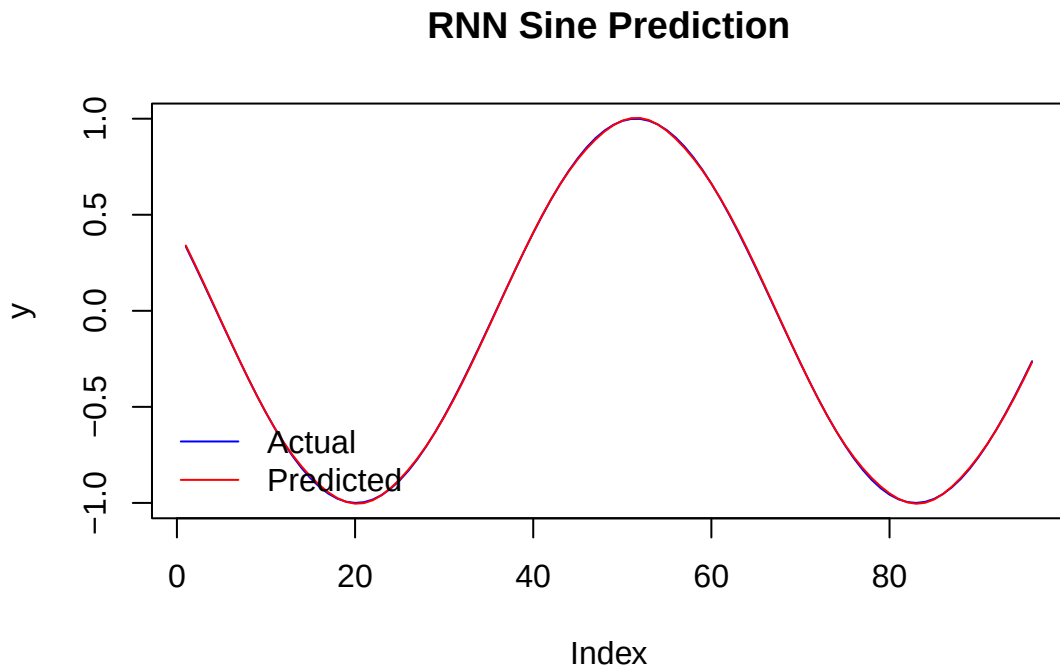
```

preds <- predict(model_rnn, x_test)

```

3/3 - 0s - 11ms/step

```
plot(y_test, type='l', main="RNN Sine Prediction", col='blue', ylab="y")
lines(preds, col='red')
legend("bottomleft", legend=c("Actual","Predicted"), col=c("blue","red"), lty=1, bty="n")
```



9.5 4. Convolutional Neural Network (CNN)

CNNs are built for **spatial pattern recognition** — ideal for images.

The **MNIST** dataset contains 70,000 grayscale images of handwritten digits (0–9), each 28×28 pixels.

These lines prepare the data for use in a CNN.

```
#library(keras3)

# Load and preview the MNIST data
mnist <- dataset_mnist()
```

```

# We only use a subset (5k train, 1k test) for quick demonstration
x_train <- mnist$train$x[1:5000,,]
y_train <- mnist$train$y[1:5000]
x_test  <- mnist$test$x[1:1000,,]
y_test  <- mnist$test$y[1:1000]

# Normalize pixel values: 0-255 -> 0-1
x_train <- x_train / 255
x_test  <- x_test  / 255

# Reshape to [samples, height, width, channels]
# MNIST images are grayscale, so channels = 1
x_train <- array_reshape(x_train, c(nrow(x_train), 28, 28, 1))
x_test  <- array_reshape(x_test,  c(nrow(x_test),  28, 28, 1))

# Convert labels to categorical one-hot vectors (10 classes for digits 0-9)
y_train_cat <- to_categorical(y_train, 10)
y_test_cat  <- to_categorical(y_test,  10)

```

9.5.1 Explanation of the MNIST Preprocessing

1. **Load** the built-in dataset from Keras.
2. **Subsample** for faster training (5,000 train / 1,000 test).
3. **Normalize** pixel intensities to the 0–1 range.
4. **Reshape** to the 4D structure required by CNNs: [samples, height, width, channels].
5. **One-hot encode** labels, converting a digit like 3 into a vector [0,0,0,1,0,0,0,0,0,0].

These steps turn raw pixel data into numeric tensors suitable for learning.

9.5.2 4.1 Train and Evaluate the CNN


```

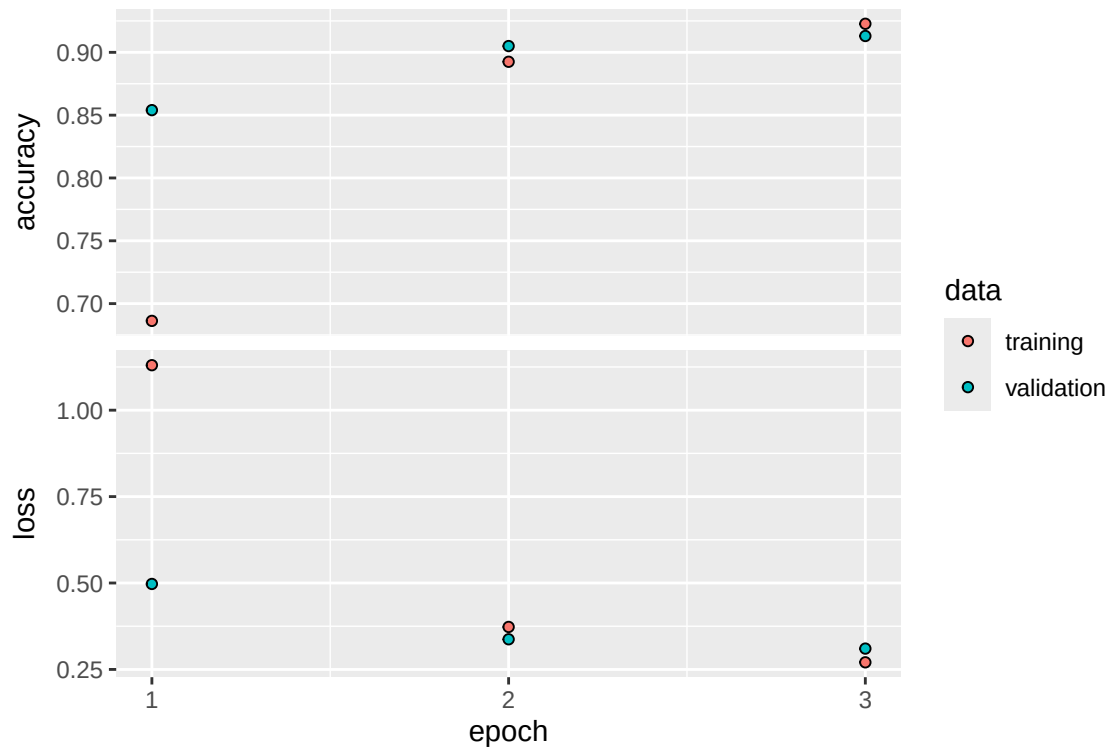
model_cnn <- keras_model_sequential() |>
  layer_conv_2d(filters = 16, kernel_size = c(3,3), activation = "relu", input_shape = c(28,28,1)) |>
  layer_max_pooling_2d(pool_size = c(2,2)) |>
  layer_flatten() |>
  layer_dense(units = 32, activation = "relu") |>
  layer_dense(units = 10, activation = "softmax")

compile(model_cnn, optimizer = "adam",
        loss = "categorical_crossentropy",
        metrics = "accuracy")

history_cnn <- fit(model_cnn, x_train, y_train_cat,
                  epochs = 3, batch_size = 64, validation_split = 0.2, verbose = 0)

plot(history_cnn)

```



```
score <- evaluate(model_cnn, x_test, y_test_cat)
```

32/32 - 0s - 745us/step - accuracy: 0.9050 - loss: 0.3117

```
score
```

```
$accuracy  
[1] 0.905
```

```
$loss  
[1] 0.3116771
```

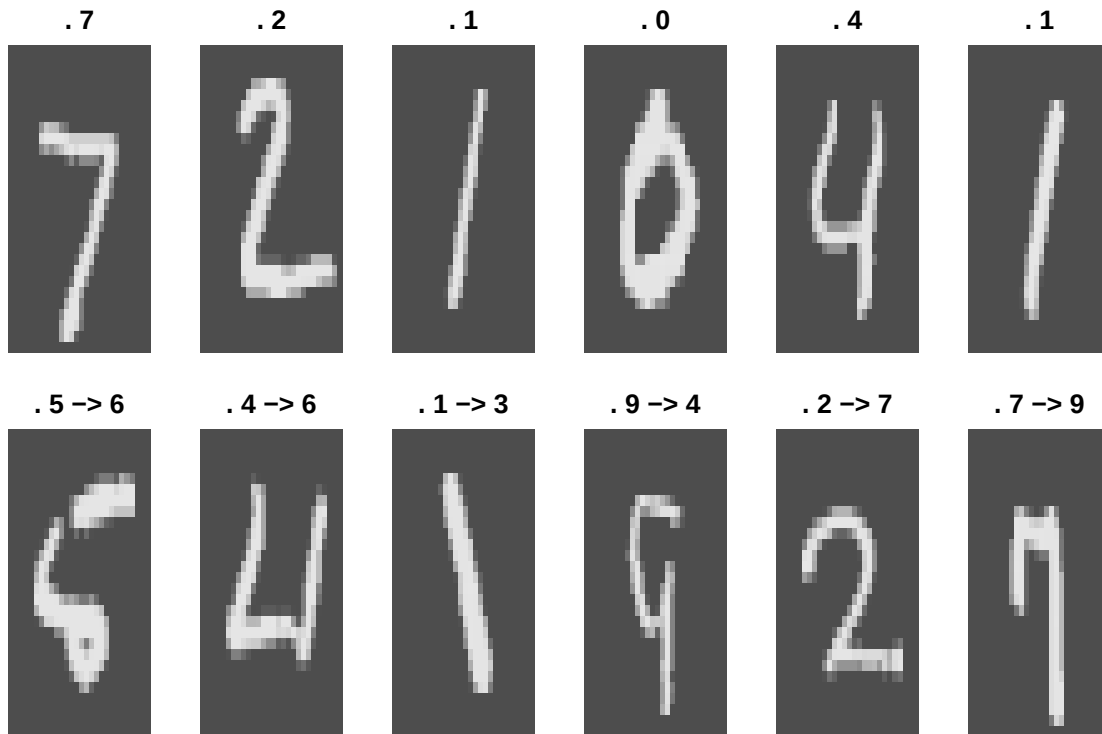
9.5.3 4.2 Visualizing Correct and Incorrect Predictions

We visualize examples of digits that the CNN classified correctly and incorrectly.

```
pred_probs <- predict(model_cnn, x_test)
```

```
32/32 - 0s - 1ms/step
```

```
pred_classes <- apply(pred_probs, 1, which.max) - 1  
  
correct_idx <- which(pred_classes == y_test)[1:6]  
incorrect_idx <- which(pred_classes != y_test)[1:6]  
  
par(mfrow=c(2,6), mar=c(1,1,2,1))  
for (i in correct_idx) {  
  img <- x_test[i,,,1]  
  image(1:28, 1:28, t(apply(img, 2, rev)), col=gray.colors(255), axes=FALSE,  
        main=paste(" ", y_test[i]))  
}  
for (i in incorrect_idx) {  
  img <- x_test[i,,,1]  
  image(1:28, 1:28, t(apply(img, 2, rev)), col=gray.colors(255), axes=FALSE,  
        main=paste(" ", y_test[i], "→", pred_classes[i]))  
}
```



9.6 Summary

- **Feed-forward networks:** handle static, tabular data (e.g., housing prices).
- **RNNs:** handle sequential data with temporal dependencies (e.g., sine wave, stock prices, text, timer series).
- **CNNs:** handle image and spatial data by learning patterns like edges and textures.

10 Feedforward Neural Network - Python

10.1 Feedforward Neural Network (MLP) Using Ames Housing (Python)

This chapter fits a feed-forward neural network to the Ames Housing subset used in the R version.

RMSE is computed manually for compatibility with older scikit-learn versions.

10.1.1 1. Load and Prepare Data

```
import numpy as np
import pandas as pd

from sklearn.datasets import fetch_openml
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPRegressor
from sklearn.metrics import mean_squared_error

np.random.seed(12345)

ames = fetch_openml(name="house_prices", as_frame=True)
df = ames.frame

selected_cols = [
    "SalePrice", "BedroomAbvGr", "YearBuilt", "MoSold", "LotArea",
    "Street", "CentralAir", "1stFlrSF", "2ndFlrSF", "FullBath",
    "HalfBath", "Fireplaces", "GarageArea", "GrLivArea", "TotRmsAbvGrd"
]

df_sub = df[selected_cols].dropna()
```

```
df_sub = pd.get_dummies(df_sub, columns=["Street","CentralAir"], drop_first=True)

y = df_sub["SalePrice"]
X = df_sub.drop("SalePrice", axis=1)
```

10.1.2 2. Train/Test Split & Scaling

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=12345
)

scaler = StandardScaler()
scaler.fit(X_train)

X_train_s = scaler.transform(X_train)
X_test_s = scaler.transform(X_test)
```

10.1.3 3. Simple MLP Model

```
nn_simple = MLPRegressor(
    solver="lbfgs",
    alpha=1e-5,
    hidden_layer_sizes=(5,),
    random_state=12345,
    max_iter=5000
)

nn_simple.fit(X_train_s, y_train)
pred_simple = nn_simple.predict(X_test_s)

rmse_simple = mean_squared_error(y_test, pred_simple) ** 0.5
rmse_simple
```

38907.904754996634

10.1.4 4. Grid Search Tuning

```
param_grid = {
    "hidden_layer_sizes": [(h,) for h in [3,4,5,6,7]],
    "alpha": [0.00005, 0.0005],
    "solver": ["lbfgs"]
}

nn_base = MLPRegressor(max_iter=5000, random_state=12345)

grid = GridSearchCV(
    estimator=nn_base,
    param_grid=param_grid,
    cv=10,
    scoring="neg_mean_squared_error",
    n_jobs=-1
)

grid.fit(X_train_s, y_train)

grid.best_params_

{'alpha': 5e-05, 'hidden_layer_sizes': (5,), 'solver': 'lbfgs'}

best_model = grid.best_estimator_
pred_gs = best_model.predict(X_test_s)

rmse_gs = mean_squared_error(y_test, pred_gs) ** 0.5
rmse_gs
```

39176.951700141195

11 Naive Bayes Classification R version

11.1 1. Overview and Learning Objectives

In this chapter, we will:

1. Use the **Ames Housing** data to create a **binary classification** problem:
 - Predict whether a house is **high price** vs **low price** based on its features..
 2. Understand how Naive Bayes handles:
 - **Categorical vs numeric** predictors
 - **Laplace correction** (additive smoothing)
 - **Kernel density estimation** for numeric variables
 3. Use **caret** (`train(method = "nb")`) to:
 - **Tune** key hyperparameters (`usekernel`, `fL`)
 - Evaluate performance with **cross-validation**
 4. Interpret:
 - Class priors
 - Conditional feature distributions
 - Predictions & confusion matrix
 - ROC curve and AUC
-

11.2 2. Data Setup and Binary Target Creation

```
library(tidyverse)
library(caret)
library(AmesHousing)
library(pROC)

set.seed(4321)

ames <- make_ordinal_ames() %>% mutate(id = row_number())

training <- ames %>% sample_frac(0.7)
testing  <- anti_join(ames, training, by = "id")

predictors <- c(
  "Sale_Price", "Bedroom_AbvGr", "Year_Built", "Mo_Sold", "Lot_Area",
  "Street", "Central_Air", "First_Flr_SF", "Second_Flr_SF", "Full_Bath",
  "Half_Bath", "Fireplaces", "Garage_Area", "Gr_Liv_Area", "TotRms_AbvGrd"
)

training <- training %>% select(all_of(predictors))
testing  <- testing  %>% select(all_of(predictors))
```

We turn `Sale_Price` into a **classification target**:

- "High" if above the training-set median
- "Low" otherwise

This keeps the task aligned with Naive Bayes as a **classification** model.

```
price_median <- median(training$Sale_Price)

training <- training %>% mutate(Price_High = factor(ifelse(Sale_Price > price_median, "High", "Low")))
testing  <- testing  %>% mutate(Price_High = factor(ifelse(Sale_Price > price_median, "High", "Low")))

training_nb <- training %>% select(-Sale_Price)
testing_nb  <- testing  %>% select(-Sale_Price)
```


11.3 3. Naive Bayes Intuition

Naive Bayes is a **generative classifier** based on Bayes' theorem:

$$[P(Y = k | X) = P(Y=k) \prod_{j=1}^p P(x_j | Y=k)]$$

Key assumptions:

11.3.0.1 1. Conditional Independence

Given the class, features are treated as independent:

$$[P(x_1, x_2, \dots, x_p | Y=k) = \prod_j P(x_j | Y=k)]$$

This is rarely exactly true, yet often works surprisingly well.

11.3.0.2 2. Modeling Numeric Features

Naive Bayes supports two approaches:

- **Gaussian (Normal):** assumes each numeric feature is normally distributed within each class.
- **Kernel Density Estimation:** more flexible, handles skewness and multimodal distributions.

11.3.0.3 3. Laplace Correction

Prevents zero probabilities when a category never appears in a given class:

$$P(x=v | Y=k) = (\text{count} + 1) / (N + K)$$

This keeps the model from collapsing.

11.4 4. Fitting Naive Bayes (Baseline Model)

```

set.seed(12345)

nb.ames <- naiveBayes(
  Price_High ~ .,
  data = training_nb,
  laplace = 0,
  usekernel = TRUE
)

nb.ames

```

Naive Bayes Classifier for Discrete Predictors

Call:

```
naiveBayes.default(x = X, y = Y, laplace = laplace, usekernel = TRUE)
```

A-priori probabilities:

Y	High	Low
	0.4987811	0.5012189

Conditional probabilities:

	Bedroom_AbvGr	
Y	[,1]	[,2]
High	2.948192	0.8218245
Low	2.770428	0.8207514

	Year_Built	
Y	[,1]	[,2]
High	1989.604	23.12763
Low	1953.780	25.88135

	Mo_Sold	
Y	[,1]	[,2]
High	6.353861	2.732694
Low	6.157588	2.691989

	Lot_Area	
Y	[,1]	[,2]
High	11726.913	10345.651
Low	8673.011	4673.732

Street
Y Grvl Pave
High 0.0009775171 0.9990224829
Low 0.0058365759 0.9941634241

Central_Air
Y N Y
High 0.008797654 0.991202346
Low 0.122568093 0.877431907

First_Flr_SF
Y [,1] [,2]
High 1333.7087 408.4059
Low 986.6527 264.8090

Second_Flr_SF
Y [,1] [,2]
High 447.7625 493.0434
Low 222.7490 317.7090

Full_Bath
Y [,1] [,2]
High 1.913001 0.4415516
Low 1.233463 0.4412614

Half_Bath
Y [,1] [,2]
High 0.5259042 0.5168999
Low 0.2324903 0.4406718

Fireplaces
Y [,1] [,2]
High 0.8670577 0.6271108
Low 0.3180934 0.5359368

Garage_Area
Y [,1] [,2]
High 581.0430 173.7891
Low 362.3016 190.0580

Gr_Liv_Area
Y [,1] [,2]

```

High 1783.837 473.5654
Low  1215.255 342.4179

      TotRms_AbvGrd
Y      [,1]      [,2]
High  7.043988 1.516711
Low   5.825875 1.373640

```

11.4.0.1 Explanation of key settings

- **laplace = 0**
No smoothing; probability estimates reflect raw counts.
Zero probabilities may occur for rare categorical levels.
- **usekernel = TRUE**
Numeric features are modeled using **kernel density estimates** instead of assuming normality.

11.4.0.2 What the output means

- **A-priori probabilities:**
Estimated from class frequencies (High vs Low).
- **Conditional tables:**
 - For categorical features: probability of each level given class.
 - For numerical features: kernel-based density info (not printed in summary).

11.5 5. Predicting on Test Data

```

test_pred_class <- predict(nb.ames, testing_nb)
test_pred_prob  <- predict(nb.ames, testing_nb, type="raw")
head(test_pred_prob)

```

	High	Low
[1,]	0.9999965	3.462116e-06
[2,]	0.9850961	1.490391e-02
[3,]	0.9830462	1.695380e-02
[4,]	0.5936144	4.063856e-01
[5,]	0.9437367	5.626333e-02
[6,]	0.9998253	1.746816e-04

- test_pred_class: predicted High/Low label
- test_pred_prob: estimated probability for each class

11.6 6. Confusion Matrix Interpretation

```
conf_matrix <- confusionMatrix(
  test_pred_class,
  testing_nb$Price_High,
  positive="High"
)

conf_matrix
```

Confusion Matrix and Statistics

	Reference	
Prediction	High	Low
High	373	49
Low	55	402

Accuracy : 0.8817
 95% CI : (0.8585, 0.9023)
 No Information Rate : 0.5131
 P-Value [Acc > NIR] : <2e-16

 Kappa : 0.7631

 McNemar's Test P-Value : 0.6239

```
Sensitivity : 0.8715
Specificity : 0.8914
Pos Pred Value : 0.8839
Neg Pred Value : 0.8796
Prevalence : 0.4869
Detection Rate : 0.4243
Detection Prevalence : 0.4801
Balanced Accuracy : 0.8814

'Positive' Class : High
```

11.6.0.1 What the confusion matrix tells us

- **Accuracy:** overall correctness
- **Sensitivity (Recall for High):** proportion of expensive homes correctly identified
- **Specificity (Recall for Low):** proportion of cheaper homes correctly identified
- **Balanced Accuracy:** average of sensitivity & specificity
- **Kappa:** agreement beyond chance

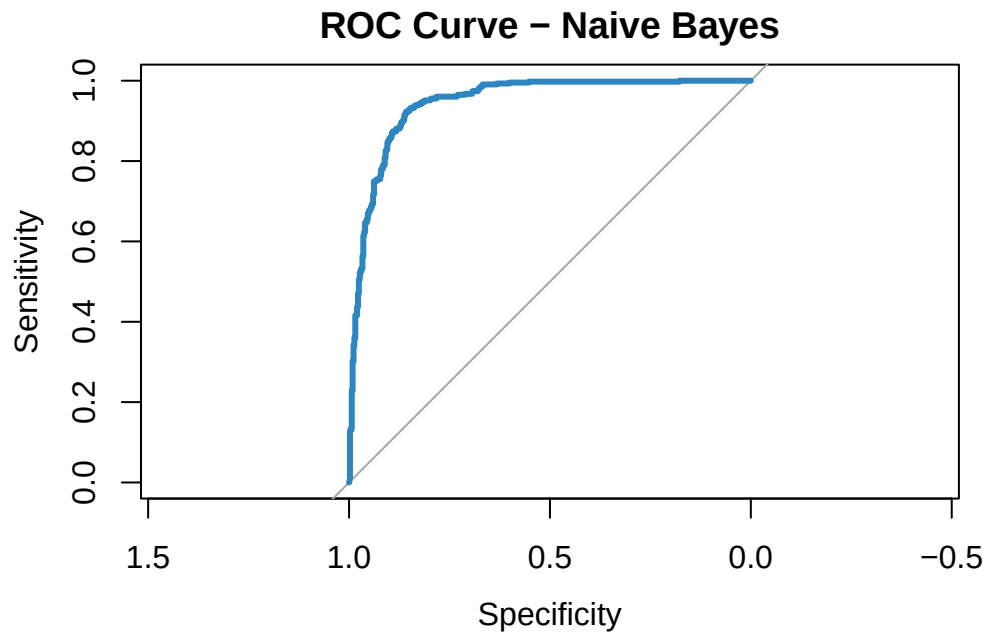
This is the primary evaluation for classification problems.

11.7 7. ROC Curve & AUC

ROC curves evaluate model ranking ability — how well it separates High from Low across thresholds.

```
roc_obj <- roc(
  response = testing_nb$Price_High,
  predictor = test_pred_prob[, "High"],
  levels = c("Low", "High")
)

plot(roc_obj, col="#2E86C1", lwd=3, main="ROC Curve - Naive Bayes")
```



```
auc_value <- auc(roc_obj)
auc_value
```

Area under the curve: 0.9457

11.7.0.1 Interpretation

- **ROC Curve:** visual representation of sensitivity vs (1 - specificity)
- **AUC:** single summary metric
 - 0.5 = no better than random
 - 0.7 = decent
 - 0.8 = strong
 - 0.9 = outstanding

11.8 8. Tuning Naive Bayes Using `caret::train()`

We now tune:

- `usekernel` {TRUE, FALSE}
- `fL` {0, 0.5, 1} (*fractional Laplace*)

```
set.seed(12345)

tune_grid <- expand.grid(
  usekernel = c(TRUE, FALSE),
  fL        = c(0, 0.5, 1),
  adjust    = 1
)

ctrl <- trainControl(
  method = "cv",
  number = 10,
  classProbs = TRUE,
  summaryFunction = twoClassSummary
)

training_nb$Price_High <- relevel(training_nb$Price_High, ref="High")
testing_nb$Price_High  <- relevel(testing_nb$Price_High, ref="High")

nb.ames.caret <- train(
  Price_High ~ .,
  data = training_nb,
  method = "nb",
  tuneGrid = tune_grid,
  trControl = ctrl,
  metric = "ROC"
)

nb.ames.caret
```

Naive Bayes

```
2051 samples
14 predictor
2 classes: 'High', 'Low'
```


Resampling results across tuning parameters:

usekernel	fL	ROC	Sens	Spec
FALSE	0.0	0.9325587	0.9217484	0.8453794
FALSE	0.5	0.9325587	0.9217484	0.8453794
FALSE	1.0	0.9325587	0.9217484	0.8453794
TRUE	0.0	0.9549057	0.9197316	0.8589473
TRUE	0.5	0.9549057	0.9197316	0.8589473
TRUE	1.0	0.9549057	0.9197316	0.8589473

The final values used for the model were `fL = 0`, `usekernel = TRUE` and `adjust = 1`.

\$\$\$ How tuning works

- **Cross-validation** evaluates each combination of hyperparameters.
- Best model chosen by the metric (ROC).
- caret automatically handles class probabilities and ROC scoring.

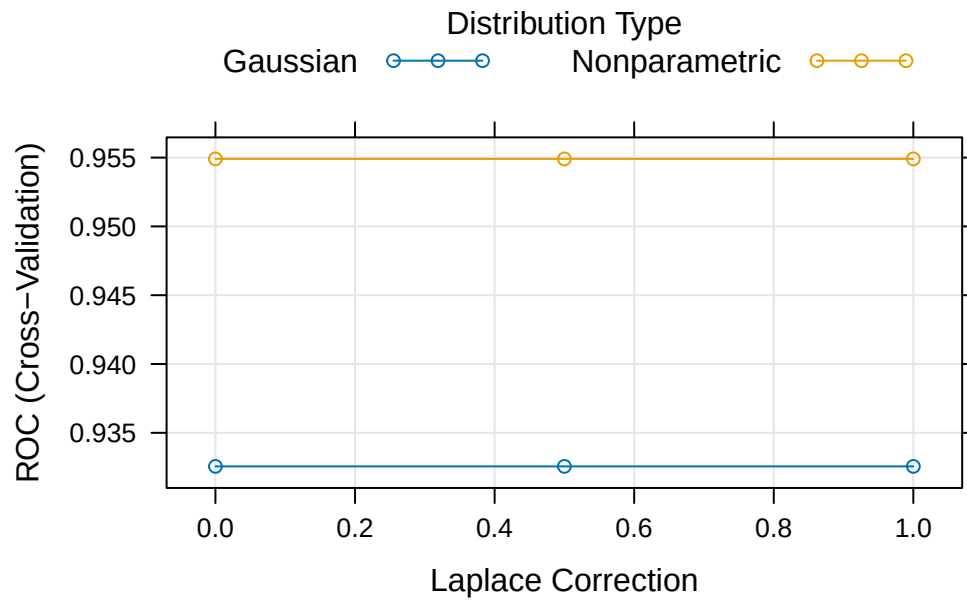
```
nb.ames.caret$results
```

[illegible]

```
nb.ames.caret$bestTune
```

```
  fL usekernel adjust  
4  0      TRUE      1
```

```
plot(nb.ames.caret)
```



11.9 9. Evaluating the Tuned Model on the Test Set

```
test_pred_class_tuned <- predict(nb.ames.caret, testing_nb)  
test_pred_prob_tuned <- predict(nb.ames.caret, testing_nb, type="prob")  
  
confusionMatrix(  
  test_pred_class_tuned,  
  testing_nb$Price_High,  
  positive="High"  
)
```

Confusion Matrix and Statistics

```

      Reference
Prediction High Low
      High  398  77
      Low   30 374

      Accuracy : 0.8783
      95% CI : (0.8548, 0.8992)
      No Information Rate : 0.5131
      P-Value [Acc > NIR] : < 2.2e-16

      Kappa : 0.7571

      Mcnemar's Test P-Value : 8.708e-06

      Sensitivity : 0.9299
      Specificity : 0.8293
      Pos Pred Value : 0.8379
      Neg Pred Value : 0.9257
      Prevalence : 0.4869
      Detection Rate : 0.4528
      Detection Prevalence : 0.5404
      Balanced Accuracy : 0.8796

      'Positive' Class : High
```

11.10 10. Summary

- Naive Bayes is a **probabilistic, generative classifier**.
- It models **class priors + conditional likelihoods** for each predictor.
- Numeric predictors can use:
 - **Gaussian distributions**, or
 - **Kernel density estimates** (`usekernel = TRUE`).

- **Laplace smoothing** protects against zero probabilities.
- **ROC** and **AUC** evaluate ranking performance.
- **Confusion matrix** evaluates classification performance.
- **caret** enables systematic tuning over smoothing and density options.

12 Naive Bayes Classification - Python Version

12.1 1. Overview and Learning Objectives

In this Python chapter, we will:

1. Load the **Ames Housing** dataset from OpenML.
 2. Create a **High vs Low** price classification target.
 3. Fit **Gaussian Naive Bayes**.
 4. Produce predictions and class probabilities.
 5. Compute a **correct ROC curve and AUC**.
 6. Mirror the structure of the R version.
-

12.2 2. Setup and Data Preparation

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import confusion_matrix, classification_report, roc_curve, auc
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import fetch_openml

# Load Ames Housing dataset
ames = fetch_openml(name="house_prices", as_frame=True)
df = ames.frame.copy()

df.rename(columns={"SalePrice": "Sale_Price"}, inplace=True)

df.head()
```

	Id	MSSubClass	MSZoning	LotFrontage	LotArea	Street	Alley	LotShape	LandContour	Utilities
0	1	60	RL	65.0	8450	Pave	NaN	Reg	Lvl	AllPu
1	2	20	RL	80.0	9600	Pave	NaN	Reg	Lvl	AllPu
2	3	60	RL	68.0	11250	Pave	NaN	IR1	Lvl	AllPu
3	4	70	RL	60.0	9550	Pave	NaN	IR1	Lvl	AllPu
4	5	60	RL	84.0	14260	Pave	NaN	IR1	Lvl	AllPu

12.3 2.1 Select Variables of Interest

```

predictors = [
    "Sale_Price",
    "BedroomAbvGr",
    "YearBuilt",
    "MoSold",
    "LotArea",
    "Street",
    "CentralAir",
    "1stFlrSF",
    "2ndFlrSF",
    "FullBath",
    "HalfBath",
    "Fireplaces",
    "GarageArea",
    "GrLivArea",
    "TotRmsAbvGrd"
]

df = df[predictors].dropna()

```

12.4 2.2 Create a Binary Target Variable

```

median_price = df["Sale_Price"].median()

df["Price_High"] = np.where(df["Sale_Price"] > median_price, "High", "Low")
df["Price_High"] = df["Price_High"].astype("category")

df["Price_High"].value_counts(normalize=True)

```

```

Price_High
Low      0.50137
High     0.49863
Name: proportion, dtype: float64

```

12.5 2.3 Train/Test Split

```

X = df.drop(columns=["Sale_Price", "Price_High"])
y = df["Price_High"]

# Encode categorical predictors
X = pd.get_dummies(X, drop_first=True)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.30, random_state=4321, stratify=y
)

X_train.shape, X_test.shape

```

```
((1022, 14), (438, 14))
```

12.6 3. Gaussian Naive Bayes Model

```
gnb = GaussianNB()
gnb.fit(X_train, y_train)
```

GaussianNB()

12.7 4. Predictions

```
y_pred = gnb.predict(X_test)

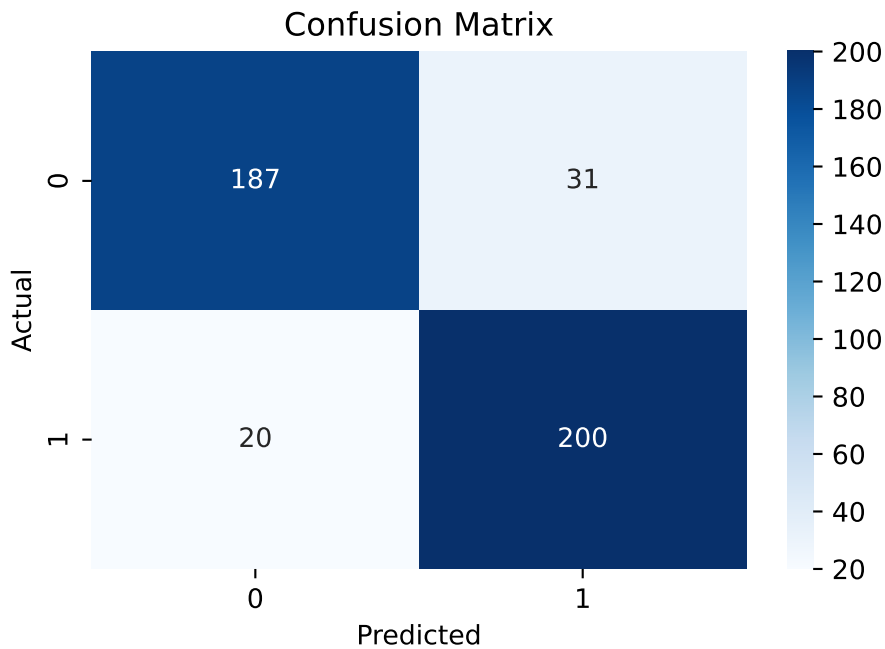
# Extract the probability of the TRUE positive class ("High")
pos_index = np.where(gnb.classes_ == "High")[0][0]
y_prob_high = gnb.predict_proba(X_test)[:, pos_index]
```

12.8 5. Confusion Matrix & Classification Report

```
cm = confusion_matrix(y_test, y_pred)
cm
```

```
array([[187,  31],
       [ 20, 200]])
```

```
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues")
plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```

```
print(classification_report(y_test, y_pred))
```

	precision	recall	f1-score	support
High	0.90	0.86	0.88	218
Low	0.87	0.91	0.89	220
accuracy			0.88	438
macro avg	0.88	0.88	0.88	438
weighted avg	0.88	0.88	0.88	438

12.9 6. ROC Curve (Corrected) and AUC

```
# Convert y_test to binary (1 = High, 0 = Low)
y_test_binary = (y_test == "High").astype(int)

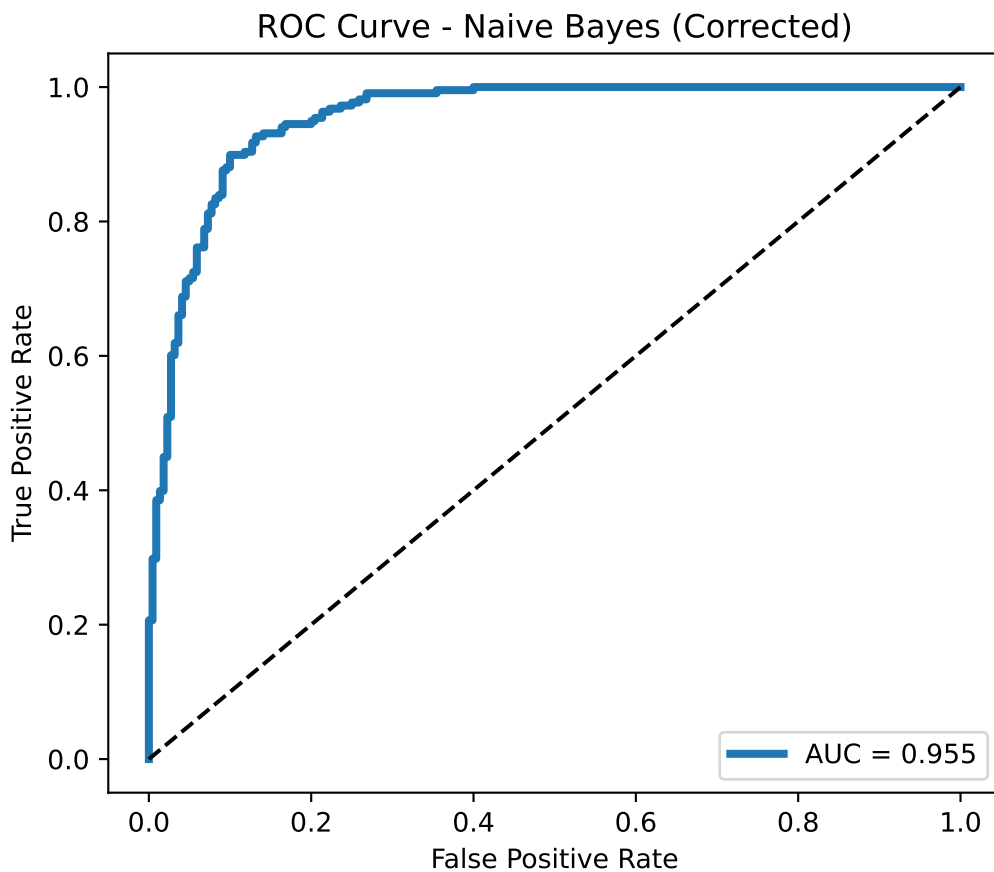
# Compute ROC using correct positive class probability
```

```
fpr, tpr, thresholds = roc_curve(y_test_binary, y_prob_high)
roc_auc = auc(fpr, tpr)

roc_auc
```

```
np.float64(0.9547122602168474)
```

```
plt.figure(figsize=(6,5))
plt.plot(fpr, tpr, label=f"AUC = {roc_auc:.3f}", linewidth=3)
plt.plot([0,1], [0,1], "k--")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve - Naive Bayes (Corrected)")
plt.legend()
plt.show()
```



12.10 7. Interpretation Notes

12.10.1 Naive Bayes Assumptions

- Predictors are independent given the class.
- Numeric features follow Gaussian distributions.
- One-hot encoded categorical variables behave like Bernoulli variables.

12.10.2 Strengths

- Fast and simple.
- Works well as a baseline.
- Highly interpretable.

12.10.3 Weaknesses

- Independence assumption rarely holds.
 - Gaussian assumption is an approximation.
-

12.11 8. Summary

This chapter:

- Loaded the Ames Housing dataset.
- Created a binary classification target.
- Fit Gaussian Naive Bayes.
- Correctly computed ROC & AUC.
- Provided fully aligned interpretation with the R version.

13 Model-Agnostic Interpretability - R Version

```
# Core data wrangling and modeling
library(tidyverse)
library(AmesHousing)
library(ranger)
library(xgboost)

# Model-agnostic interpretability
library(iml)      # permutation importance, PD, ICE, ALE, local models, Shapley

# Alternatives / model-specific tools
library(randomForest)
```

13.1 Learning Objectives

By the end of this section you should be able to:

- Explain what *model-agnostic* interpretability means and why it is useful.
- Compute and interpret **permutation feature importance** for several models.
- Use **partial dependence plots (PDP)** and **individual conditional expectation (ICE)** plots.
- Use **accumulated local effects (ALE)** plots and understand how they differ from PDP.
- Explain **local surrogate** ideas in the spirit of **LIME** and interpret a local explanation.
- Compute **Shapley value**-based explanations for individual predictions.
- Compare how these tools behave for:
 - a **regression** target (Sale Price), and
 - a **binary classification** target (High vs Low price, based on the median).

13.2 1. Data and Modeling Setup

We will reuse the Ames housing data and the same subset of predictors used earlier in the book.

```
set.seed(4321)

ames <- make_ordinal_ames() |>
  mutate(
    id = row_number()
  ) |>
  select(
    id, Sale_Price,
    Bedroom_AbvGr,
    Year_Built,
    Mo_Sold,
    Lot_Area,
    Street,
    Central_Air,
    First_Flr_SF,
    Second_Flr_SF,
    Full_Bath,
    Half_Bath,
    Fireplaces,
    Garage_Area,
    Gr_Liv_Area,
    TotRms_AbvGrd
  )

# Create a binary classification target: High vs Low price
median_price <- median(ames$Sale_Price)
ames <- ames |>
  mutate(
    HighPrice = if_else(Sale_Price >= median_price, "High", "Low"),
    HighPrice = factor(HighPrice, levels = c("Low", "High"))
  )

# Train / test split (same seed for reproducibility)
train_ids <- sample(ames$id, size = 0.7 * nrow(ames))
train <- ames |> filter(id %in% train_ids)
test  <- ames |> filter(!id %in% train_ids)
```

```

# Predictors only (same X for regression and classification)
X_train <- train |>
  select(-id, -Sale_Price, -HighPrice)
X_test  <- test |>
  select(-id, -Sale_Price, -HighPrice)

# Targets
y_train_reg <- train$Sale_Price
y_test_reg  <- test$Sale_Price

y_train_cls <- train$HighPrice
y_test_cls  <- test$HighPrice
y_train_cls_num <- if_else(y_train_cls == "High", 1, 0)

```

13.3 2. Models to Interpret

We will fit three models:

- A **linear regression** model (simple, transparent baseline).
- A **Random Forest** (nonlinear, interaction-rich).
- An **XGBoost** model (gradient-boosted trees).

We fit each model for the **regression** task. Then we fit classification versions for the **high/low** price target.

13.3.1 2.1 Regression Models

```

# Linear regression
lm_reg <- lm(Sale_Price ~ . - id - HighPrice, data = train)

# Random Forest regression
rf_reg <- ranger(
  Sale_Price ~ . - id - HighPrice,
  data      = train,
  num.trees = 500,
  mtry      = 4,
  importance = "permutation",

```

```

    seed      = 4321
  )

# XGBoost regression: use model.matrix on the predictor set
xgb_train_reg <- model.matrix(~ ., data = X_train)

xgb_reg <- xgboost(
  data      = xgb_train_reg,
  label     = y_train_reg,
  nrounds   = 200,
  objective = "reg:squarederror",
  max_depth = 4,
  eta       = 0.1,
  subsample = 0.8,
  colsample_bytree = 0.8,
  verbose   = 0
)

```

13.3.2 2.2 Classification Models (High vs Low Price)

```

# Logistic regression (glm)
glm_cls <- glm(
  HighPrice ~ . - id - Sale_Price,
  data      = train,
  family    = binomial
)

# Random Forest classification
rf_cls <- ranger(
  HighPrice ~ . - id - Sale_Price,
  data      = train,
  num.trees = 500,
  mtry      = 4,
  probability = TRUE,
  seed      = 4321
)

# XGBoost classification (probability of "High")
xgb_train_cls <- xgb_train_reg # same predictors as regression
xgb_cls <- xgboost(

```

```

data      = xgb_train_cls,
label     = y_train_cls_num,
nrounds   = 200,
objective = "binary:logistic",
max_depth = 4,
eta       = 0.1,
subsample = 0.8,
colsample_bytree = 0.8,
verbose   = 0
)

```

13.4 3. Wrapping Models in `iml::Predictor`

The `iml` package provides a common interface to models. Once a model is wrapped in a `Predictor`, we can apply:

- permutation importance
- PDP, ICE, ALE
- local surrogate models (LIME-style)
- Shapley value explanations

13.4.1 3.1 Regression Predictors

```

# Predictors for regression models
pred_lm_reg <- Predictor$new(
  model = lm_reg,
  data  = X_train,
  y     = y_train_reg
)

pred_rf_reg <- Predictor$new(
  model = rf_reg,
  data  = X_train,
  y     = y_train_reg,
  predict.function = function(m, newdata) {
    predict(m, data = newdata)$predictions
  }
)

```



```
)

pred_xgb_reg <- Predictor$new(
  model = xgb_reg,
  data = X_train,
  y = y_train_reg,
  predict.function = function(m, newdata) {
    new_mat <- model.matrix(~ ., data = newdata)
    predict(m, new_mat)
  }
)
```

13.4.2 3.2 Classification Predictors (Probability of High Price)

For classification, we will interpret the **predicted probability that a home is “High” price** (above the median).

```
# Logistic regression: probability of "High"
pred_glm_cls <- Predictor$new(
  model = glm_cls,
  data = X_train,
  y = y_train_cls_num,
  predict.function = function(m, newdata) {
    predict(m, newdata, type = "response")
  }
)

# Random Forest classification: use probability for "High"
pred_rf_cls <- Predictor$new(
  model = rf_cls,
  data = X_train,
  y = y_train_cls_num,
  predict.function = function(m, newdata) {
    preds <- predict(m, data = newdata)$predictions
    preds[, "High"]
  }
)

# XGBoost classification: probability of "High"
pred_xgb_cls <- Predictor$new(
  model = xgb_cls,
```

```

data = X_train,
y     = y_train_cls_num,
predict.function = function(m, newdata) {
  new_mat <- model.matrix(~ ., data = newdata)
  predict(m, new_mat)
}
)

```

13.4.3 3.3 Why Some Models Need predict.function

The `iml` package expects the wrapped model to behave like a simple prediction function:

input: a data frame of new observations
output: a numeric vector of predictions

Different R model objects behave differently:

- `lm()` already satisfies this contract:
`predict(lm_reg, newdata)` returns a numeric vector of fitted values.
 → No custom `predict.function` is needed.
- **ranger (Random Forest)** returns a **list**, whose `$predictions` element contains the numeric predictions.
 → We supply a `predict.function` that calls `predict()` and pulls out `$predictions`.
- **xgboost** expects a **numeric matrix**, not a data frame, and does not understand formulas or factors.
 → We supply a `predict.function` that first converts `newdata` to a design matrix via `model.matrix()`, then calls `predict()`.

```

# Linear model works out of the box:
# predict(lm_reg, newdata) -> numeric vector
pred_lm_reg <- Predictor$new(
  model = lm_reg,
  data = X_train,
  y     = y_train_reg
)

# ranger: extract $predictions from the list
pred_rf_reg <- Predictor$new(
  model = rf_reg,
  data = X_train,
  y     = y_train_reg,

```

```

    predict.function = function(m, newdata) {
      predict(m, data = newdata)$predictions
    }
  )

# xgboost: convert data frame to numeric matrix
pred_xgb_reg <- Predictor$new(
  model = xgb_reg,
  data  = X_train,
  y      = y_train_reg,
  predict.function = function(m, newdata) {
    new_mat <- model.matrix(~ ., data = newdata)
    predict(m, new_mat)
  }
)

```

13.5 4. Permutation Feature Importance

Idea: Shuffle one feature at a time and see how much model performance degrades. Larger performance loss → feature was more important.

We'll start with the **Random Forest regression** model.

```

set.seed(4321)
imp_rf_reg <- FeatureImp$new(
  predictor = pred_rf_reg,
  loss      = "rmse"
)

plot(imp_rf_reg)

```

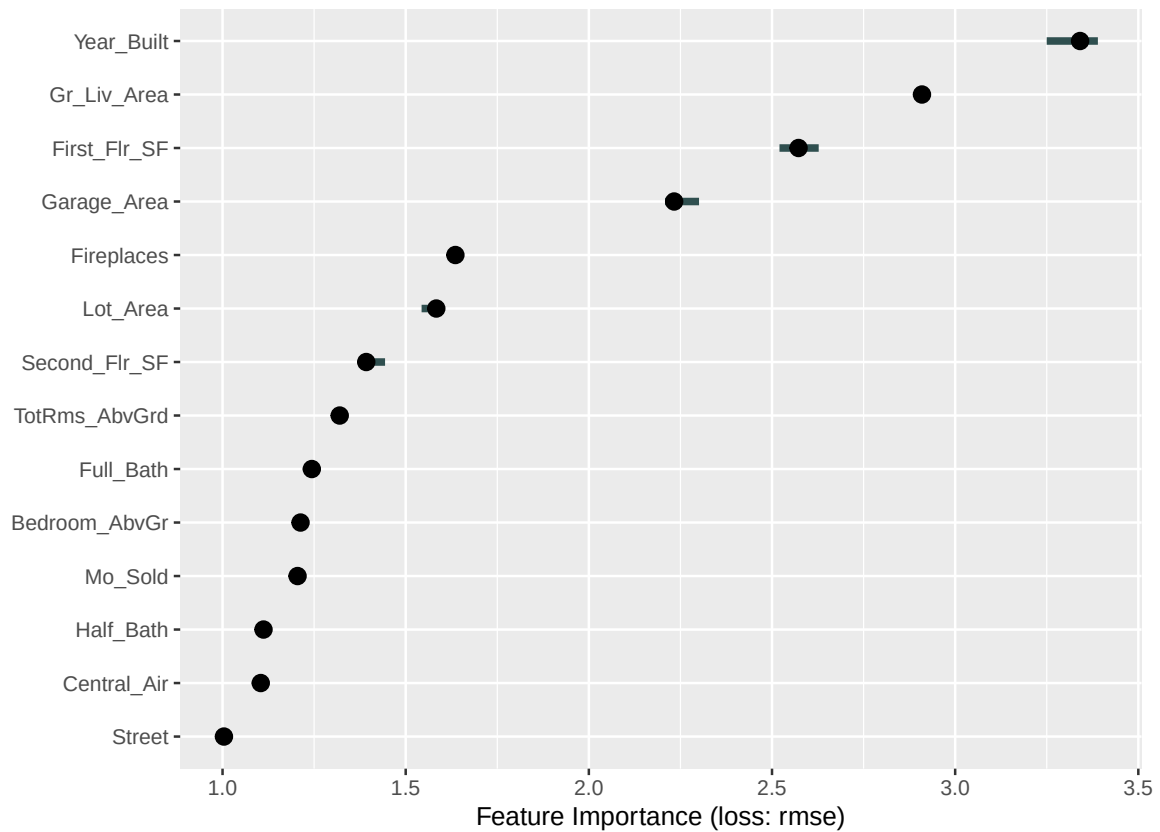


Figure 13.1: Permutation feature importance for Random Forest regression.

You can repeat this for XGBoost:

```
set.seed(4321)
imp_xgb_reg <- FeatureImp$new(
  predictor = pred_xgb_reg,
  loss      = "rmse"
)

plot(imp_xgb_reg)
```

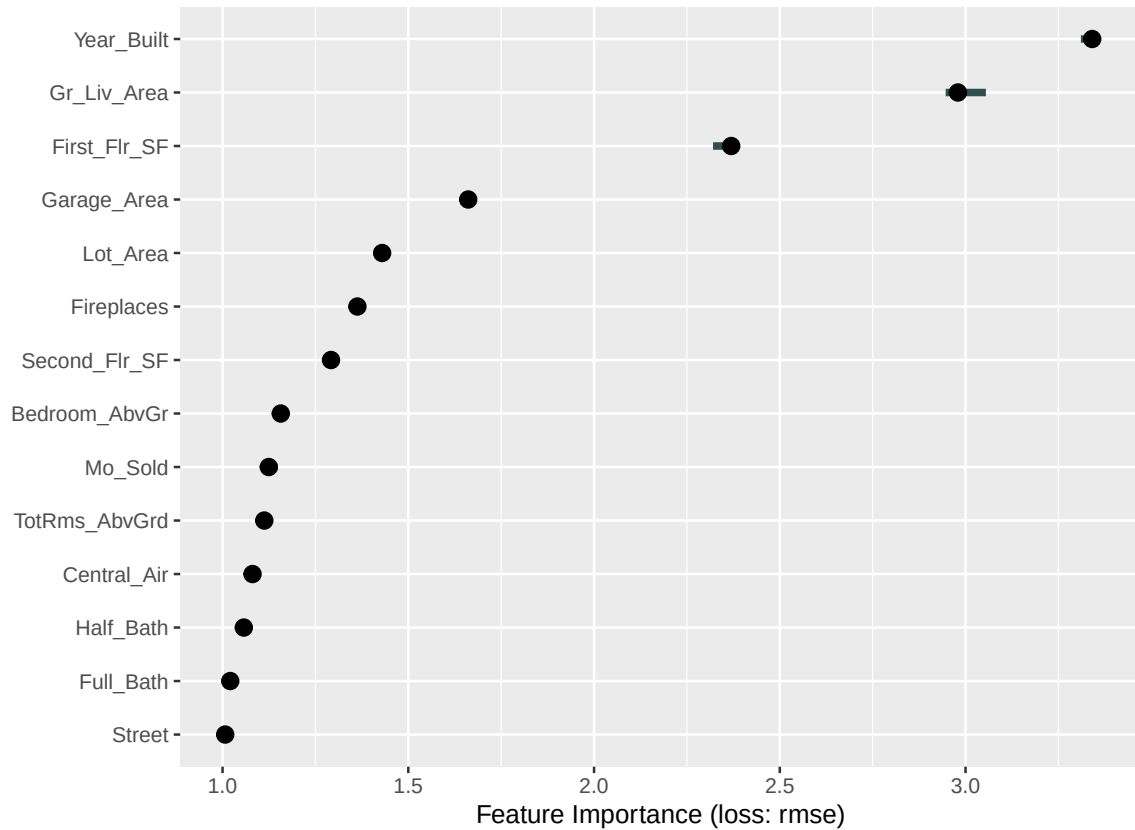


Figure 13.2: Permutation feature importance for XGBoost regression.

13.5.1 4.1 Classification: Importance for Probability of High Price

Now compare to the **Random Forest classification** model, where the outcome is a *probability*.

```
set.seed(4321)
imp_rf_cls <- FeatureImp$new(
  predictor = pred_rf_cls,
  loss      = "ce" # cross-entropy for probabilities
)

plot(imp_rf_cls)
```

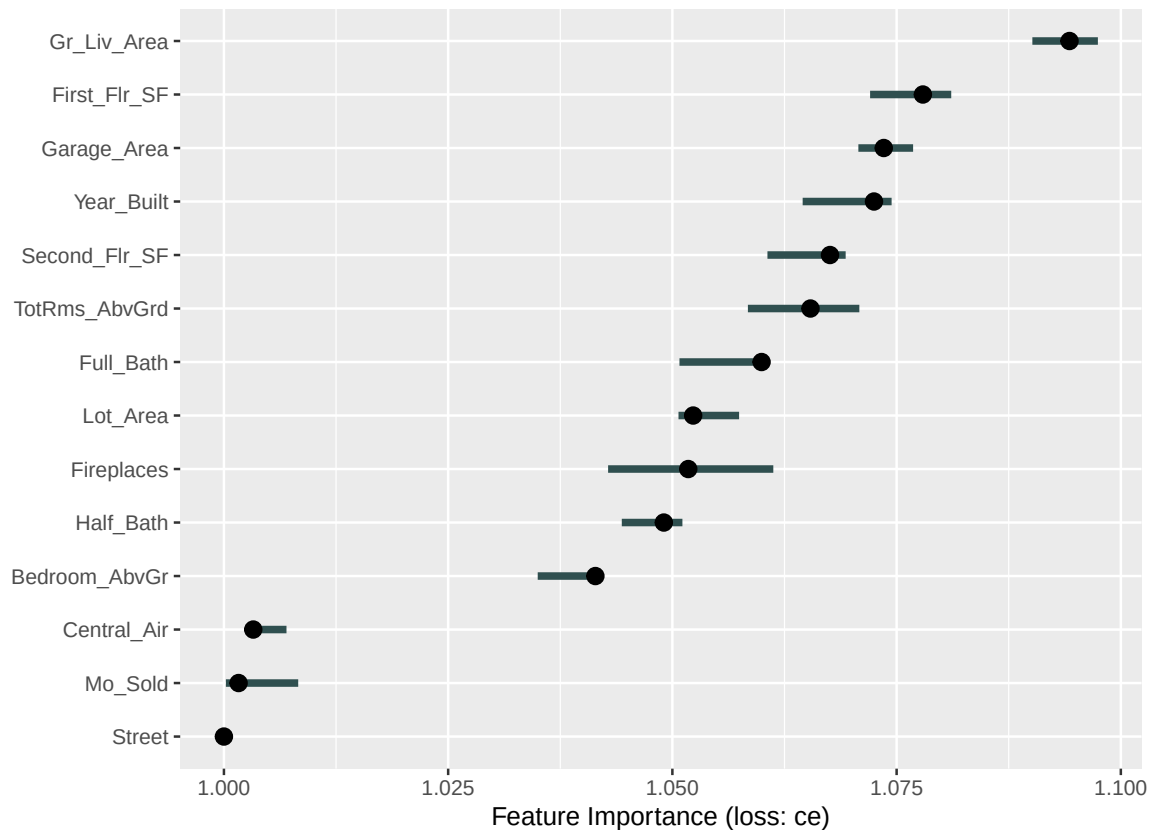


Figure 13.3: Permutation feature importance for RF probability of High price.

13.6 5. Partial Dependence and ICE

Partial Dependence Plot (PDP):

Average prediction as we vary one feature, holding others “as observed” and averaging over them.

Individual Conditional Expectation (ICE):

One curve per observation—shows heterogeneity across individual observations.

We’ll focus on **Gr_Liv_Area** (above-ground living area) as a key driver.

13.6.1 5.1 PDP for RF Regression

```
pdp_rf_reg <- FeatureEffect$new(  
  predictor = pred_rf_reg,  
  feature   = "Gr_Liv_Area",  
  method    = "pdp"  
)  
  
plot(pdp_rf_reg)
```

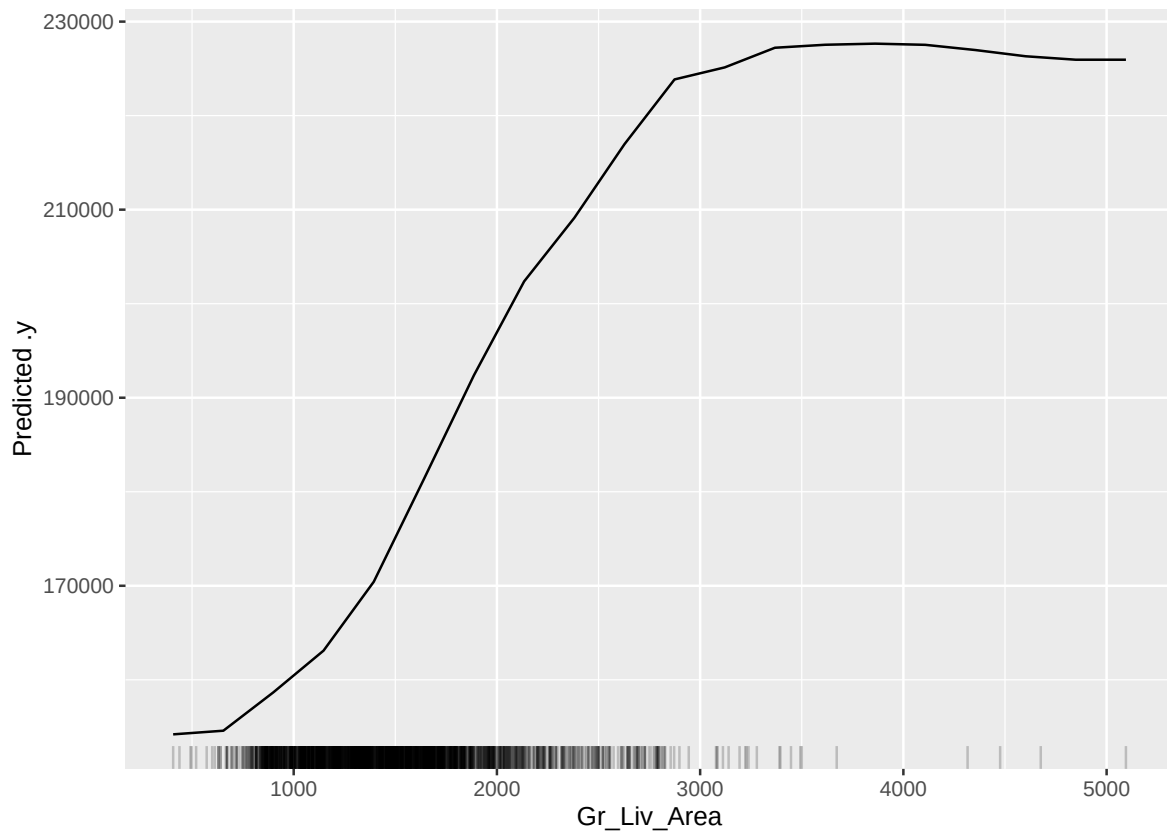


Figure 13.4: Partial dependence of Sale Price on Gr_Liv_Area (Random Forest regression).

13.6.2 5.2 ICE for RF Regression

```
ice_rf_reg <- FeatureEffect$new(
  predictor = pred_rf_reg,
  feature   = "Gr_Liv_Area",
  method    = "ice"
)

plot(ice_rf_reg)
```

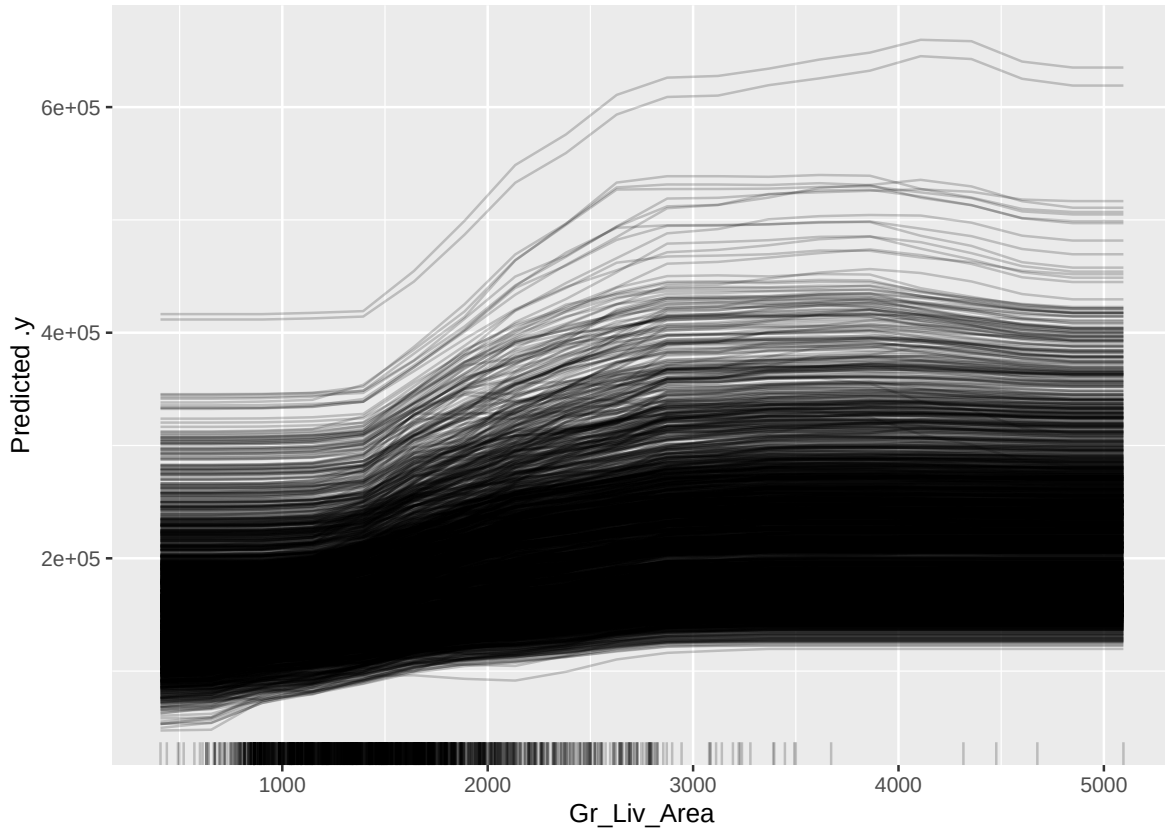


Figure 13.5: ICE curves of Sale Price vs Gr_Liv_Area (Random Forest regression).

Interpretation point:

- PDP: one smooth curve (average effect).
- ICE: many curves, one per home.

Differences between ICE curves indicate **interactions** or **nonlinearities** that the average PDP hides.

13.6.3 5.3 Classification: PDP for Probability of High Price

Now, look at the **probability the home is High price** vs. **Gr_Liv_Area** for RF classification.

```
pdp_rf_cls <- FeatureEffect$new(  
  predictor = pred_rf_cls,  
  feature   = "Gr_Liv_Area",  
  method    = "pdp"  
)  
  
plot(pdp_rf_cls)
```

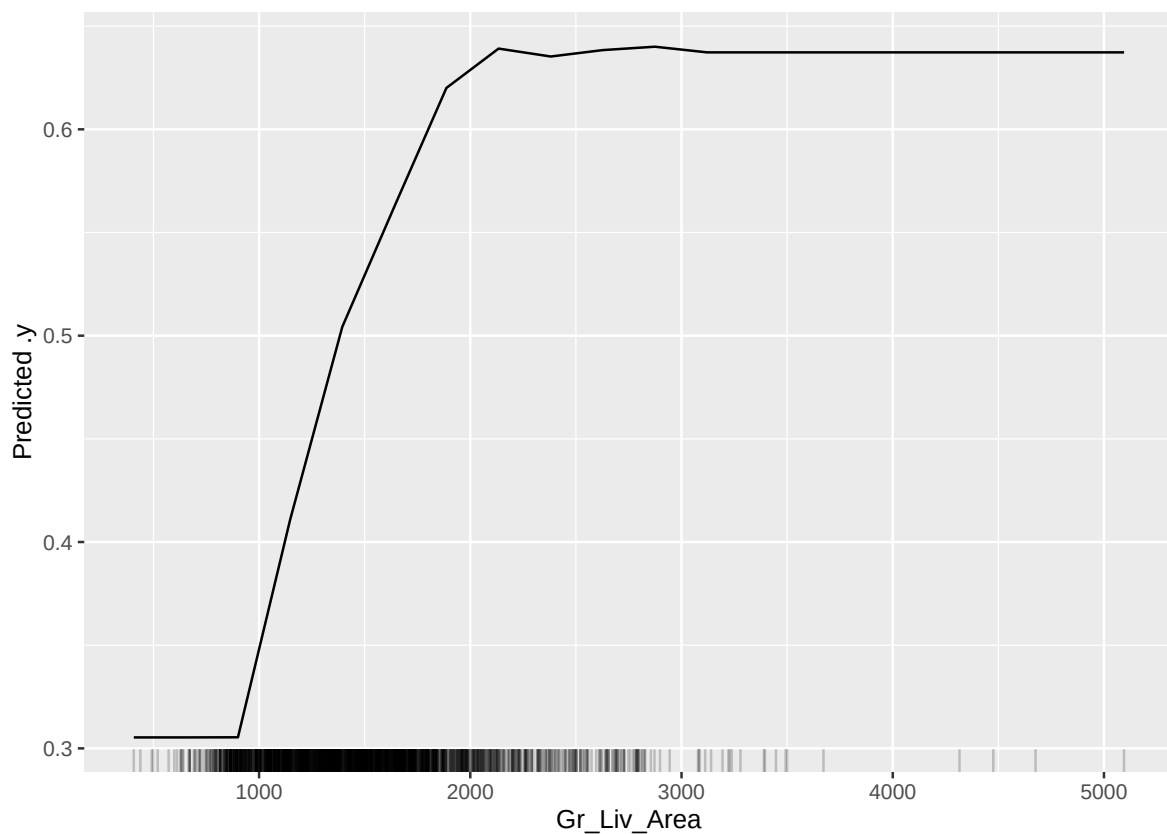


Figure 13.6: PDP of probability(HighPrice) vs Gr_Liv_Area (Random Forest classification).

13.7 6. Accumulated Local Effects (ALE)

ALE plots estimate the effect of a feature in a more *local* way:

- Instead of averaging over unrealistic combinations of features (a PDP problem),
- ALE looks at changes in predictions as we move within small intervals of the feature.

We'll compute ALE for `Gr_Liv_Area` for the RF regression model.

```
ale_rf_reg <- FeatureEffect$new(  
  predictor = pred_rf_reg,  
  feature   = "Gr_Liv_Area",  
  method    = "ale"  
)  
  
plot(ale_rf_reg)
```

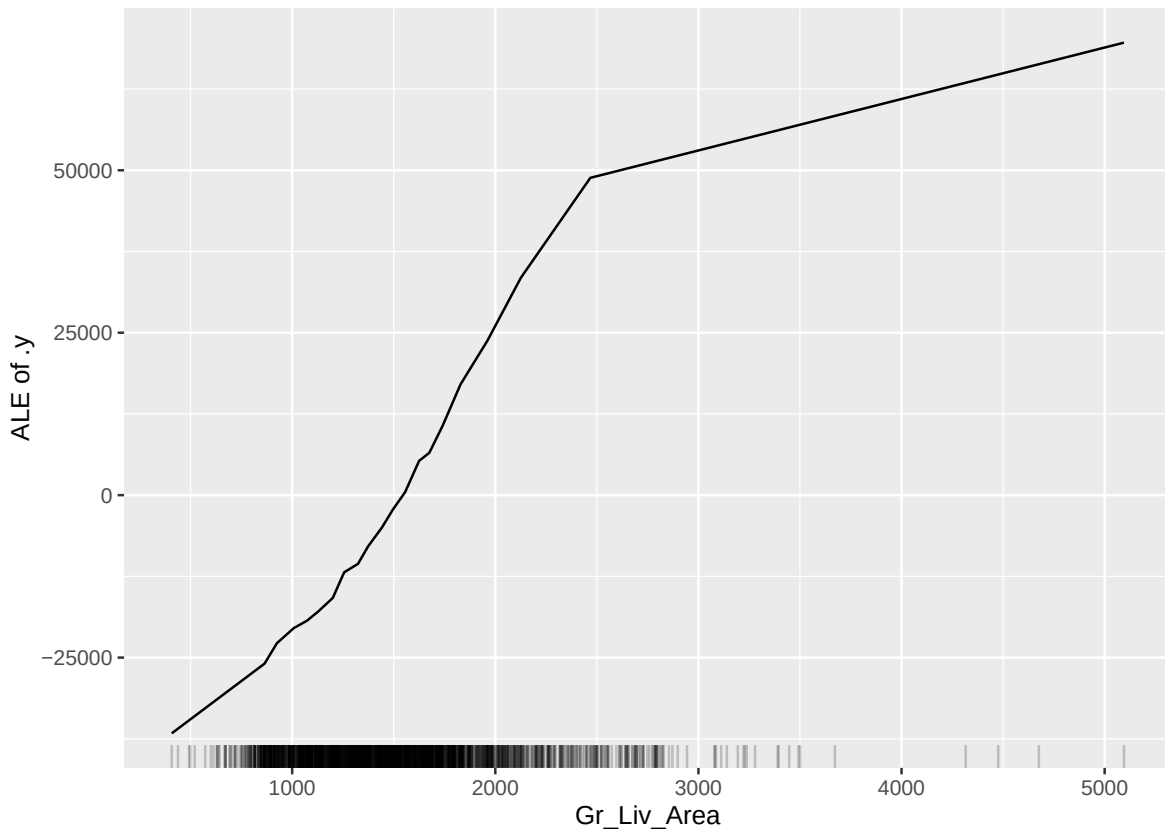


Figure 13.7: ALE plot for `Gr_Liv_Area` (Random Forest regression).

Teaching point:

- PDP assumes features are independent; ALE relaxes this by focusing on local neighborhoods.
- ALE is often more reliable when predictors are correlated (as they often are in housing data).

13.8 7. Local Surrogate Models (LIME-Style)

LIME idea: For one specific observation:

1. Generate small perturbations around that point.
2. Get model predictions for the perturbed points.
3. Fit a simple, local surrogate model (e.g., a linear regression) to approximate the complex model nearby.
4. Use the surrogate's coefficients as a local explanation.

`iml::LocalModel` provides a LIME-style explanation.

13.8.1 7.1 Choose a House to Explain

```
# Pick a test-set observation to explain
obs_index <- 10
x_interest <- X_test[obs_index, , drop = FALSE]
x_interest

# A tibble: 1 x 14
  Bedroom_AbvGr Year_Built Mo_Sold Lot_Area Street Central_Air First_Flr_SF
    <int>      <int>   <int>   <int> <fct>   <fct>          <int>
1         3      1971     3    1680 Pave    Y             525
# i 7 more variables: Second_Flr_SF <int>, Full_Bath <int>, Half_Bath <int>,
#   Fireplaces <int>, Garage_Area <dbl>, Gr_Liv_Area <int>, TotRms_AbvGrd <int>
```

13.8.2 7.2 Local Explanation for RF Regression

```
local_rf_reg <- LocalModel$new(  
  predictor = pred_rf_reg,  
  x.interest = x_interest  
)  
  
plot(local_rf_reg)
```

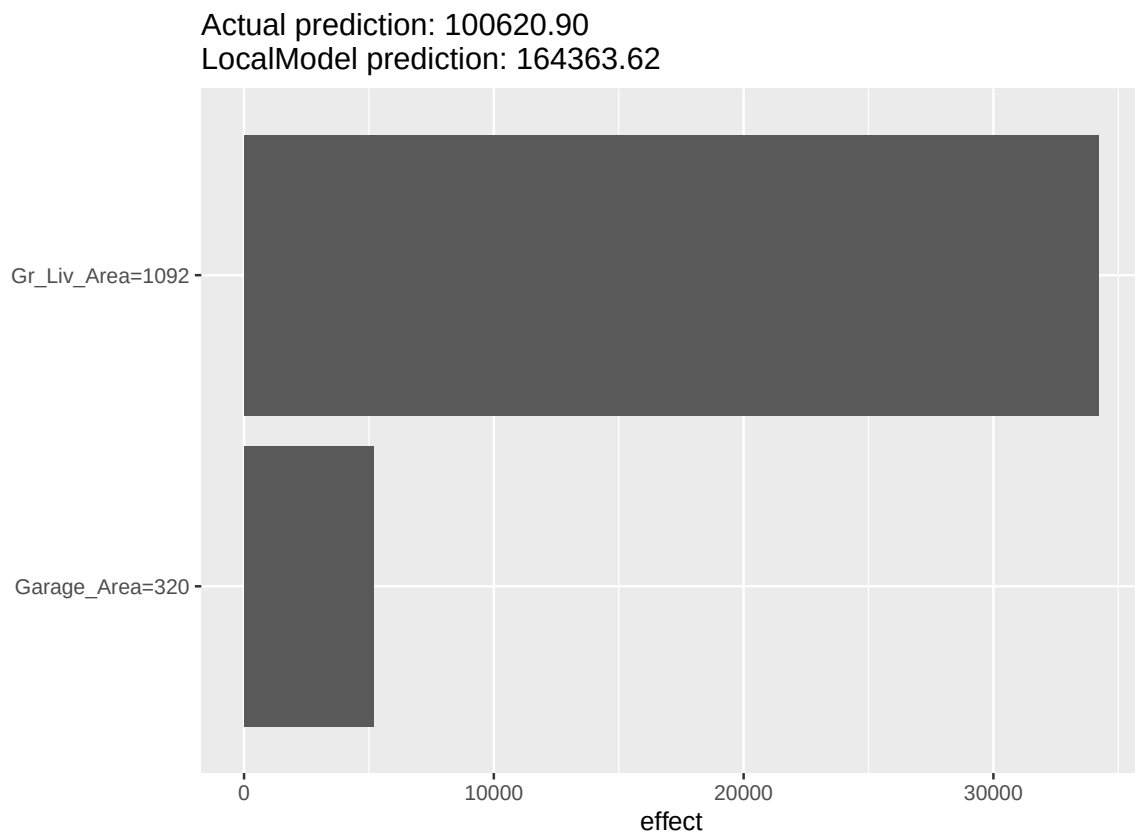


Figure 13.8: Local surrogate explanation (LIME-style) for RF regression prediction.

13.8.3 7.3 Local Explanation for RF Classification (Probability of High Price)

```
local_rf_cls <- LocalModel$new(  
  predictor = pred_rf_cls,
```

```

    x.interest = x_interest
)

plot(local_rf_cls)

```

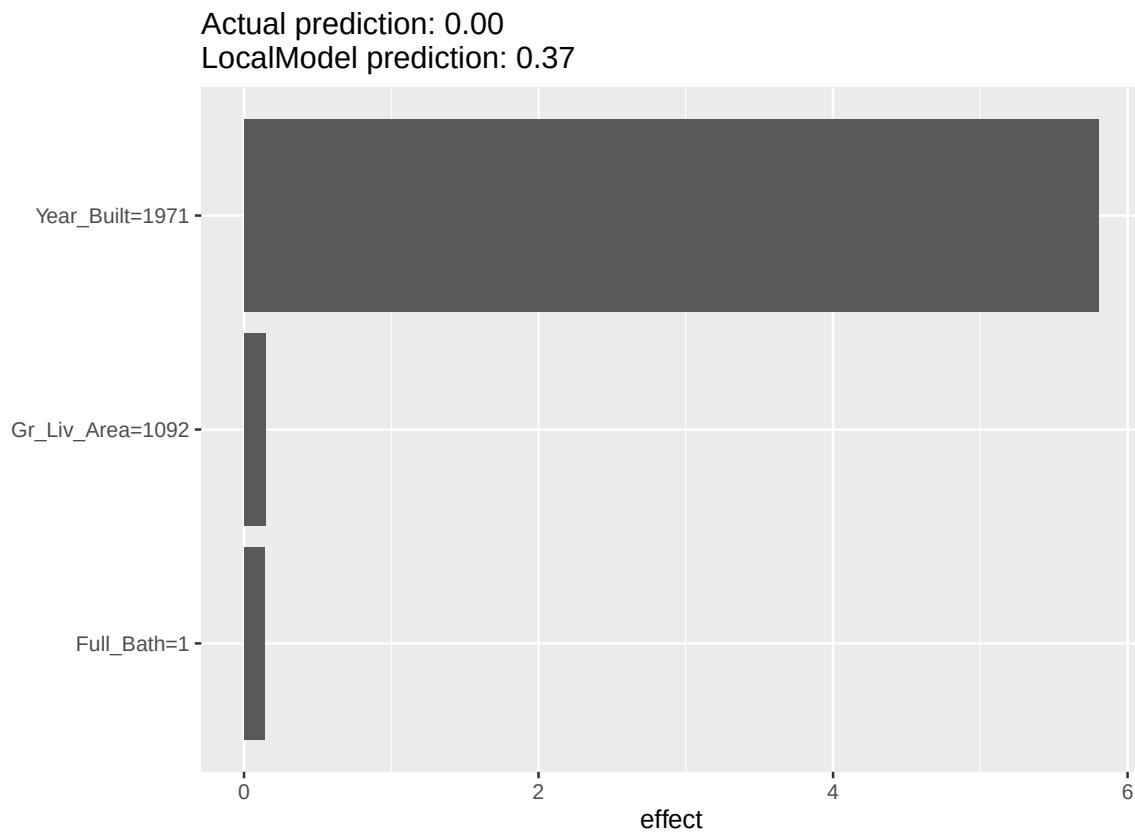


Figure 13.9: Local surrogate explanation (LIME-style) for RF classification probability of HighPrice.

Note:

This is the same home, explained two different ways: - contribution to **predicted price**,
- contribution to **probability of being High price**.

13.9 8. Shapley Values

Shapley values come from cooperative game theory:

- Think of each feature as a “player” in a game whose payoff is the model prediction.
- Shapley values allocate the difference between the prediction and a baseline fairly across features.
- They are **local** explanations for one observation, but with nice global averaging properties.

13.9.1 8.1 Shapley Values for RF Regression

```
set.seed(4321)
shap_rf_reg <- Shapley$new(
  predictor = pred_rf_reg,
  x_interest = x_interest
)
plot(shap_rf_reg)
```

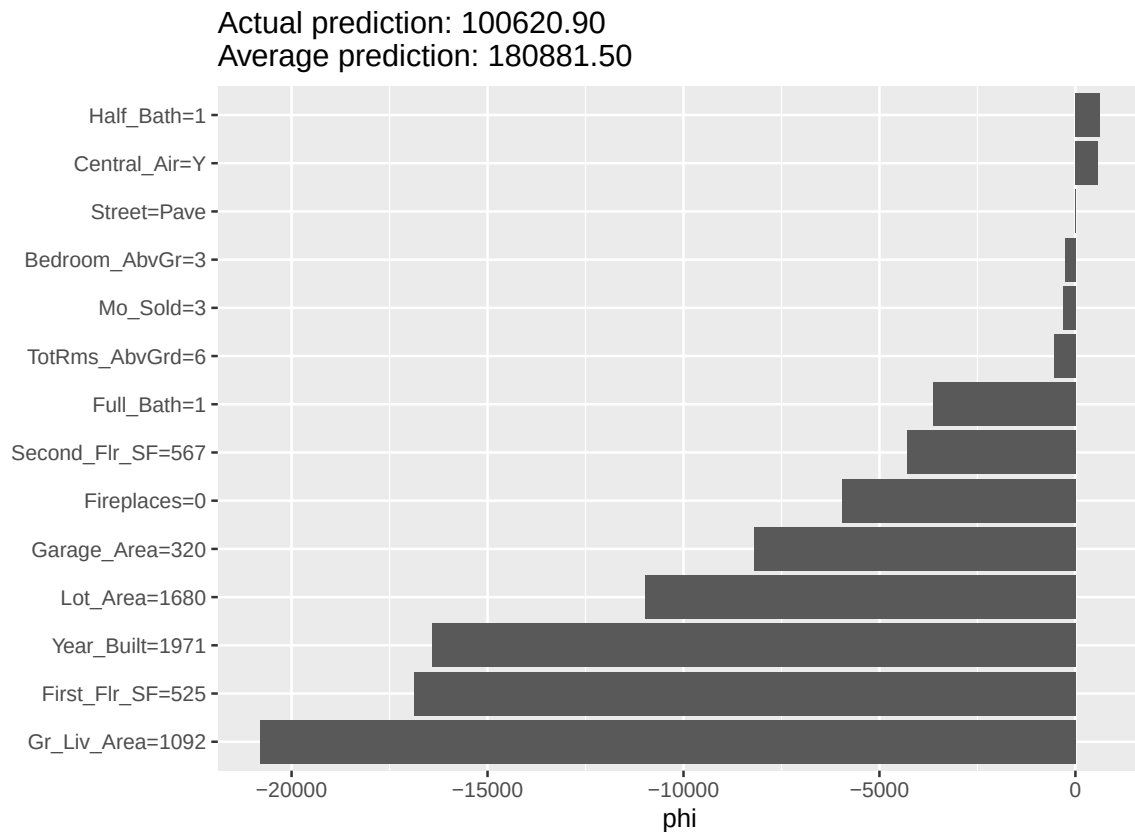


Figure 13.10: Shapley value explanation for RF regression prediction.

13.9.2 8.2 Shapley Values for RF Classification

Now explain the **probability of High price** for the same home.

```
set.seed(4321)
shap_rf_cls <- Shapley$new(
  predictor = pred_rf_cls,
  x.interest = x_interest
)

plot(shap_rf_cls)
```

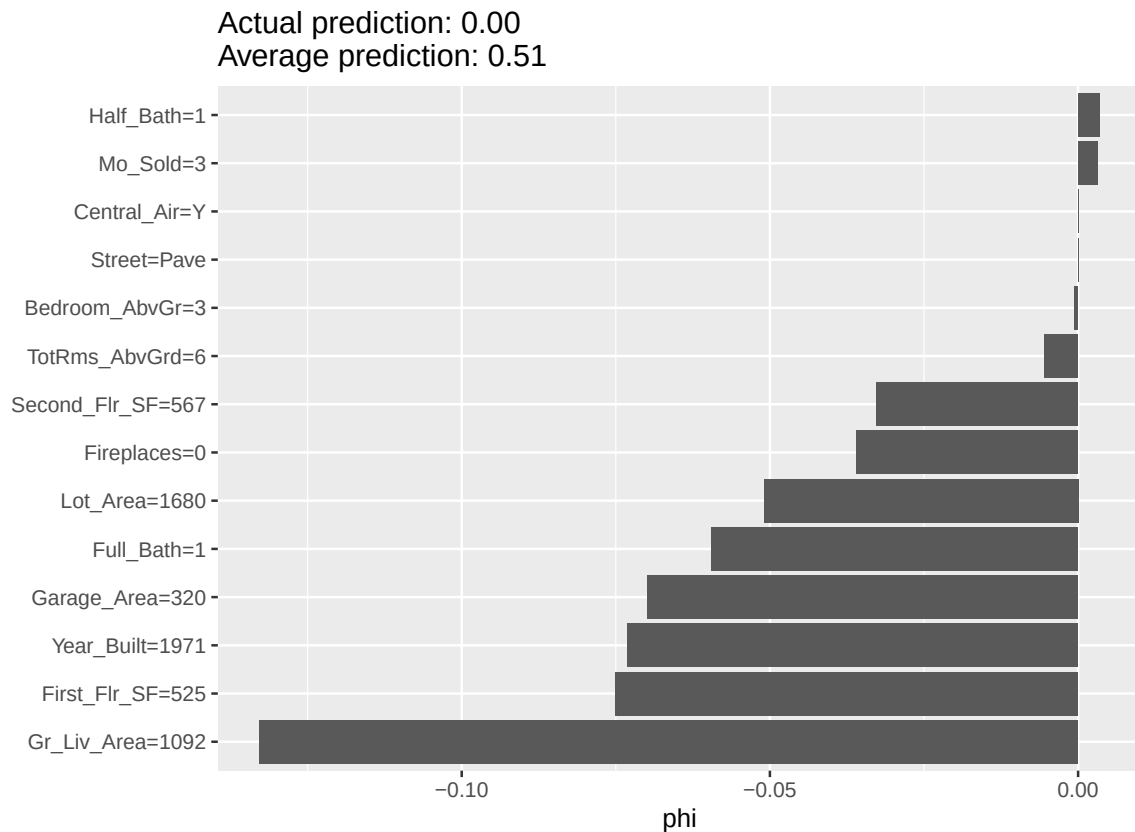


Figure 13.11: Shapley value explanation for RF classification probability of HighPrice.

Interpretation:

- Shapley values are additive: the sum of contributions (plus baseline) equals the prediction.
- Comparing regression vs classification Shapley results shows how the “story” changes when we move from dollar predictions to probabilities.

13.10 9. Alternatives and Model-Specific Tools

There are some other tools to use in R.

13.10.1 9.1 Random Forest Variable Importance with randomForest

Here we refit a Random Forest using the `randomForest` package and use its built-in variable importance tools.

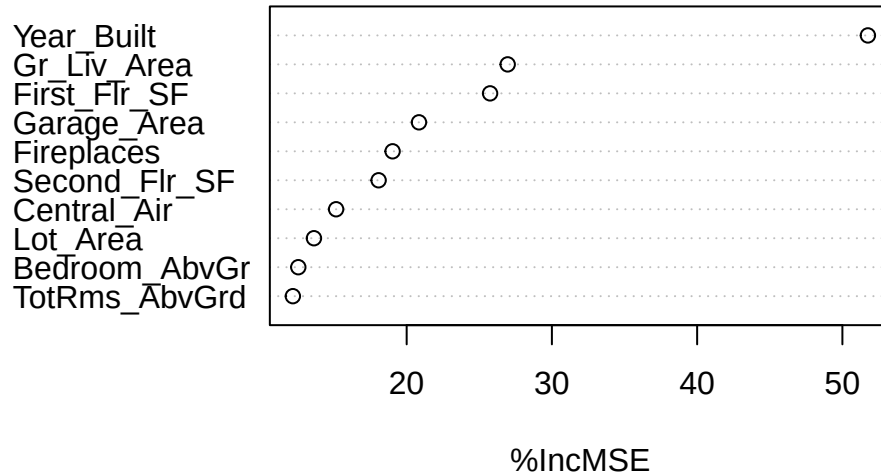
```
# Create a training data frame consistent with earlier setup
training <- train |>
  select(
    Sale_Price,
    Bedroom_AbvGr,
    Year_Built,
    Mo_Sold,
    Lot_Area,
    Street,
    Central_Air,
    First_Flr_SF,
    Second_Flr_SF,
    Full_Bath,
    Half_Bath,
    Fireplaces,
    Garage_Area,
    Gr_Liv_Area,
    TotRms_AbvGrd
  )

training.df <- as.data.frame(training)

set.seed(12345)
rf.ames <- randomForest(
  Sale_Price ~ .,
  data      = training.df,
  ntree     = 250,
  importance = TRUE
)

# Look at variable importance
varImpPlot(
  rf.ames,
  sort = TRUE,
  n.var = 10,
  main = "Top 10 - Variable Importance",
  type = 1
)
```

Top 10 – Variable Importance



```
importance(rf.ames)
```

	%IncMSE	IncNodePurity
Bedroom_AbvGr	12.539754	1.882775e+11
Year_Built	51.755973	2.825361e+12
Mo_Sold	2.341797	1.865486e+11
Lot_Area	13.616999	5.594206e+11
Street	4.048947	4.826931e+09
Central_Air	15.143605	8.973762e+10
First_Flr_SF	25.743101	2.087213e+12
Second_Flr_SF	18.068988	5.829487e+11
Full_Bath	10.525856	5.160484e+11
Half_Bath	11.073376	1.102183e+11
Fireplaces	19.031528	5.394140e+11
Garage_Area	20.848758	1.967501e+12
Gr_Liv_Area	26.954713	2.461302e+12
TotRms_AbvGrd	12.164865	3.738709e+11

Comparison:

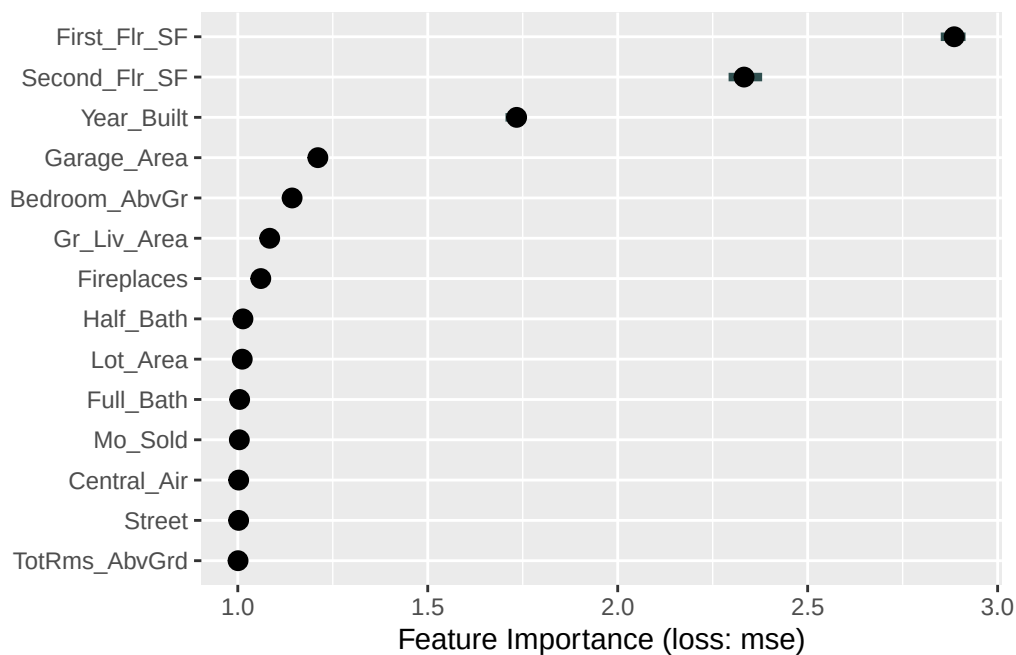
- `randomForest::varImpPlot()` is *model-specific* and convenient.
- `iml::FeatureImp` is *model-agnostic* and can be applied to any model with a suitable prediction function.

13.10.2 9.2 Permutation Importance for Linear Regression (Again with `iml`)

```
lm.ames <- lm(Sale_Price ~ ., data = training)

linear_pred <- Predictor$new(
  lm.ames,
  data = training[,-1],
  y = training$Sale_Price,
  type = "response"
)

plot(FeatureImp$new(linear_pred, loss = "mse"))
```



13.10.3 9.3 ICE / PDP / ALE via `FeatureEffects`

The `FeatureEffects` (plural) helper in `iml` creates multiple types of effects in one object and then uses a `$plot()` method.

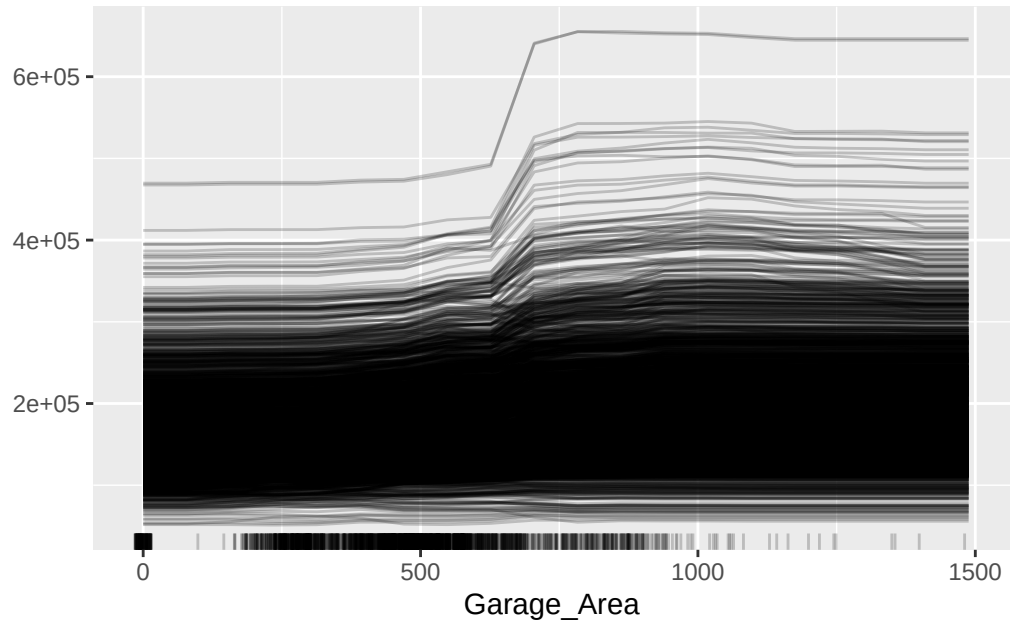
```
forest_pred <- Predictor$new(
  rf.ames,
  data = training[,-1],
```

```

y      = training$Sale_Price,
type = "response"
)

# ICE
ice_plot <- FeatureEffects$new(forest_pred, method = "ice")
ice_plot$plot("Garage_Area")

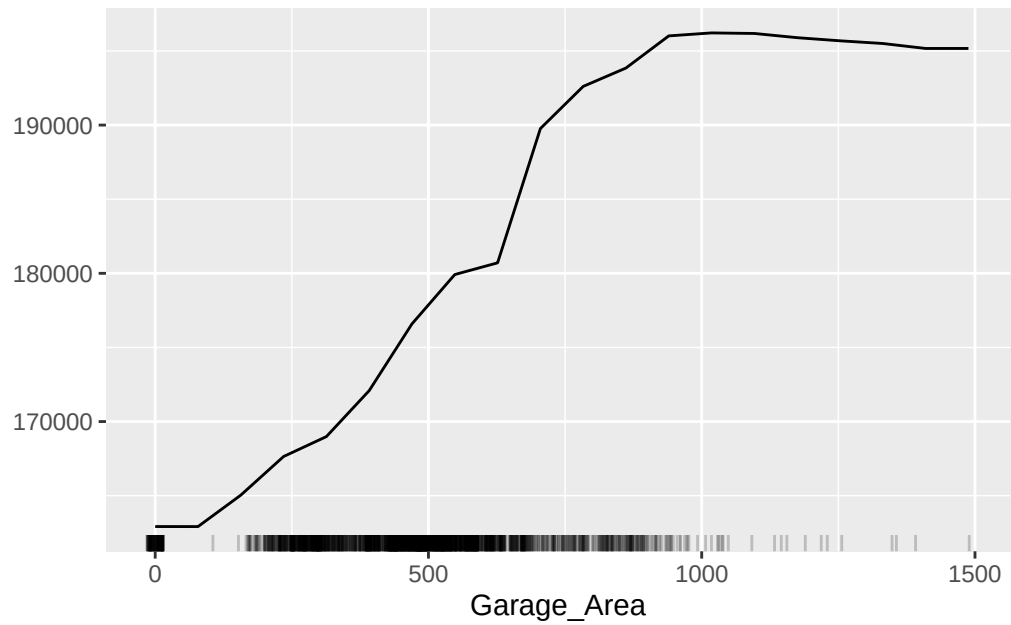
```



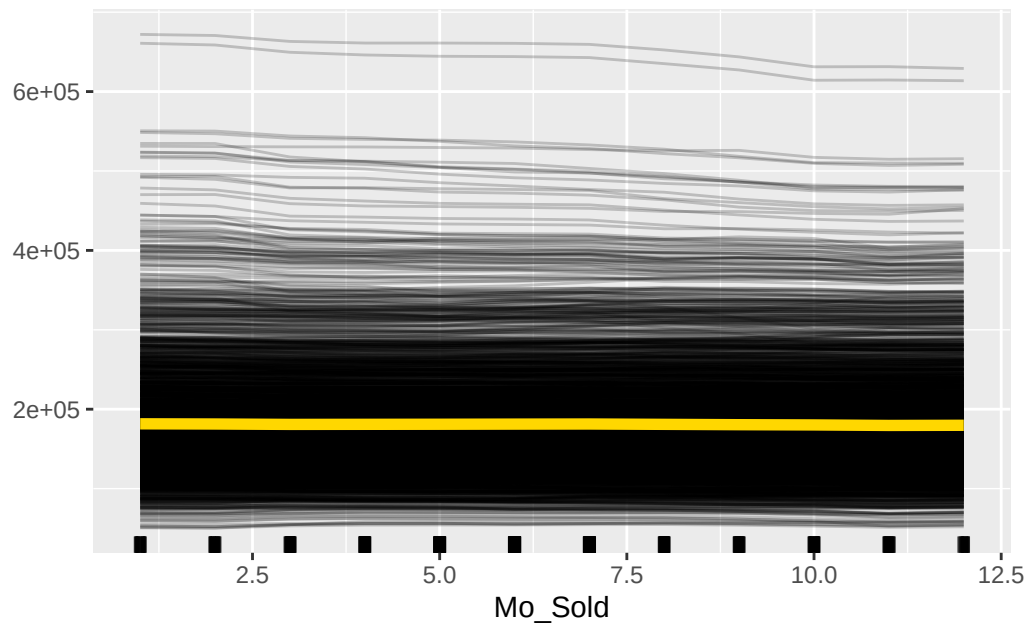
```

# PDP
pd_plot <- FeatureEffects$new(forest_pred, method = "pdp")
pd_plot$plot("Garage_Area")

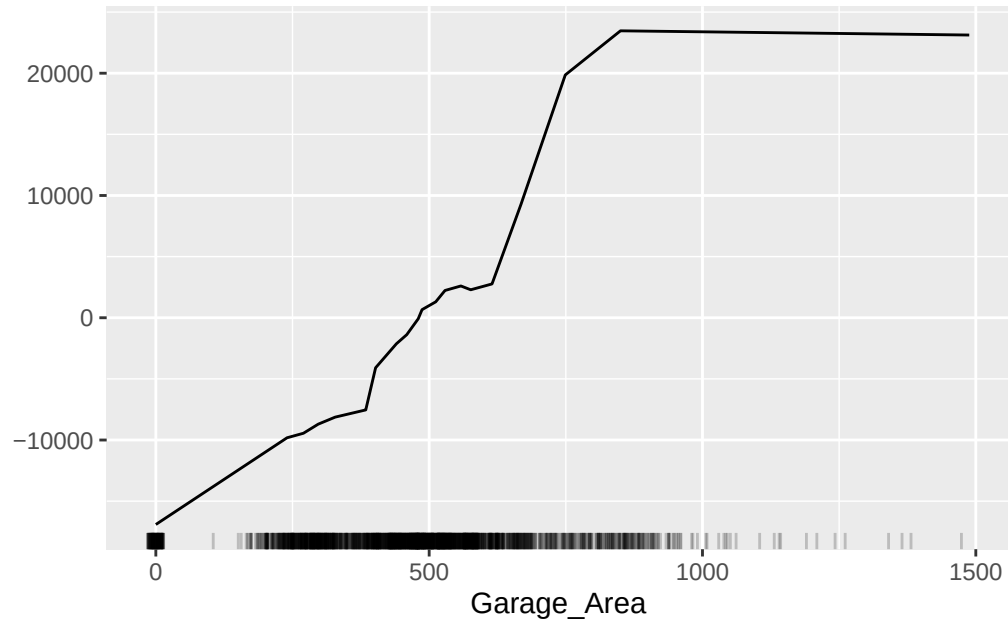
```



```
# PDP + ICE overlay
pdice_plot <- FeatureEffects$new(forest_pred, method = "pdp+ice")
pdice_plot$plot("Mo_Sold")
```



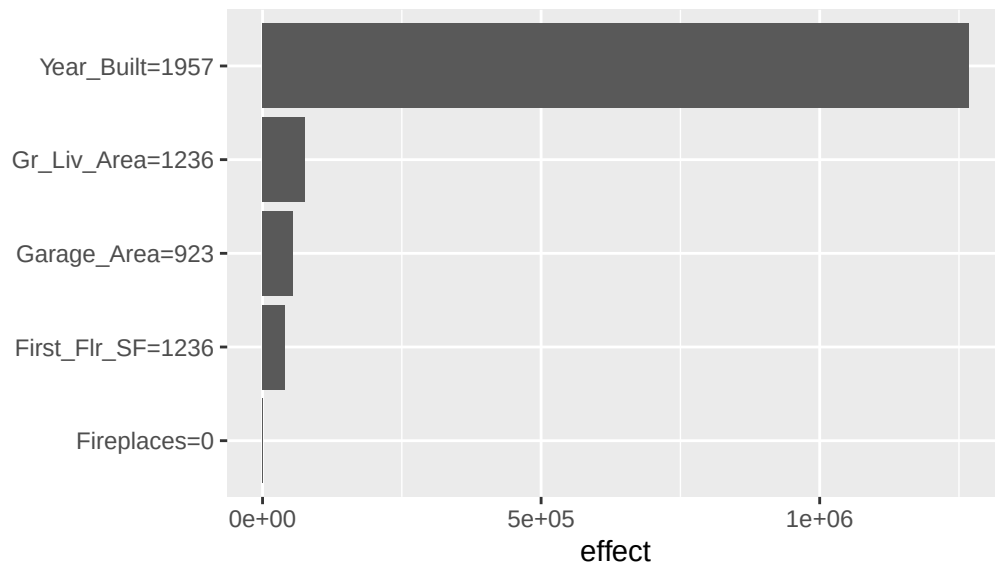
```
# ALE
ale_plot <- FeatureEffects$new(forest_pred, method = "ale")
ale_plot$plot("Garage_Area")
```



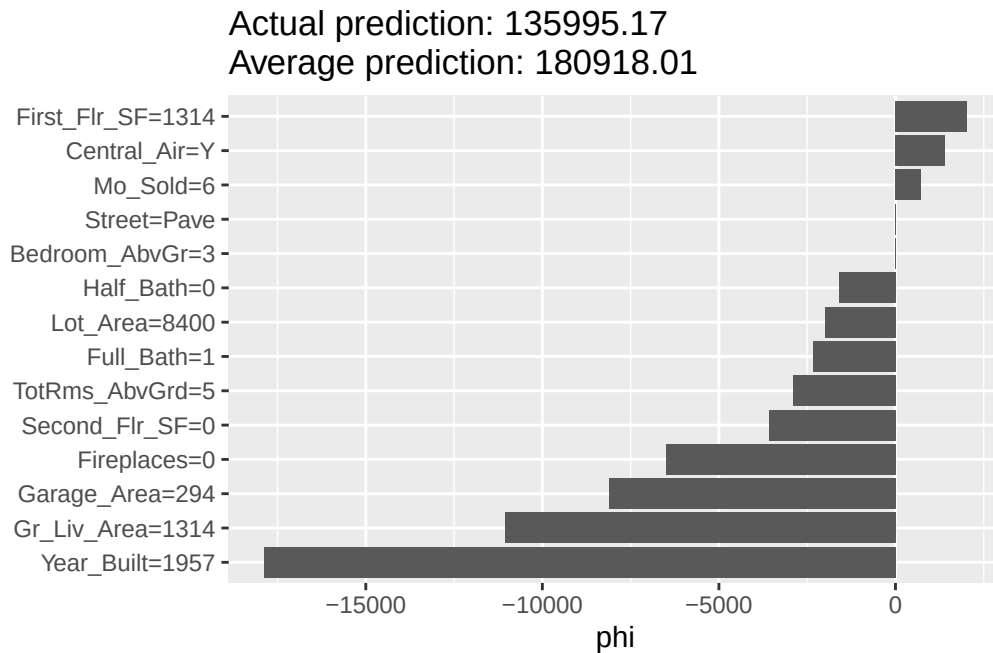
13.10.4 9.4 LIME-Style LocalModel and Shapley (Alternative Examples)

```
# LIME-style explanation for a particular point
point <- 132
lime.explain <- LocalModel$new(
  forest_pred,
  x.interest = training[point,-1],
  k = 5
)
plot(lime.explain)
```

Actual prediction: 147656.71
LocalModel prediction: 180309.23



```
# Shapley values for a particular point
point <- 100
shap_alt <- Shapley$new(
  forest_pred,
  x.interest = training[point,-1]
)
shap_alt$plot()
```



Teaching point:

These examples show that: - You can work *entirely* within `iml` for a consistent, model-agnostic interface, or
- You can mix `iml` tools with model-specific utilities like `randomForest::varImpPlot()` depending on your audience and goals.

13.11 10. Summary and Comparison

Key takeaways:

- **Permutation importance** tells us which features matter *globally* for predictions.
- **PDP and ICE** show how predictions change as one feature moves:
 - PDP: average effect
 - ICE: individual-level variation
- **ALE** is often preferred when predictors are correlated.
- **Local surrogate (LIME-style) and Shapley values:**

- provide *local*, observation-level explanations,
 - but can be averaged to create *global* stories.
- Model-specific tools (like `varImpPlot`) are convenient but tie you to a particular algorithm, whereas `iml` gives you a common language across very different models.

14 Model-Agnostic Interpretability - Python Version

15 Setup: Libraries

```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression, LogisticRegression
from sklearn.ensemble import RandomForestRegressor, RandomForestClassifier
from xgboost import XGBRegressor, XGBClassifier

from sklearn.inspection import permutation_importance, PartialDependenceDisplay
from sklearn.preprocessing import OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline

import matplotlib.pyplot as plt

# Python ALE + SHAP
#from pyALE import ale
#import shap
```

15.1 Load the Ames Housing Dataset

```
from sklearn.datasets import fetch_openml

ames = fetch_openml(name="house_prices", as_frame=True)
df = ames.frame.copy()

# Rename to match R version
df.rename(columns={"SalePrice": "Sale_Price"}, inplace=True)

cols = [
    "Sale_Price", "BedroomAbvGr", "YearBuilt", "MoSold", "LotArea", "Street",
```

```

    "CentralAir", "1stFlrSF", "2ndFlrSF", "FullBath", "HalfBath", "Fireplaces",
    "GarageArea", "GrLivArea", "TotRmsAbvGrd"
]
df = df[cols].dropna()

median_price = df["Sale_Price"].median()
df["HighPrice"] = (df["Sale_Price"] >= median_price).astype(int)

```

15.2 Train/Test Split

```

X = df.drop(columns=["Sale_Price", "HighPrice"])
y_reg = df["Sale_Price"]
y_cls = df["HighPrice"]

X_train, X_test, y_train_reg, y_test_reg = train_test_split(
    X, y_reg, test_size=0.3, random_state=4321
)

_, _, y_train_cls, y_test_cls = train_test_split(
    X, y_cls, test_size=0.3, random_state=4321
)

```

15.3 Preprocessing

```

categorical = ["Street", "CentralAir"]
numeric = [c for c in X.columns if c not in categorical]

preprocess = ColumnTransformer(
    transformers=[
        ("num", "passthrough", numeric),
        ("cat", OneHotEncoder(handle_unknown="ignore"), categorical)
    ]
)

```

15.4 Regression Models

```
lm_reg = Pipeline([
    ("prep", preprocess),
    ("model", LinearRegression())
]).fit(X_train, y_train_reg)

rf_reg = Pipeline([
    ("prep", preprocess),
    ("model", RandomForestRegressor(
        n_estimators=500, max_features=4, random_state=4321))
]).fit(X_train, y_train_reg)

xgb_reg = Pipeline([
    ("prep", preprocess),
    ("model", XGBRegressor(
        n_estimators=200, max_depth=4, subsample=0.8,
        colsample_bytree=0.8, eta=0.1, random_state=4321))
]).fit(X_train, y_train_reg)
```

15.5 Classification Models

```
log_cls = Pipeline([
    ("prep", preprocess),
    ("model", LogisticRegression(max_iter=500))
]).fit(X_train, y_train_cls)

rf_cls = Pipeline([
    ("prep", preprocess),
    ("model", RandomForestClassifier(
        n_estimators=500, max_features=4, random_state=4321))
]).fit(X_train, y_train_cls)

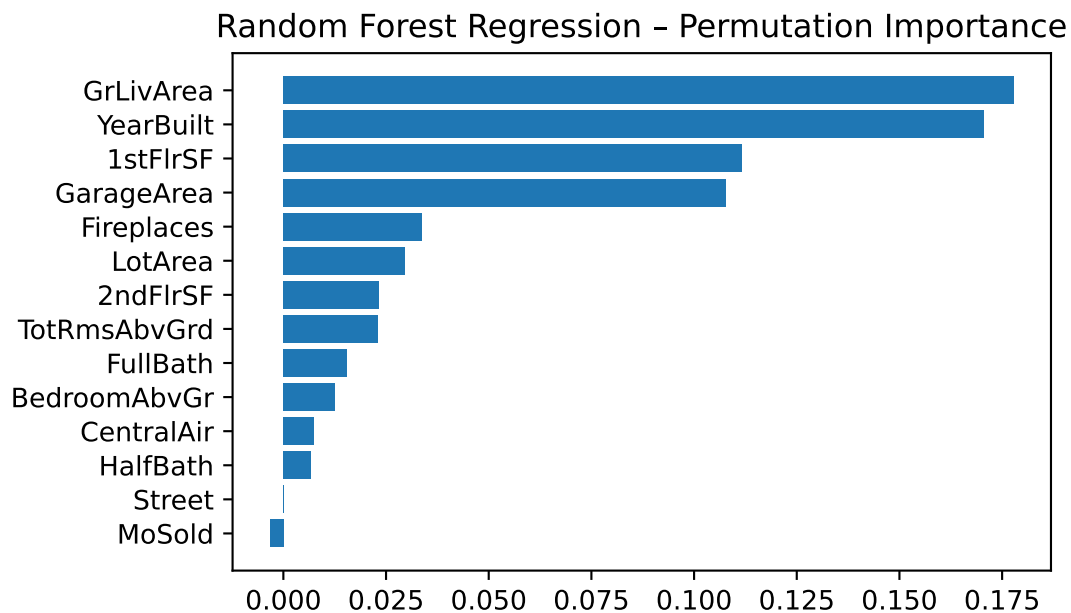
xgb_cls = Pipeline([
    ("prep", preprocess),
    ("model", XGBClassifier(
        n_estimators=200, max_depth=4, subsample=0.8,
        colsample_bytree=0.8, eta=0.1, random_state=4321,
```

```
eval_metric="logloss"))
]).fit(X_train, y_train_cls)
```

15.6 Permutation Importance

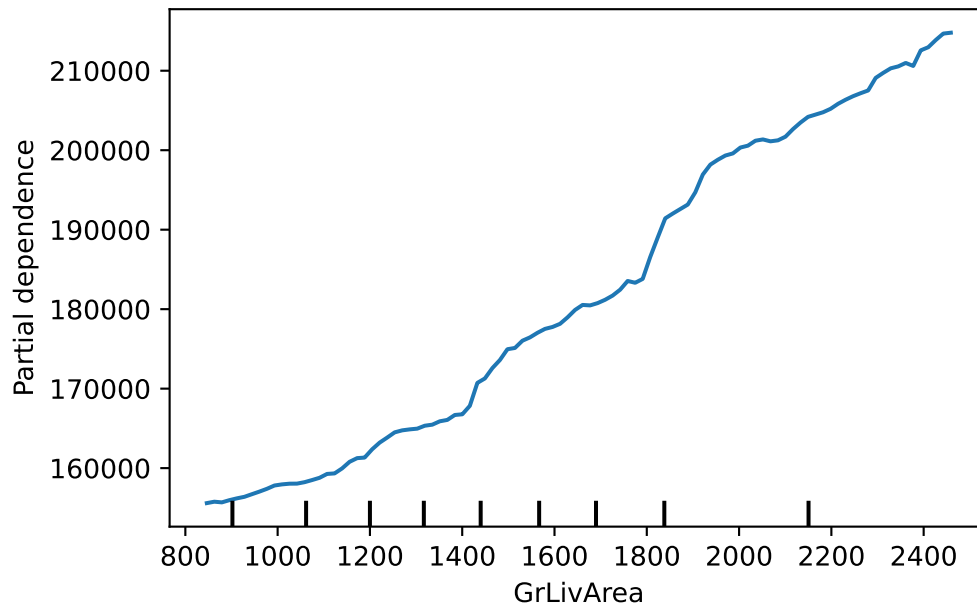
```
r = permutation_importance(rf_reg, X_test, y_test_reg, n_repeats=10, random_state=4321)

sorted_idx = r.importances_mean.argsort()
plt.barh(np.array(X.columns)[sorted_idx], r.importances_mean[sorted_idx])
plt.title("Random Forest Regression - Permutation Importance")
plt.show()
```



15.7 PDP Example

```
fig = PartialDependenceDisplay.from_estimator(
    rf_reg, X_train, features=["GrLivArea"]
)
plt.show()
```



15.8 ALE Example (pyALE)

```
# ale_df = ale(  
#     X=X_train,  
#     model=rf_reg,  
#     feature="GrLivArea",  
#     include_CI=False  
# )  
# plt.plot(ale_df["GrLivArea"], ale_df["ale"])  
# plt.xlabel("GrLivArea")  
# plt.ylabel("ALE Effect")  
# plt.title("ALE Plot - GrLivArea (Random Forest)")  
# plt.show()
```

15.9 SHAP Example

```
# X_small = X_train.sample(50, random_state=1)  
#  
# explainer = shap.KernelExplainer(  

```

```
#     rf_reg.predict,  
#     X_small  
# )  
#  
# shap_values = explainer.shap_values(X_test.iloc[0:1])  
#  
# shap.plots.waterfall(shap_values[0])
```


16 Clustering with K-means, DBSCAN, and Hierarchical R Version

16.1 Learning Objectives

By the end of this section, you will be able to:

- Apply K-means, DBSCAN, and Hierarchical clustering.
 - Interpret common graphs.
 - Produce parallel coordinates plots for each clustering solution.
 - Compare cluster assignments to true Iris species.
-

16.2 Setup

```
library(tidyverse)
library(factoextra)
library(cluster)
library(dbSCAN)
library(NbClust)
library(GGally) # for parallel coordinate plots
set.seed(4321)
```

16.3 Dataset

```
data(iris)
iris_num <- iris %>% select(where(is.numeric))
iris_scaled <- scale(iris_num)
head(iris_scaled)
```

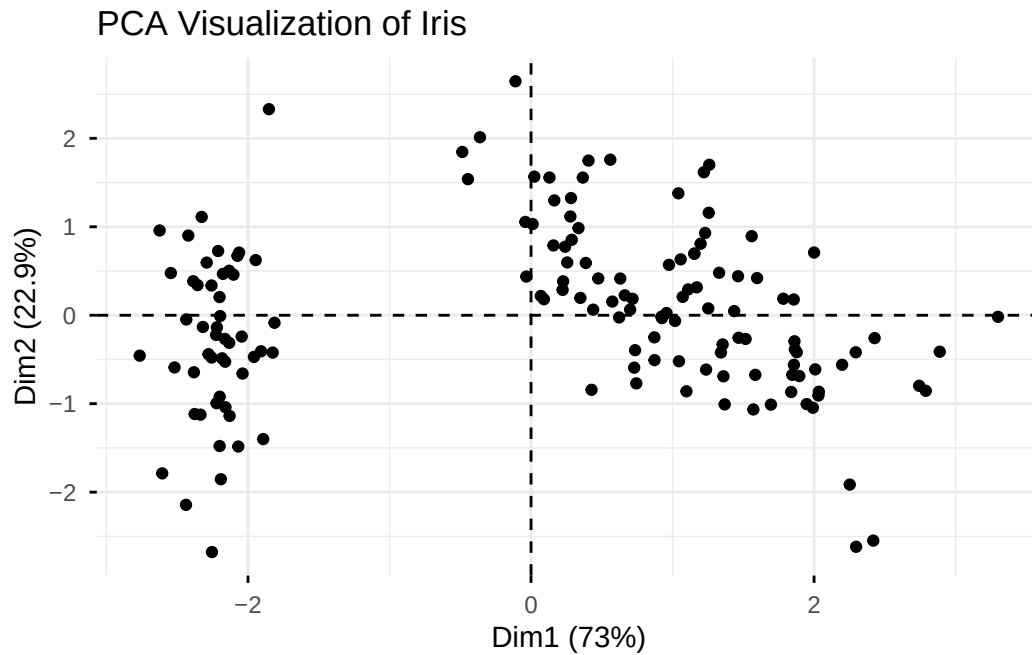
	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width
[1,]	-0.8976739	1.01560199	-1.335752	-1.311052
[2,]	-1.1392005	-0.13153881	-1.335752	-1.311052
[3,]	-1.3807271	0.32731751	-1.392399	-1.311052
[4,]	-1.5014904	0.09788935	-1.279104	-1.311052
[5,]	-1.0184372	1.24503015	-1.335752	-1.311052
[6,]	-0.5353840	1.93331463	-1.165809	-1.048667

16.3.1 Explanation

We extract the **numeric features** of the Iris dataset and scale them so that all variables contribute equally. Scaling is crucial because distance-based clustering is sensitive to variable magnitude.

16.4 PCA Visualization

```
pca <- prcomp(iris_scaled)
fviz_pca_ind(pca, geom="point", title="PCA Visualization of Iris")
```



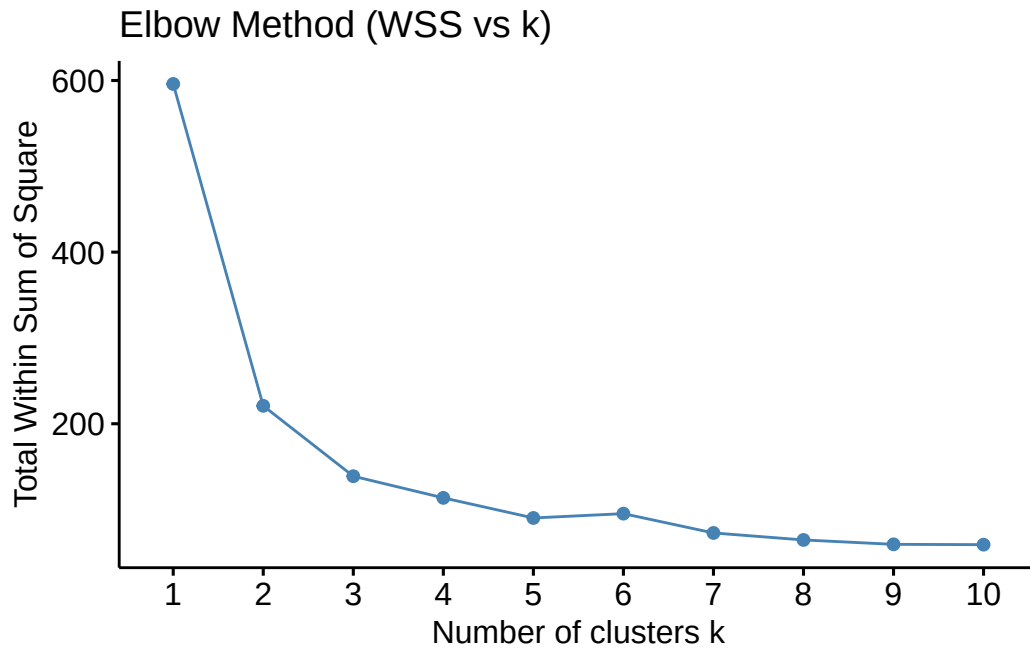
16.4.1 Explanation

A PCA scatterplot allows us to see the **structure of high-dimensional data** in two dimensions. If clusters exist naturally, PCA often reveals them visually.

16.5 PART 1 — Determining k for K-means

16.5.1 Elbow Plot

```
fviz_nbclust(iris_scaled, kmeans, method="wss") +  
  labs(title="Elbow Method (WSS vs k)")
```

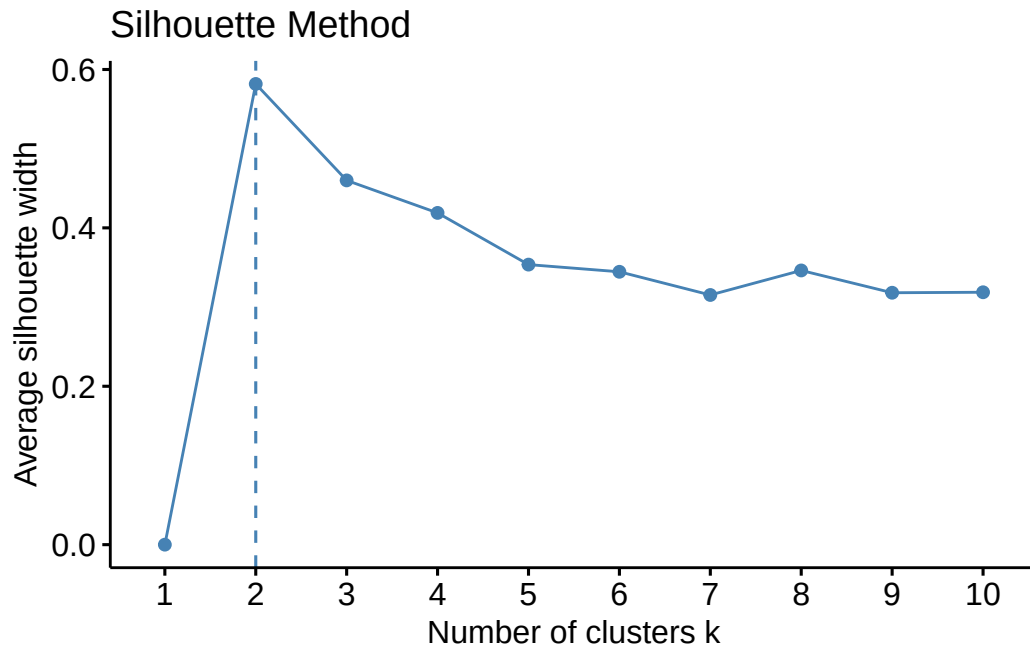


16.5.1.1 Explanation

The elbow plot shows how total within-cluster variation decreases as k increases. We select the k where improvement slows down — the “elbow.”

16.5.2 Silhouette Plot

```
fviz_nbclust(iris_scaled, kmeans, method="silhouette") +  
  labs(title="Silhouette Method")
```

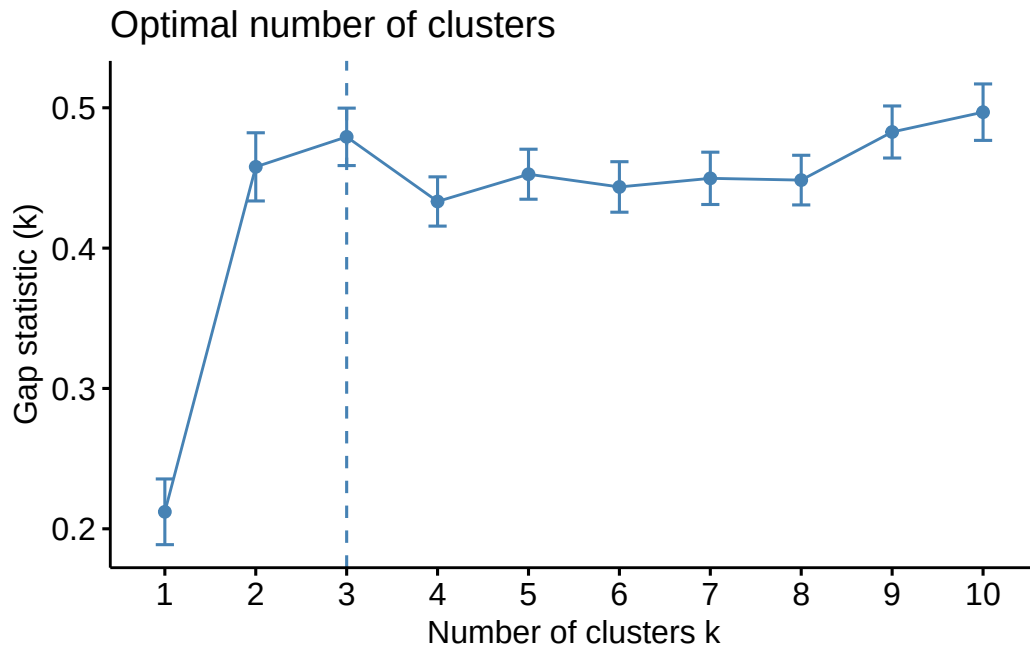


16.5.2.1 Explanation

The silhouette method evaluates how well-separated clusters are. Higher values indicate clearer boundaries. The optimal k maximizes silhouette width.

16.5.3 Gap Statistic

```
set.seed(4321)
gap <- clusGap(iris_scaled, FUN=kmeans, nstart=25, K.max=10, B=30)
fviz_gap_stat(gap)
```



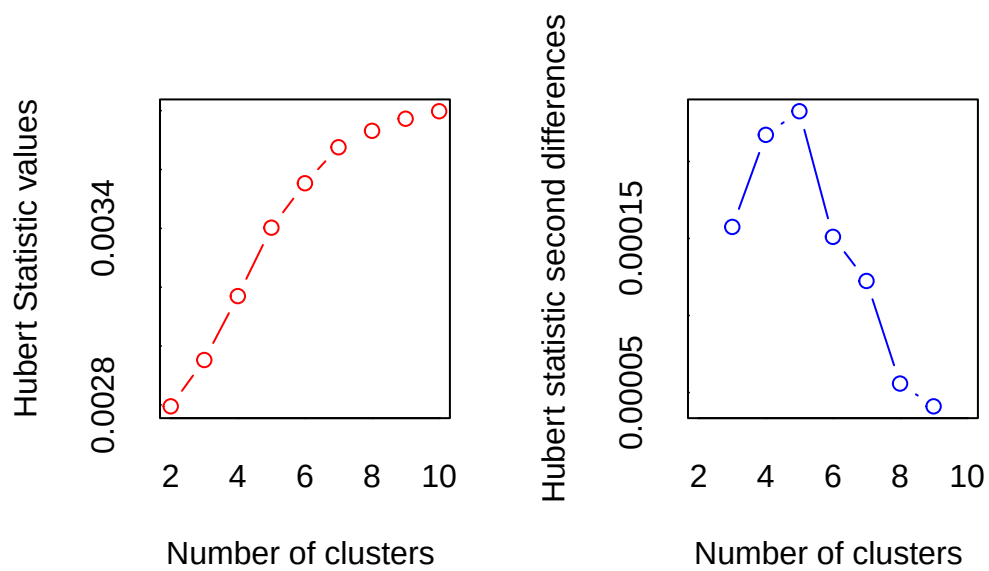
16.5.3.1 Explanation

The gap statistic compares cluster dispersion to what would be expected if no clustering existed.

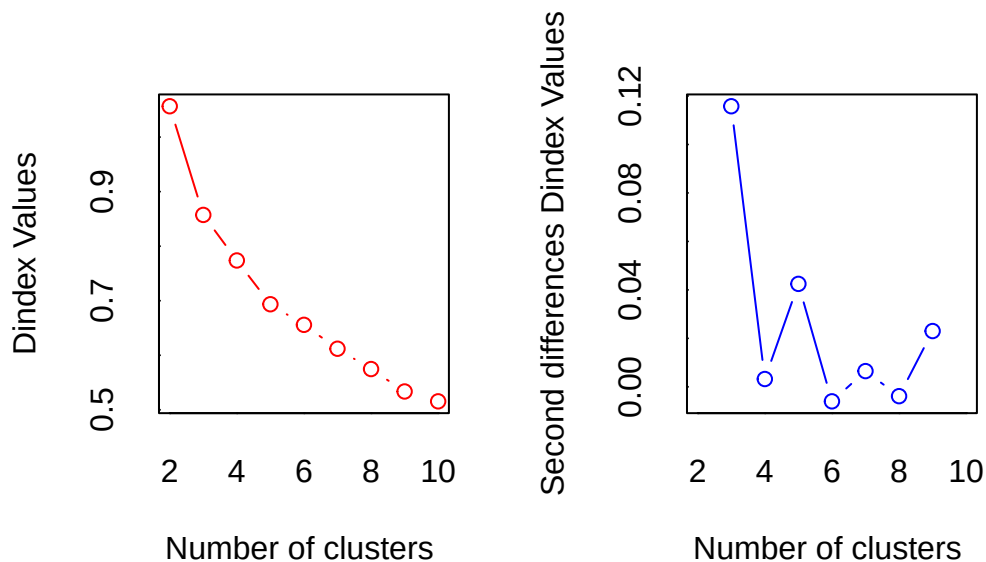
A large “gap” indicates strong underlying cluster structure.

16.5.4 NbClust

```
set.seed(4321)
nb_res <- NbClust(iris_scaled, distance="euclidean",
                  min.nc=2, max.nc=10, method="kmeans")
```



*** : The Hubert index is a graphical method of determining the number of clusters. In the plot of Hubert index, we seek a significant knee that corresponds to a significant increase of the value of the measure i.e the significant peak in index second differences plot.



*** : The D index is a graphical method of determining the number of clusters.

In the plot of D index, we seek a significant knee (the significant peak in the second differences plot) that corresponds to a significant increase of the value of the measure.

* Among all indices:

* 11 proposed 2 as the best number of clusters

* 10 proposed 3 as the best number of clusters

* 1 proposed 4 as the best number of clusters

* 1 proposed 6 as the best number of clusters

* 1 proposed 10 as the best number of clusters

***** Conclusion *****

* According to the majority rule, the best number of clusters is 2

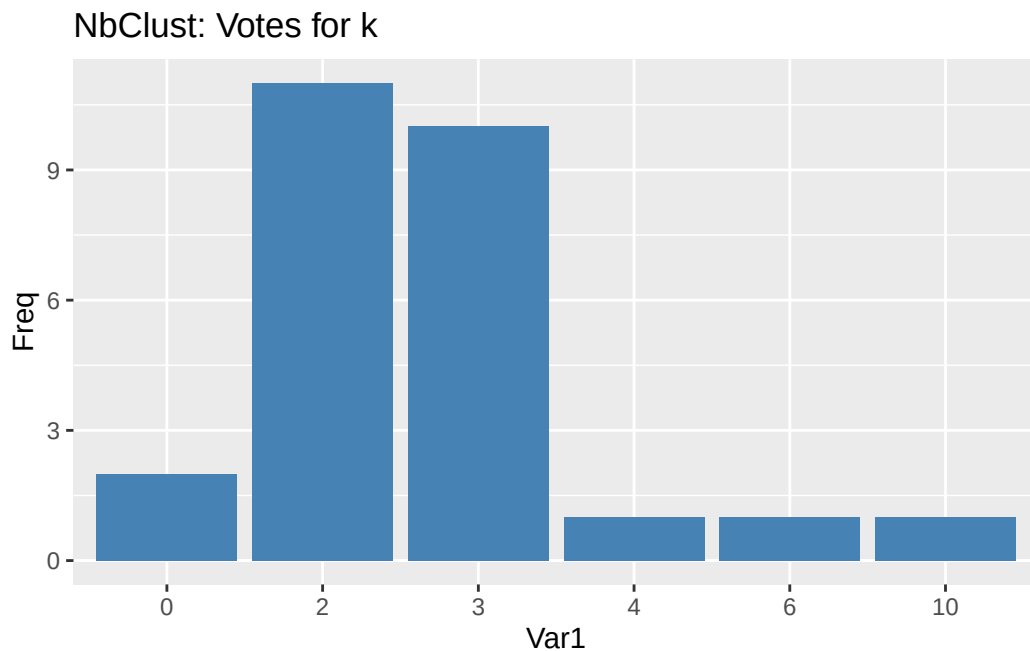
```
best_k <- nb_res$Best.nc[1]
best_k
```



```
[1] 3
```

```
vote_dist <- as.data.frame(table(nb_res$Best.nc[1, ]))

ggplot(vote_dist, aes(Var1, Freq)) +
  geom_col(fill="steelblue") +
  labs(title="NbClust: Votes for k")
```



16.5.4.1 Explanation

NBClust computes 30+ indices to recommend the number of clusters. We visualize the vote distribution and use the **majority vote**.

16.6 K-means Clustering

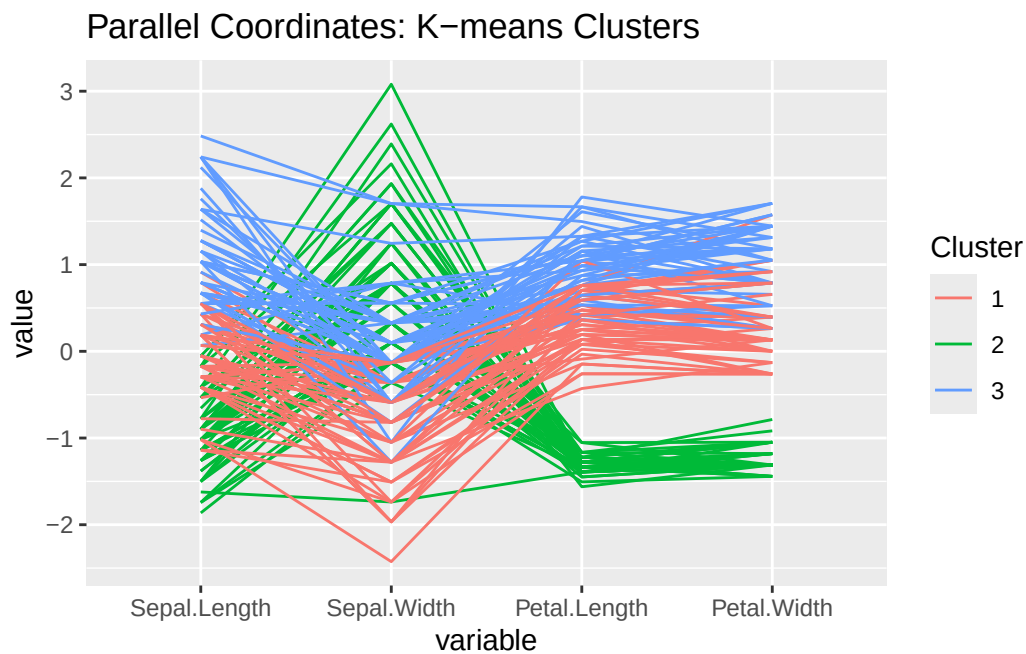
```
kmeans_fit <- kmeans(iris_scaled, centers = best_k, nstart=30)
fviz_cluster(kmeans_fit, iris_scaled,
  main=paste("K-means Clusters (k=", best_k, ")", sep=""))
```


16.7.1 Explanation

This table shows how well K-means clusters match the true species.
K-means typically identifies Setosa perfectly but overlaps Versicolor and Virginica.

16.8 Parallel Coordinates for K-means

```
iris_km <- iris_num %>% mutate(Cluster = factor(kmeans_fit$cluster))
GGally::ggparcoord(iris_km, columns = 1:4, groupColumn = "Cluster",
                    title="Parallel Coordinates: K-means Clusters")
```



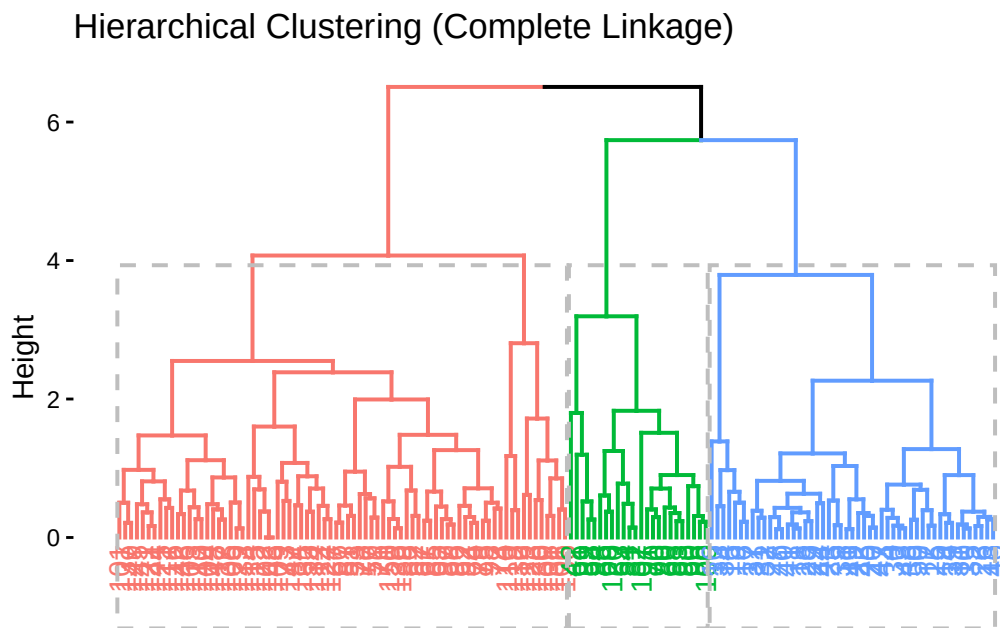
16.8.1 Explanation

Parallel coordinate plots show **variable profiles** across clusters.
K-means clusters should show distinct shape patterns if clustering is meaningful.

16.9 PART 2 — Hierarchical Clustering

16.9.1 Dendrogram

```
d <- dist(iris_scaled)
hc_complete <- hclust(d, method="complete")
fviz_dend(hc_complete, k=3, rect=TRUE,
          main="Hierarchical Clustering (Complete Linkage)")
```



16.9.1.1 Explanation

Hierarchical clustering builds a **tree structure** showing merging of points/clusters. Cutting the dendrogram at height = 3 yields three clusters.

16.9.2 Cut into 3 Clusters and Compare

```
hc_clusters <- cutree(hc_complete, k=3)
table(Hierarchical = hc_clusters, Species = iris$Species)
```

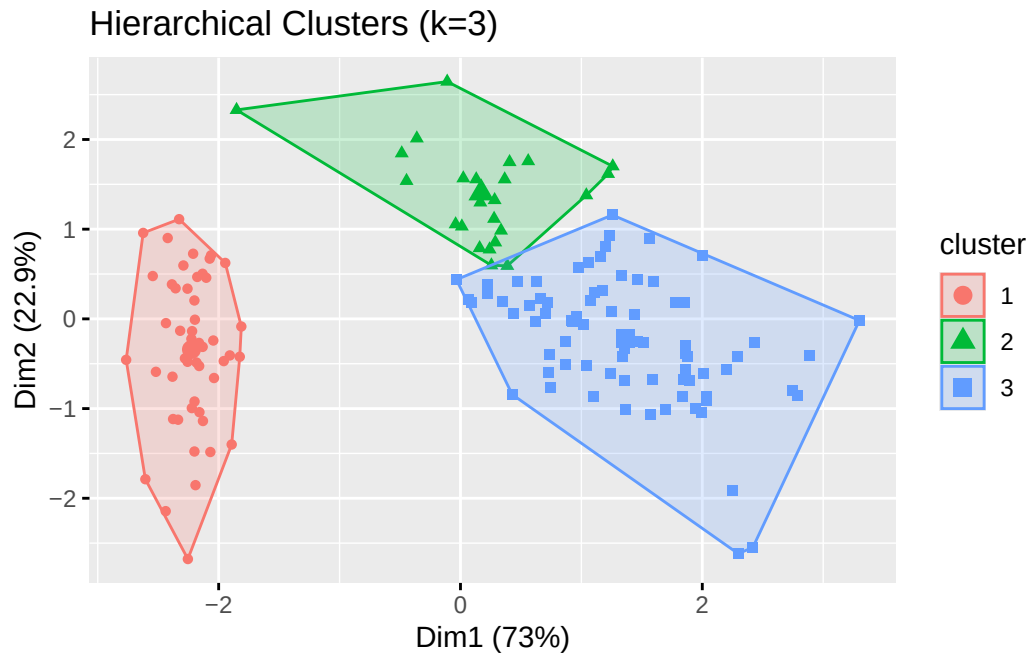
	Species		
Hierarchical	setosa	versicolor	virginica
1	49	0	0
2	1	21	2
3	0	29	48

16.9.2.1 Explanation

Hierarchical clustering does not require pre-specifying k — we “cut” the dendrogram. This table shows how hierarchical groupings align with real species.

16.9.3 PCA Visualization of Hierarchical Clusters

```
fviz_cluster(list(data=iris_scaled, cluster=hc_clusters),
              geom="point",
              main="Hierarchical Clusters (k=3)")
```

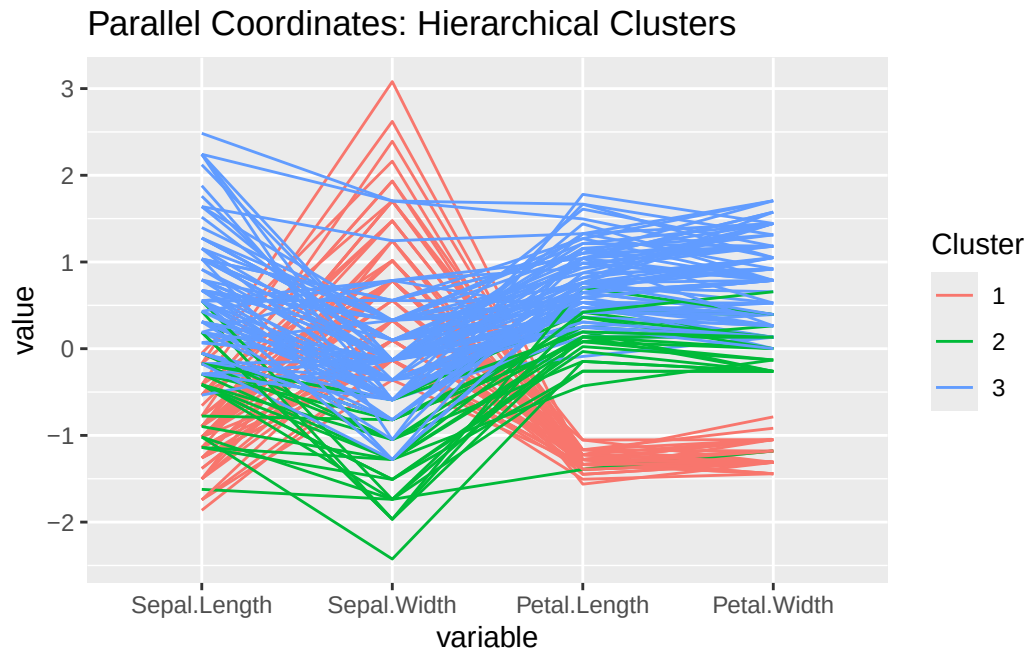


16.9.3.1 Explanation

A PCA scatterplot shows how hierarchical clusters spread in low-dimensional space. Complete linkage tends to produce compact clusters.

16.9.4 Parallel Coordinates for Hierarchical

```
iris_hc <- iris_num %>% mutate(Cluster = factor(hc_clusters))
GGally::ggparcoord(iris_hc, columns = 1:4, groupColumn = "Cluster",
  title="Parallel Coordinates: Hierarchical Clusters")
```



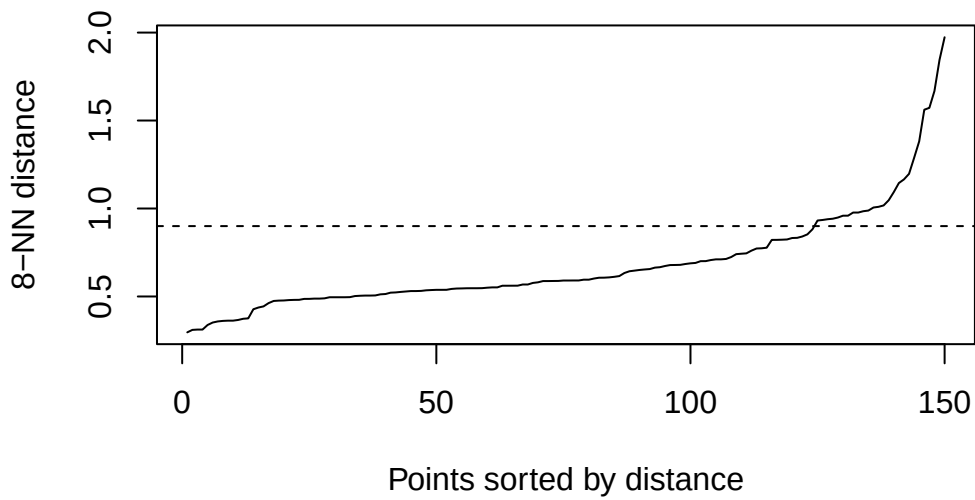
16.9.5 Explanation

Parallel coordinates allow us to compare variable patterns across hierarchical clusters.

16.10 PART 3 — DBSCAN

16.10.1 kNN Distance Plot

```
minPts <- 2 * ncol(iris_scaled)
kNNdistplot(iris_scaled, k=minPts)
abline(h=0.9, lty=2)
```



16.10.1.1 Explanation

The kNN plot helps choose eps by showing the distance to each point's k-th neighbor. The “knee” in the curve is the suggested eps threshold.

16.10.2 Fit DBSCAN

```
db_fit <- dbscan(iris_scaled, eps=0.9, minPts=minPts)
db_fit
```

DBSCAN clustering for 150 objects.
Parameters: eps = 0.9, minPts = 8
Using euclidean distances and borderpoints = TRUE
The clustering contains 2 cluster(s) and 7 noise points.

```
0 1 2
7 48 95
```

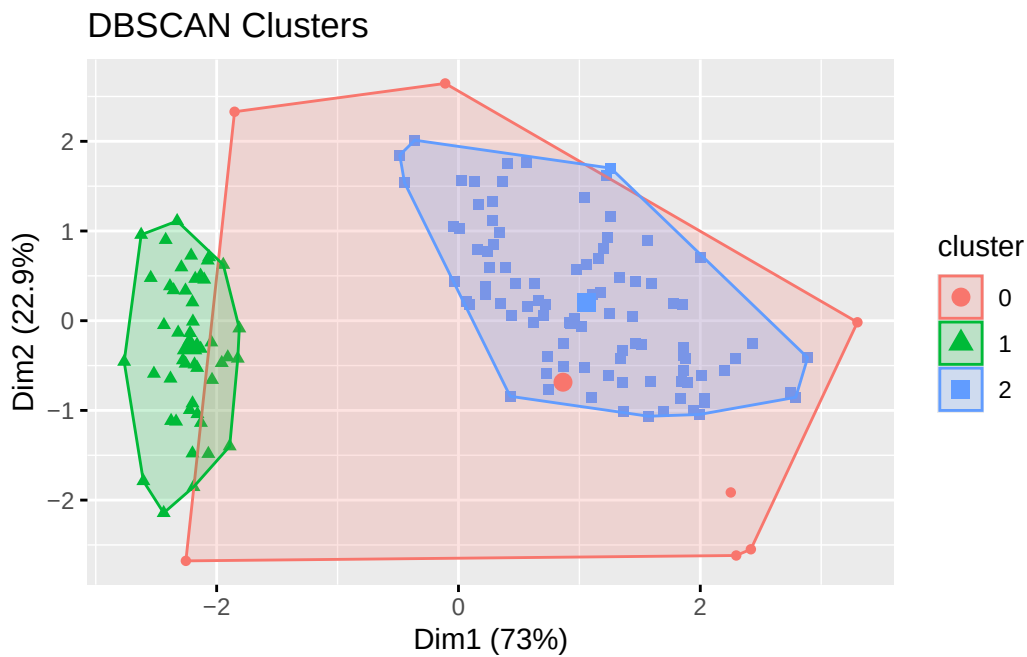
Available fields: cluster, eps, minPts, metric, borderPoints

16.10.2.1 Explanation

DBSCAN groups high-density regions and labels low-density points as **noise (cluster 0)**.

16.10.3 DBSCAN Clusters in PCA Space

```
fviz_cluster(list(data=iris_scaled, cluster=db_fit$cluster),  
             geom="point",  
             ellipse.type="convex",  
             main="DBSCAN Clusters")
```



16.10.3.1 Explanation

DBSCAN finds **arbitrary-shaped clusters** that density-based methods capture better than K-means.

16.10.4 Compare DBSCAN to True Species

```
table(DBSCAN = db_fit$cluster, Species = iris$Species)
```

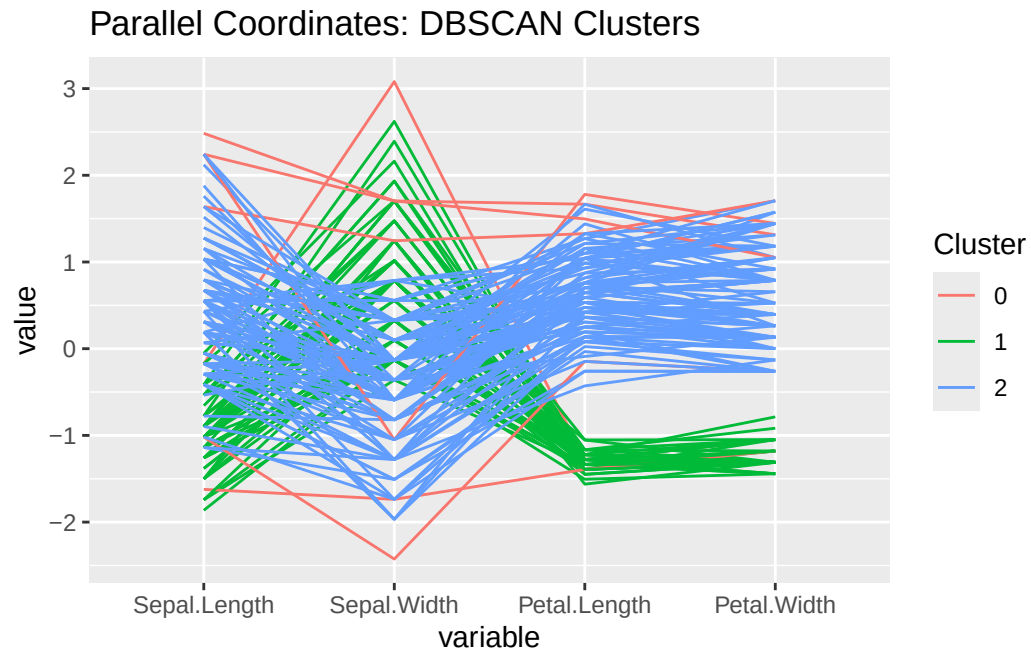
	Species		
DBSCAN	setosa	versicolor	virginica
0	2	1	4
1	48	0	0
2	0	49	46

16.10.4.1 Explanation

DBSCAN often segments Setosa well but may merge Versicolor and Virginica depending on eps.

16.10.5 Parallel Coordinates for DBSCAN

```
iris_db <- iris_num %>% mutate(Cluster = factor(db_fit$cluster))
GGally::ggparcoord(iris_db, columns = 1:4, groupColumn = "Cluster",
                    title="Parallel Coordinates: DBSCAN Clusters")
```



16.10.5.1 Explanation

Parallel coordinates show how DBSCAN clusters differ in variable space and highlight noise points.

17 Isolation Forest - R Version

17.1 Learning Objectives

After this notebook, you will be able to:

- Understand Isolation Forest intuition
- Generate synthetic data with injected anomalies
- Fit Isolation Forest

17.2 Setup

```
library(tidyverse)
library(isotree)
set.seed(4321)
```

17.2.1 1. Isolation Forest Overview

Isolation Forest finds anomalies by isolating points with random partitions:

- Normal points require **many** splits
 - Outliers isolate **quickly**
 - Fewer splits → **higher anomaly score**
-

17.2.2 2. Generate Synthetic Data (Two Clusters + Outliers)

```
n <- 500

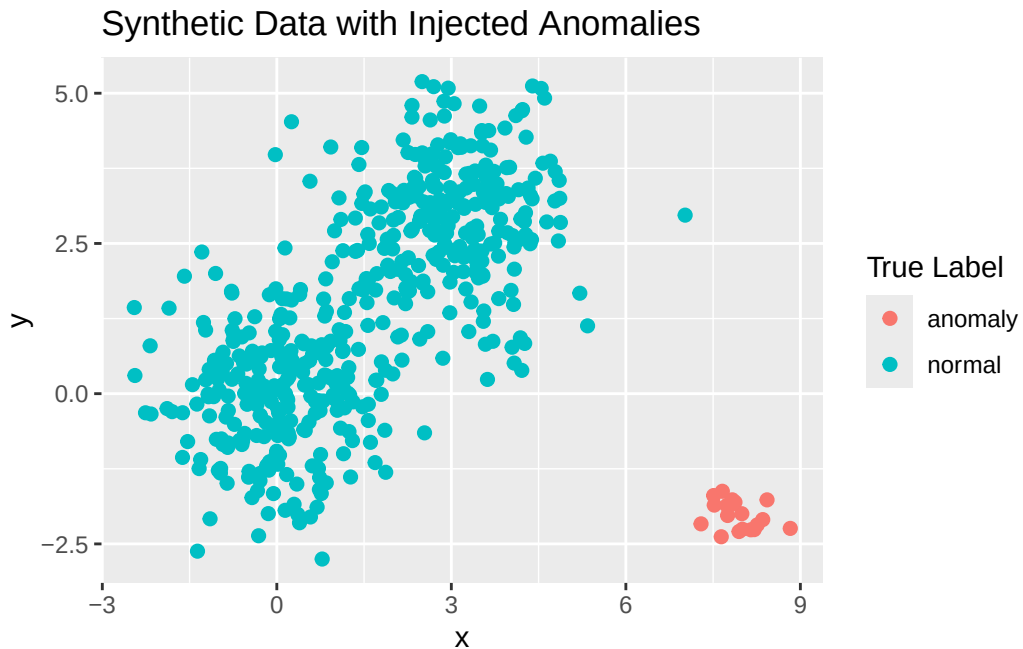
cluster1 <- tibble(
  x = rnorm(n/2, 0, 1),
  y = rnorm(n/2, 0, 1),
  true_type = "normal"
)

cluster2 <- tibble(
  x = rnorm(n/2, 3, 1),
  y = rnorm(n/2, 3, 1),
  true_type = "normal"
)

# Inject anomalies
anoms <- tibble(
  x = rnorm(20, 8, 0.4),
  y = rnorm(20, -2, 0.4),
  true_type = "anomaly"
)

df <- bind_rows(cluster1, cluster2, anoms)

ggplot(df, aes(x, y, color = true_type)) +
  geom_point(size = 2) +
  labs(title = "Synthetic Data with Injected Anomalies",
       color = "True Label")
```



17.2.3 3. Fit Isolation Forest

```
iso_model <- isolation.forest(df[, c("x", "y")], ntrees = 500)

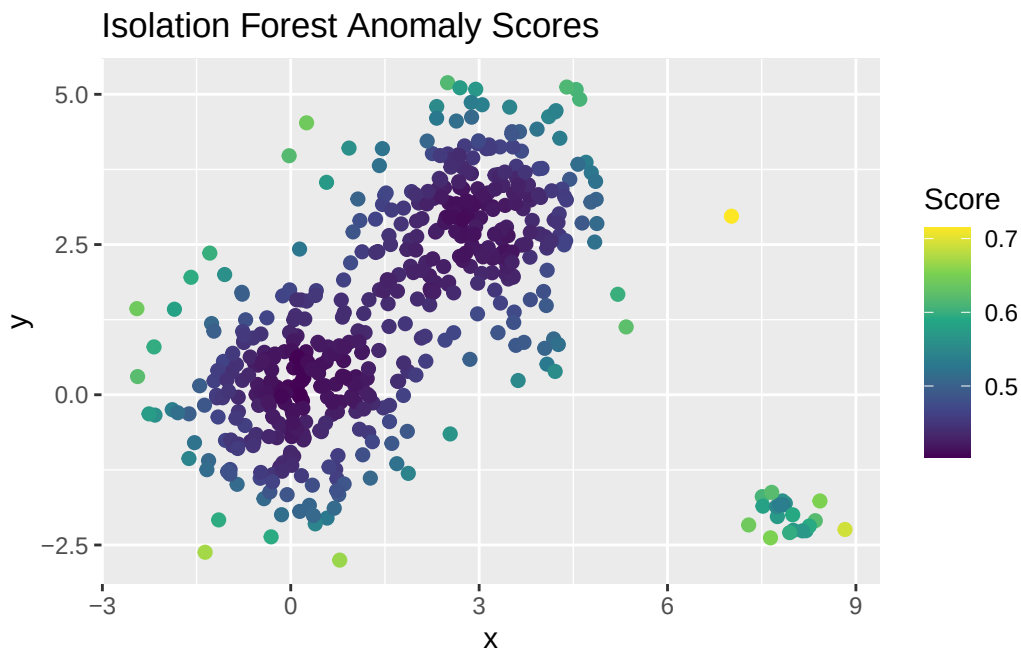
df$score <- predict(iso_model,
                    df[, c("x", "y")],
                    type = "score")

summary(df$score)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.4033	0.4227	0.4400	0.4608	0.4760	0.7149

17.2.4 4. Visualize Anomaly Scores

```
ggplot(df, aes(x, y, color = score)) +  
  geom_point(size = 2) +  
  scale_color_viridis_c() +  
  labs(title = "Isolation Forest Anomaly Scores",  
        color = "Score")
```

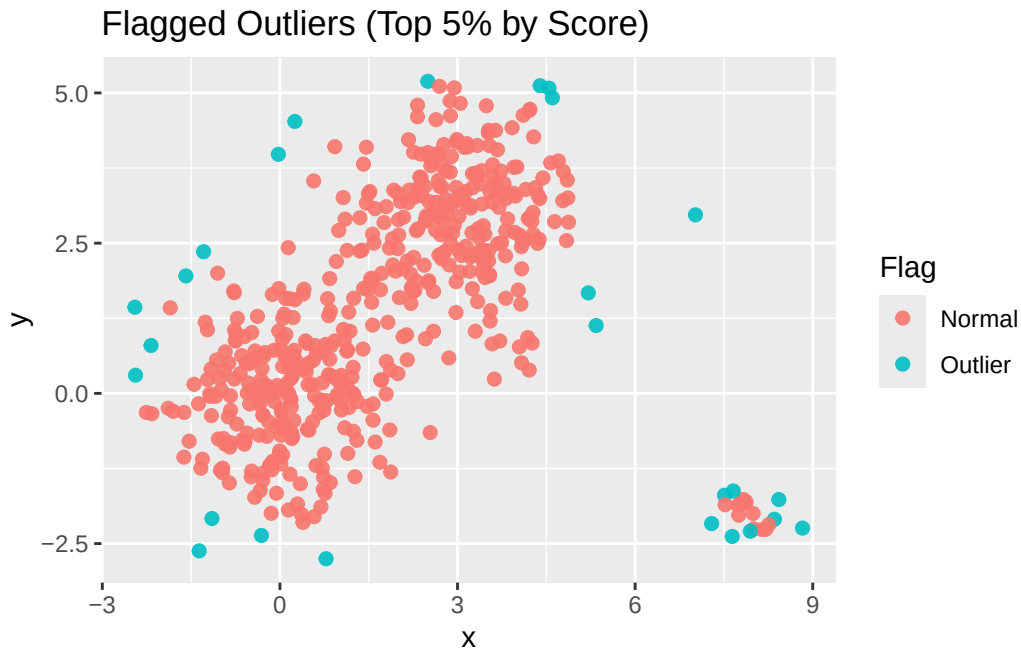


17.2.5 5. Flag Outliers (Top 5%)

```
cutoff <- quantile(df$score, 0.95)  
df$flag <- ifelse(df$score >= cutoff, "Outlier", "Normal")  
  
table(df$flag)
```

Normal	Outlier
494	26

```
ggplot(df, aes(x, y, color = flag)) +
  geom_point(size = 2, alpha = 0.9) +
  labs(title = "Flagged Outliers (Top 5% by Score)",
       color = "Flag")
```



17.2.6 6. Print Top Anomalies

```
df %>%
  arrange(desc(score)) %>%
  slice(1:10)
```

```
# A tibble: 10 x 5
      x     y true_type score flag
  <dbl> <dbl> <chr>    <dbl> <chr>
1  7.02  2.97 normal    0.715 Outlier
2  8.83 -2.24 anomaly    0.691 Outlier
3 -1.36 -2.62 normal    0.671 Outlier
4  0.780 -2.75 normal    0.665 Outlier
```


5	8.43	-1.77	anomaly	0.653	Outlier
6	7.64	-2.38	anomaly	0.653	Outlier
7	0.252	4.53	normal	0.641	Outlier
8	-2.45	1.43	normal	0.641	Outlier
9	7.29	-2.17	anomaly	0.641	Outlier
10	5.34	1.13	normal	0.624	Outlier

17.2.7 7. Benchmarked Accuracy (for synthetic data)

```
table(Predicted = df$flag, Truth = df$true_type)
```

	Truth	
Predicted	anomaly	normal
Normal	12	482
Outlier	8	18

17.2.8 8. Parallel Coordinates

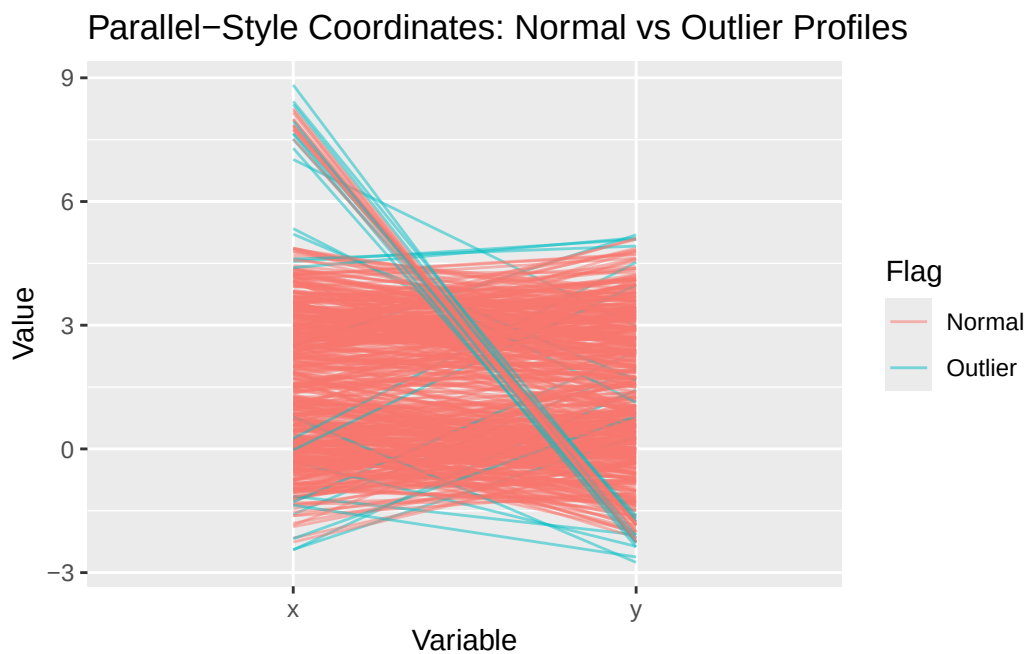
This version is **100% Quarto-safe**, with no `row_number()` inside `aes()` and no `ggparcoord` dependency.

```
# Clean factor
df_pc <- df %>%
  mutate(flag = factor(flag, levels = c("Normal", "Outlier")))

# Create explicit, stable row ID
df_pc <- df_pc %>%
  mutate(id = row_number())

# Long format for parallel-style plot
df_long <- df_pc %>%
  pivot_longer(cols = c(x, y),
               names_to = "Variable",
               values_to = "Value")
```

```
# Plot
ggplot(df_long,
       aes(x = Variable,
           y = Value,
           group = id,
           color = flag)) +
geom_line(alpha = 0.5) +
labs(title = "Parallel-Style Coordinates: Normal vs Outlier Profiles",
     color = "Flag")
```



17.2.9 Interpretation

This shows how outliers differ from normal points across variables.

If outliers consistently sit higher/lower on certain dimensions, they appear as distinct paths.

17.3 PART 2 — Business-Context Example

17.4 Simulated Credit Transactions Dataset (Multidimensional)

We simulate **fraud-like behavior** with:

- Unusually large transaction amounts
 - Transactions at unusual times
 - Rare merchant categories
 - Abnormal distances between transactions
 - Too-many transactions in a short window
 - High-risk device scores
-

17.4.1 1. Generate Synthetic Credit Transaction Data

```
set.seed(1234)

n <- 2000

# Normal transactions
norm <- tibble(
  amount = rgamma(n, shape=2, scale=40),
  hour = sample(6:22, n, replace=TRUE),
  merchant_risk = rbeta(n, 2, 8),
  device_score = rnorm(n, mean=50, sd=10),
  distance_km = rgamma(n, shape=2, scale=3),
  trans_per_24h = rpois(n, lambda=3),
  true_type = "normal"
)

# Fraudulent transactions
fraud <- tibble(
  amount = rgamma(60, shape=9, scale=60), # large
```

```

hour = sample(c(0,1,2,3,4), 60, replace=TRUE), # late night
merchant_risk = rbeta(60, 8, 1), # high risk
device_score = rnorm(60, mean=20, sd=5), # bad device score
distance_km = rgamma(60, shape=1, scale=20), # far distances
trans_per_24h = rpois(60, lambda=12), # high velocity
true_type = "fraud"
)

df <- bind_rows(norm, fraud)

```

17.4.2 2. Fit Isolation Forest

```

iso <- isolation.forest(df %>% select(-true_type), ntrees=600)

df$score <- predict(iso, df %>% select(-true_type), type="score")

summary(df$score)

```

```

      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.3653  0.3969  0.4150  0.4279  0.4423  0.7326

```

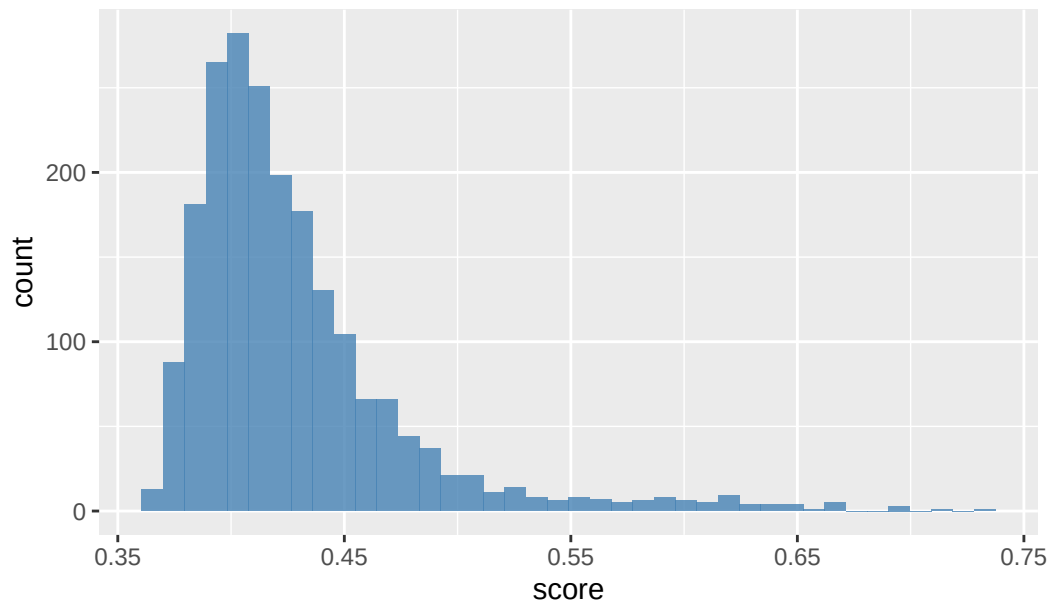
17.4.3 3. Visualize Score Distribution

```

ggplot(df, aes(score)) +
  geom_histogram(bins=40, fill="steelblue", alpha=0.8) +
  labs(title="Anomaly Score Distribution (Credit Transactions)")

```

Anomaly Score Distribution (Credit Transactions)



17.4.4 4. Flag Top 5% as Suspected Fraud

```
cutoff <- quantile(df$score, 0.95)
df$flag <- ifelse(df$score >= cutoff, "Suspected Fraud", "Legit")

table(df$flag)
```

```
Legit Suspected Fraud
1957          103
```

17.4.5 5. Compare Predictions to True Fraud Labels

```
table(Predicted=df$flag, Truth=df$true_type)
```

Predicted	Truth	
	fraud	normal
Legit	0	1957
Suspected Fraud	60	43

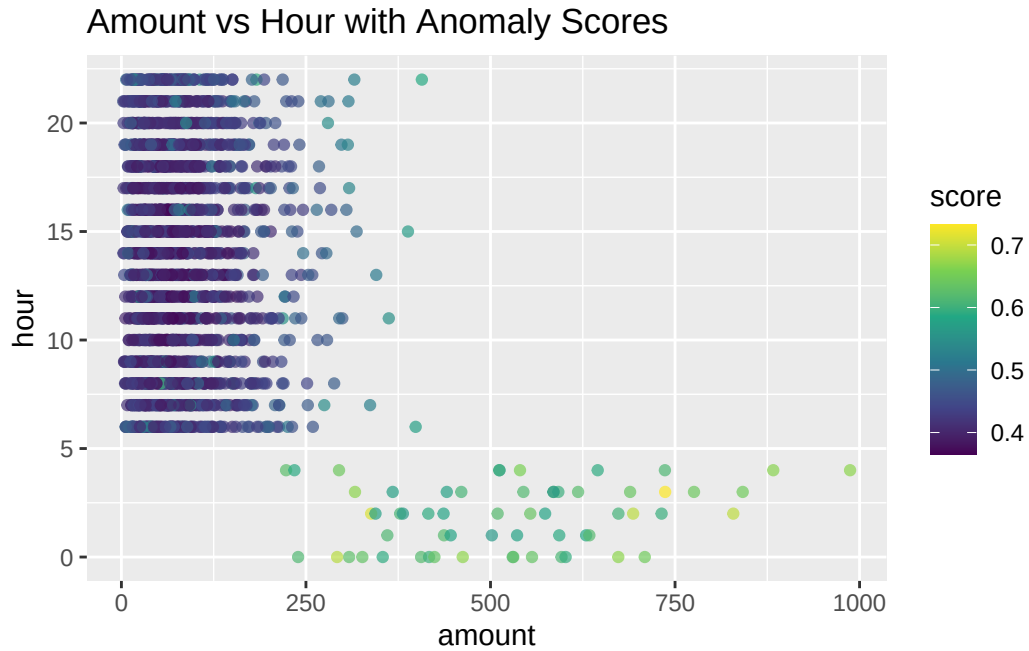
This gives insight into:

- **True positives** (fraud detected)
 - **False positives** (legit transactions flagged)
 - **False negatives** (missed fraud)
-

17.4.6 6. Visual Diagnostics

17.4.7 Scatterplot: Amount vs Hour

```
ggplot(df, aes(amount, hour, color=score)) +  
  geom_point(alpha=0.7) +  
  scale_color_viridis_c() +  
  labs(title="Amount vs Hour with Anomaly Scores")
```



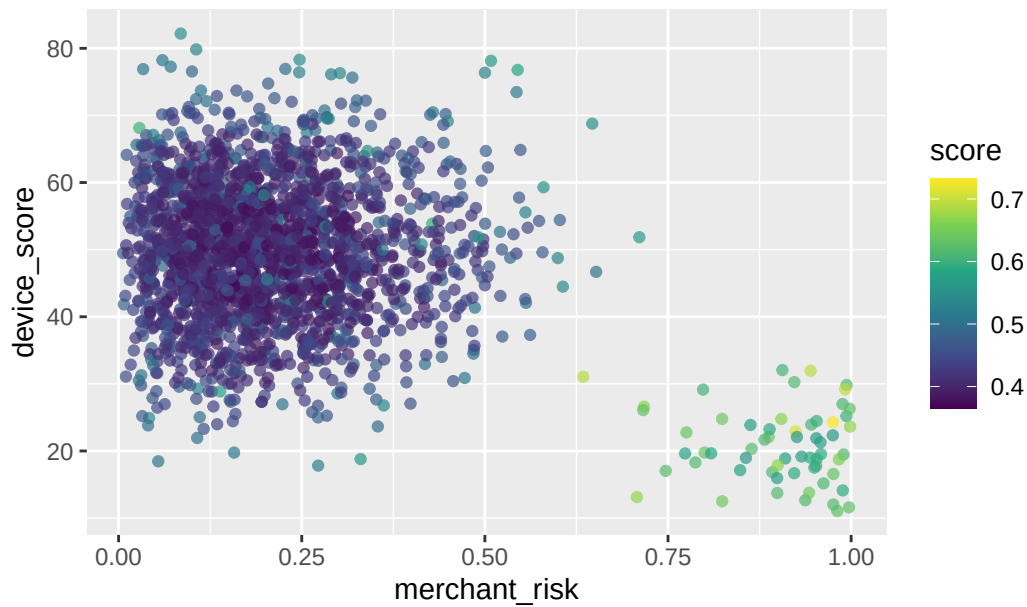
Fraud transactions cluster at:

- Large amounts
- Extreme nighttime hours (0–4 AM)

17.4.8 Scatterplot: Merchant Risk vs Device Score

```
ggplot(df, aes(merchant_risk, device_score, color=score)) +
  geom_point(alpha=0.7) +
  scale_color_viridis_c() +
  labs(title="Risky Merchant vs Bad Device Score (Anomaly Colored)")
```

Risky Merchant vs Bad Device Score (Anomaly Colored)



Fraud behaviors:

- High merchant risk
- Low device trust score

17.4.9 7. Parallel Coordinates

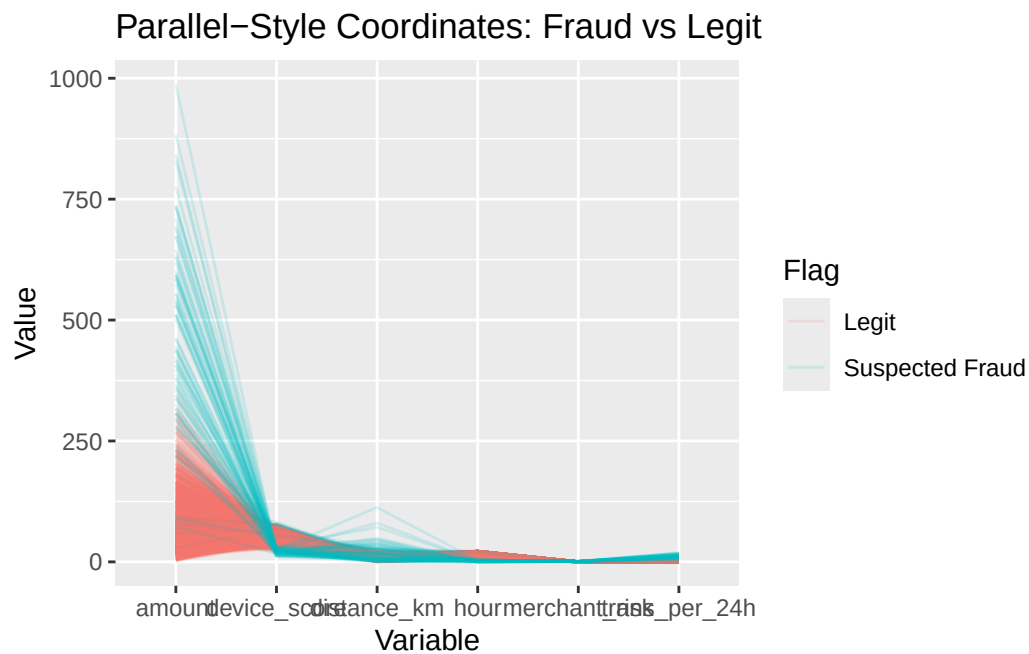
```
df_pc <- df %>%
  mutate(flag = factor(flag, levels=c("Legit","Suspected Fraud"))) %>%
  mutate(id = row_number())

df_long <- df_pc %>%
  pivot_longer(cols = c(amount, hour, merchant_risk, device_score, distance_km, trans_per_24h),
               names_to = "Variable",
               values_to = "Value")

ggplot(df_long, aes(x=Variable, y=Value, group=id, color=flag)) +
  geom_line(alpha=0.15) +
```



```
labs(title="Parallel-Style Coordinates: Fraud vs Legit",
     color="Flag")
```



This reveals fraud profiles across dimensions:

- Higher amount
- Higher merchant risk
- Out-of-bounds distance
- High transaction velocity
- Low device reputation

17.4.10 8. Print Most Suspicious Transactions

```
df %>%
  arrange(desc(score)) %>%
  slice(1:15)
```

```
# A tibble: 15 x 9
```

	amount	hour	merchant_risk	device_score	distance_km	trans_per_24h	true_type
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<int>	<chr>
1	737.	3	0.975	24.3	112.	10	fraud
2	338.	2	0.924	22.9	80.4	5	fraud
3	829.	2	0.992	29.2	0.0882	17	fraud
4	693.	2	0.945	31.9	71.8	10	fraud
5	292.	0	0.634	31.1	46.1	9	fraud
6	462.	0	0.707	13.1	25.8	15	fraud
7	987.	4	0.900	17.9	1.44	10	fraud
8	673.	0	0.999	23.6	27.3	17	fraud
9	316.	3	0.717	26.6	15.6	18	fraud
10	883.	4	0.905	24.8	32.6	7	fraud
11	540.	4	0.983	18.8	8.63	19	fraud
12	842.	3	0.943	13.8	17.8	8	fraud
13	709.	0	0.824	12.5	5.25	12	fraud
14	295.	4	0.887	22.1	2.25	18	fraud
15	776.	3	0.824	24.8	27.8	5	fraud

```
# i 2 more variables: score <dbl>, flag <chr>
```

These represent the most abnormal behavior across multiple dimensions.

17.5 Summary & Business Interpretation

Isolation Forest in fraud analytics:

- Flags **behavior that deviates from customer norms**
- Useful for pre-screening before rule-based or analyst review
- Helps detect evolving fraud patterns not captured by rules
- Imbalanced classes do not harm the model

- Supports many dimensions (amount, risk, device, geography)
-